

Open-FDD Documentation

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1 Home

2 Open-FDD

Open-FDD is an open-source **building edge platform** for BACnet integration, feather historian storage, Arrow-native fault detection, and operator dashboards — designed for **trusted LAN / OT edge** deployment behind Caddy, not direct public internet exposure.

2.1 Who it is for

Role	What you get
IT / controls engineer	Pull three GHCR images, log in, poll BACnet, view faults
Commissioning integrator	Discover points, bind BRICK model, pin FDD rules in Rule Lab
Developer / maintainer	Extend the Operator Bridge, publish images, run CI and LAN security scans

2.2 Start here

Path	Go to
Try it	[Quick Start — Run with Docker]({{ "/quick-start/docker/")
Write FDD rules	[Rule Cookbook — PyArrow & DataFusion SQL]({{ "/rule-cookbook/dual-backend-rules/")
Operate it	<u>Updating the stack</u> ({{ "/quick-start/updating/")
API reference	[Appendix — REST API]({{ "/" appendix/bridge_api/")
Develop it	<u>Developer Guide</u> ({{ "/developer/")

2.3 v3 edge stack

Three primary containers on a Linux edge host:

Container	Role
openfdd-bridge	Operator Bridge — API, dashboard, feather ingest
openfdd-commission	BACnet discover/read/write + poll loop
openfdd-mcp-rag	Doc-search sidecar for agent tools

Host **Caddy** on :80 or :443 is the normal LAN front door. BACnet UDP :47808 is bound by **commission** only — the bridge ingests samples.csv into the feather historian.

Details: [Containers](#)({{ "/architecture/containers/" | relative_url }}) · [Deployment modes](#)({{ "/architecture/deployment-modes/" | relative_url }})

LAN / OT edge only. Do not expose the Operator Bridge to the public internet without TLS, strong auth, and site review. BACnet writes are disabled by default — see [\[BACnet write guard\]](#)({{ "/security/bacnet-writes/" | relative_url }}).

2.4 Distribution

Channel	Use when
GHCR ghcr.io/bbartling/openfdd-*	Production or trial edge deploy
This repository	Custom builds, commissioning, Rule Lab development
PyPI open-fdd	Arrow runtime lint/test without full UI

2.5 License

MIT — see repository LICENSE.

3 Open-FDD Central / Portfolio / RCx reports

4 Open-FDD Central / Portfolio / RCx reports

Planning doc for the next feature branch: `feature/openfdd-central-portfolio-rcx-reports`.

4.1 Boundary

Layer	Role
Open-FDD Edge	Lives at the building. BACnet, local FDD, model/trends/faults, safe REST APIs.
Open-FDD Central	Multi-building desk over Tailscale/VPN. No direct BACnet. No equipment commands. Validation jobs, RCx reports, portfolio health.

Existing code to extend (do not rewrite):

- `portfolio/collector/` — edge login, rollup fetch, CSV history
- `portfolio/dash/` — multi-site Dash dashboard
- `scripts/portfolio_collect.py` — collector CLI
- Edge APIs: `/api/building/portfolio-rollup`, `/api/faults/status`, `/api/fdd/results`

4.2 Incremental build order

1. **Edge registry** — extend `portfolio/sites.json` schema (name, base URL, `site_id`, last check-in).
2. **Remote edge client** — reuse `portfolio/collector/edge_client.py`; add validation job endpoints.
3. **Portfolio page** — edge health, BACnet/model/fault status, last check-in.
4. **One-off validation** — trigger read-only harness against one edge (`acme_overnight_fdd_validate` pattern).

5. **24-hour validation planner** — dropdown intervals (2h / 6h / 12h), store cycle results.
6. **Fault-hour analytics** — elapsed fault hours from historian + FDD runs.
7. **Minimal DOCX generator** — python-docx + matplotlib (BytesIO charts).
8. **Report download endpoint** — query edges via REST, render .docx.
9. **Tests/fixtures** — deterministic edge mocks; no live BACnet in CI.

4.3 RCx report content (normalized roles)

- Elapsed fault hours by equipment/severity
- SAT vs setpoint, duct static vs setpoint
- Fan/motor runtime analytics
- Missing data warnings
- Duplicate/model warnings from edge audit APIs
- **Never** severity-only cards without equipment context

4.4 Prerequisites

- ACME validation PR merged and **3.0.33** deployed (equipment context on live FDD alerts).
- Follow-ups: FDD run-history equipment grouping, RTU role mapping (see [ACME validation follow-ups]({{ "/ops/acme-validation-follow-ups/" | relative_url }})).

4.5 Safety

Central is read-only toward edges. No BACnet writes, commands, or BAS changes from Central.

5 Run with Docker images

6 Run with Docker images

Deploy Open-FDD on a Linux edge host using **three published GHCR images**. No git clone required.

6.1 Images

GHCR image	Service	Role
ghcr.io/bbartling/openfdd-bridge	bridge	Operator API, dashboard, historian ingest
ghcr.io/bbartling/openfdd-commission	commission	BACnet discover/ read/write and poll loop
ghcr.io/bbartling/openfdd-mcp-rag	mcp-rag	Doc-search sidecar

Edge hosts pull **latest** from GHCR (three images: bridge, commission, mcp-rag). The retired **openfdd-bacnet-poll** image is no longer published.

6.2 CPU architecture (auto-detect)

Open-FDD GHCR images are **multi-arch** (linux/amd64 + linux/arm64) from release **3.1.6** onward. Bootstrap and site-update scripts detect the host CPU and set DOCKER_DEFAULT_PLATFORM automatically:

uname -m	Platform pulled
x86_64, amd64	linux/amd64
aarch64, arm64	linux/arm64

Verify manifests before pull (especially on Raspberry Pi):

```
cd ~/open-fdd
./scripts/openfdd_check_ghcr_platform.sh
./scripts/openfdd_check_ghcr_platform.sh --platform linux/arm64
# explicit check
OPENFDD_IMAGE_TAG=3.1.6 ./scripts/openfdd_check_ghcr_platform.sh
```

Override (QEMU / lab only):

Variable / flag	Example
OPENFDD_DOCKER_PLATFORM	OPENFDD_DOCKER_PLATFORM=linux/ amd64 docker compose pull
Bootstrap --platform	bash .../ openfdd_edge_bootstrap.sh -- start --platform linux/arm64

Variable / flag	Example
Site update	OPENFDD_DOCKER_PLATFORM=linux/ amd64 ./scripts/ openfdd_site_update.sh

ARM64 pull errors → [Raspberry Pi edge bootstrap](#)({ { "/quick-start/raspberry-pi-edge/" | relative_url } }).

6.3 1. Install and validate Docker

```
sudo systemctl enable --now docker
sudo usermod -aG docker "$USER"
newgrp docker    # or log out/in

docker --version
docker compose version
docker ps
docker run --rm hello-world
```

Expect: active / enabled for docker, docker in groups, hello-world succeeds.

6.4 2. Bootstrap the site (one script)

Downloads docker-compose.yml, creates workspace/, generates auth.env.local, and sets BACNET_BIND from the host LAN NIC (e.g. ens192 → 10.200.200.185/24:47808):

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
    https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_edge_bootstrap.sh
bash /tmp/openfdd_edge_bootstrap.sh
```

Bootstrap + pull + start in one go:

```
bash /tmp/openfdd_edge_bootstrap.sh --start
```

Auth — bootstrap writes ~/open-fdd/workspace/auth.env.local (gitignored). This file holds the API signing secret and one username/password per role. Use it for your first dashboard login:

```
cat ~/open-fdd/workspace/auth.env.local
```

Variable	Role
OFDD_AUTH_SECRET	Signs session tokens (do not share)
OFDD_OPERATOR_*	Read-only operator
OFDD_INTEGRATOR_*	

Variable	Role
	Default dashboard login — commissioning + BACnet
OFDD_AGENT_*	Agent / automation

Example shape (passwords are **random on first bootstrap only** — existing file is kept on re-runs):

```
OFDD_AUTH_SECRET=<random>
OFDD_OPERATOR_USER=operator
OFDD_OPERATOR_PASSWORD=<random>
OFDD_INTEGRATOR_USER=integrator
OFDD_INTEGRATOR_PASSWORD=<random>
OFDD_AGENT_USER=agent
OFDD_AGENT_PASSWORD=<random>
```

Sign in as **integrator** with the password from that file → First login and health check({{ "/quick-start/health-check/" | relative_url }}).

Action	Touches auth.env.local?
First bash ... bootstrap.sh --start	Creates file with random passwords (if missing)
bash ... bootstrap.sh --start again	Keeps your file — does not overwrite
bash ... bootstrap.sh --restart	Keeps your file — only restarts bridge to reload it
nano auth.env.local + --restart	Uses your edited passwords
--force-auth	Regenerates random passwords (opt-in only)

The **bridge** container reads auth.env.local at startup only. After you change passwords (manual edit or --force-auth), recreate bridge (env files are not reloaded by restart alone):

```
cd ~/open-fdd && docker compose up -d --force-recreate bridge
```

Regenerate passwords and reload (stack already running):

```
bash /tmp/openfdd_edge_bootstrap.sh --force-auth --restart --show-secrets
```

Manual edit then reload:

```
nano ~/open-fdd/workspace/auth.env.local
bash /tmp/openfdd_edge_bootstrap.sh --restart
```

Options:

Flag	Purpose
--start	docker compose pull && up -d after layout; curls http://127.0.0.1:8765/health
--image-tag TAG	default latest
--platform PLAT	auto (default), linux/arm64, linux/amd64 — or set OPENFDD_DOCKER_PLATFORM
--repo-ref BRANCH	branch for compose.edge.yml if not on master yet
--force-auth	regenerate auth.env.local (restarts bridge if stack is up)
--restart	recreate bridge to reload auth.env.local; curls /health after
--show-secrets	print passwords at end (lab only)

From a repo checkout: `./scripts/openfdd_edge_bootstrap.sh --start`

The script prints **BACnet NIC**, **BACNET_BIND**, and file paths — **validate** `commission.env` before polling OT devices:

```
nano ~/open-fdd/workspace/bacnet/commissioning/commission.env
```

If you used `--start`, the stack is already up. Next step → First login and health check({{ `"/quick-start/health-check/"` | `relative_url` }}).

6.4.1 Manual start (optional)

Use this only if you ran bootstrap **without** `--start` and want to bring the stack up yourself:

```
cd ~/open-fdd
./scripts/openfdd_check_ghcr_platform.sh
docker compose pull
docker compose up -d
docker compose ps
curl -sf http://127.0.0.1:8765/health && echo
```

Expected: **bridge**, **commission**, **mcp-rag** — all up. Then → First login and health check({{ `"/quick-start/health-check/"` | `relative_url` }}).

6.5 Long-term operation

6.5.1 Survive power cycles

All services use restart: unless-stopped. After reboot:

```
cd ~/open-fdd && docker compose ps
curl -sf http://127.0.0.1:8765/health
```

6.5.2 Start / stop / restart

```
cd ~/open-fdd
docker compose up -d          # idempotent
docker compose stop
docker compose restart commission
docker compose logs -f --tail 100 commission
```

Never run `docker compose down -v` or delete workspace/ on a live site.

6.5.3 LAN access

Put **Caddy** (or nginx) on port 80 → 127.0.0.1:8765, or expose 8765 through the firewall.

6.6 Deployment choices

Mode	Compose services	When to use
Standard OT / BACnet	bridge + commission + mcp-rag	Real buildings — commission polls BACnet into samples.csv; bridge ingests to feather
Non-BACnet / demo	bridge + mcp-rag (omit commission)	JSON API, Modbus-only, or lab without field BACnet — see Driver framework ({{ "/" drivers/
TLS on OT LAN	Same stack + host Caddy OFDD_CADDY_MODE=tls	Encrypt browser↔edge; trust self-signed CA on operator PCs — not for public internet

Do **not** run the retired `openfdd-bacnet-poll` image or legacy `openfdd-bacnet-poll` systemd unit alongside Docker **commission** — that double-polls BACnet.

6.7 Next steps

→ First login and health check({{ `"/quick-start/health-check/"` | `relative_url` }})
→ Updating the stack({{ `"/quick-start/updating/"` | `relative_url` }})
— `./scripts/openfdd_site_update.sh`

7 First login and health check

8 First login and health check

Quick checks after `docker compose up -d` on the edge host. Services use `restart: unless-stopped` — after a reboot, run `docker compose ps` once Docker is up (see [Run with Docker]({{ `"/quick-start/docker/"` | `relative_url` }}#survive-power-cycles)).

8.1 Public health

```
curl -s http://127.0.0.1:8765/health
```

Through Caddy on port 80:

```
curl -s http://127.0.0.1/health
```

Expect `"ok": true` and `"auth_required": true` when auth is configured.

8.2 Login

1. Open the dashboard (`http://<host-lan-ip>/` or `:8765`).
2. Sign in with credentials from `workspace/auth.env.local`.

```
curl -s -X POST http://127.0.0.1:8765/api/auth/login \
-H 'Content-Type: application/json' \
-d '{"username":"integrator","password":"<from-auth-env>"}
```

8.3 Smoke checklist

Check	Pass
docker compose ps	bridge, commission, mcp-rag Up
GET /health	200, ok: true
Login	Token returned
BACnet poll	samples.csv growing (tail -1 workspace/bacnet/ polls/samples.csv)
Historian	Files under workspace/data/ feather_store/

8.4 Stack detail (authenticated)

Integrator token required:

```
TOKEN=$(curl -sf -X POST http://127.0.0.1:8765/api/auth/login \  
-H 'Content-Type: application/json' \  
-d '{"username":"integrator","password":"..."}' | jq -r .token)  
curl -sf http://127.0.0.1:8765/health/stack -H "Authorization:  
Bearer $TOKEN" | jq .
```

Look for bacnet_poll green/yellow in the services list.

8.5 Troubleshooting

Symptom	Fix
401 everywhere	Configure workspace/ auth.env.local
Empty BACnet discover	Fix BACNET_BIND in commission.env — <u>network setup</u> ({{ “/ bacnet/network- setup/”
No poll rows	Commission container running? points.csv has enabled rows?
LAN browser timeout	Open firewall port 80 or 8765

Advanced probes (model graph, Rule Lab, compose profiles):
[Developer health check]({{ "/developer/health-check/" | relative_url }}).

9 Quick Start

10 Quick Start

Run Open-FDD on a Linux edge host using **three GHCR Docker images**. No git clone on the host.

Step	Page
1	Run with Docker images ({{ "/quick-start/docker/"
2	First login and health check ({{ "/quick-start/health-check/"
3	Edge site lifecycle ({{ "/quick-start/site-lifecycle/"
4	Updating the stack ({{ "/quick-start/updating/"
—	[Raspberry Pi edge] ({{ "/quick-start/raspberry-pi-edge/"

One-liner bootstrap (edge host):

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_edge_bootstrap.sh
bash /tmp/openfdd_edge_bootstrap.sh --start
```

Prerequisites: Linux, Docker enabled on boot, BACnet LAN if polling OT equipment.

BACnet writes are disabled by default. See [Write safety](#)({{ "/bacnet/write-safety/" | relative_url }}).

Stack: bridge + commission (BACnet + poll) + mcp-rag. Details: [Containers](#)({{ "/architecture/containers/" | relative_url }}).

11 Updating the stack

12 Updating the stack

Upgrade GHCR images on the edge host: **backup workspace/, safe Docker maintenance, pull new tags, validate, purge backup.**
No git pull on the host.

Full reference: [Edge site lifecycle](#)({{ "/quick-start/site-lifecycle/" | relative_url }}).

12.1 Before you start

1. SSH to the edge host (cd ~/open-fdd).
2. Pick a tag from [GitHub Packages](#) — default is **latest**.
3. Short maintenance window (containers restart briefly;
restart: unless-stopped brings them back after reboot).

12.2 One-command upgrade (recommended)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

The update script prunes unused Docker images, pulls **:latest** (or NEW_TAG), recreates containers, validates /health, and **removes the backup directory on success**.

12.3 Environment variables

Variable	Default	Use
NEW_TAG	latest	Pin GHCR tag
BACKUP_INCLUDE_POLL_SAMPLES=0	1	Fast backup (skip poll CSV history)
SKIP_DOCKER_MAINTENANCE=1	0	Skip image prune
PURGE_BACKUP_AFTER_SUCCESS=0	1	

Variable	Default	Use
RESTORE_WORKSPACE=1	0	Keep backup after upgrade Restore workspace/ from backup before recreate
RESTORE_FEATHER_MAX_GIB	200	Historian cap on restore; 0 = all data

12.4 Verify

```
docker compose ps
curl -sf http://127.0.0.1:8765/health | jq .
ls -lah workspace/data/feather_store/
```

→ [Health check]({{ "/quick-start/health-check/" | relative_url }})

12.5 Rollback

Images: restore docker-compose.yml from .bak.* snapshot or re-run with a known-good NEW_TAG.

Data: if validation failed, backup is still at ~/openfdd-backups/latest. Restore with:

```
RESTORE_WORKSPACE=1 ./scripts/openfdd_site_update.sh
```

Never run `docker compose down -v`, `docker volume prune`, or `delete workspace/` on a live site.

12.6 Next steps

→ [Edge site lifecycle](#)({{ "/quick-start/site-lifecycle/" | relative_url }})
 → [Run with Docker images](#)({{ "/quick-start/docker/" | relative_url }})
 → [Health check]({{ "/quick-start/health-check/" | relative_url }})

13 Edge site lifecycle

14 Edge site lifecycle

Professional reference for **bootstrap**, **backup**, **image update**, and **restore** on a production edge host. No git clone required on the host — scripts and GHCR images come from GitHub.

14.1 Overview

Phase	Script	What it does
First install	openfdd_edge_bootstrap.sh	Layout, compose, auth, BACnet bind, optional docker compose up
Pre-upgrade safety	openfdd_site_backup.sh	Rolling archive of workspace/ to ~/openfdd-backups/latest
Upgrade	openfdd_site_update.sh	Safe Docker prune, GHCR pull, health validation, purge backup on success
Recovery	openfdd_site_update.sh with RESTORE_WORKSPACE=1	Restore workspace/ from backup (optional historian cap)

Never on live sites: docker compose down -v, docker volume prune, or deleting workspace/. Historian data lives in the bind-mounted workspace/data/feather_store/.

14.2 1. Bootstrap (new host)

One-shot install from GHCR — does **not** clone the repository:

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \  
https://github.com/bbartling/open-fdd/raw/refs/heads/master/  
scripts/openfdd_edge_bootstrap.sh  
bash /tmp/openfdd_edge_bootstrap.sh --start
```

Creates:

Path	Purpose
~/open-fdd/docker-compose.yml	Bridge, commission, MCP stack
~/open-fdd/workspace/	Persistent site state (historian, model, auth)
~/open-fdd/scripts/openfdd_site_*.sh	Backup and update helpers

Options: --image-tag, --platform, --root, --force-auth, --restart.
See script header (--help).

Scripts auto-detect CPU (x86_64 → linux/amd64, aarch64 → linux/arm64). Verify GHCR manifests:

```
cd ~/open-fdd
./scripts/openfdd_check_ghcr_platform.sh
```

After bootstrap → [Health check]({{ "/quick-start/health-check/" | relative_url }}).

14.3 2. Backup before every upgrade

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Default destination: ~/openfdd-backups/latest (overwritten each run — one rolling copy).

Variable	Default	Meaning
BACKUP_ROOT	~/openfdd-backups/latest	Output directory
BACKUP_INCLUDE_POLL_SAMPLES	1	Set 0 to skip large workspace/bacnet/polls/
BACKUP_TIMEOUT_SECS	1800	tar timeout
BACKUP_TAR_XATTRS	0	Set 1 for --xattrs --acls

Archive: workspace-full.tgz plus backup-manifest.txt and Docker state snapshots.

14.4 3. Upgrade (recommended flow)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

openfdd_site_update.sh performs:

1. **Verify backup** — integrity check on workspace-full.tgz
2. **Safe Docker maintenance** — prune stopped containers, dangling images, and **unused images** (never volumes)
3. **Pull & recreate** — all three GHCR services for NEW_TAG (default latest)
4. **Validate** — workspace/ layout + GET /health
5. **Purge backup** — removes BACKUP_ROOT after successful validation (frees disk)

Pin a release tag:

```
export NEW_TAG=3.1.6
./scripts/openfdd_site_update.sh
```

Variable	Default	Meaning
NEW_TAG / OPENFDD_IMAGE_TAG	latest	GHCR image tag
OPENFDD_DOCKER_PLATFORM	auto	linux/arm64, linux/amd64, or auto-detect from uname -m
BACKUP_ROOT	~/openfdd- backups/latest	Backup used for validation / restore
SKIP_DOCKER_MAINTENANCE	0	Set 1 to skip prune
PURGE_BACKUP_AFTER_SUCCESS	1	Set 0 to keep backup after upgrade
REQUIRE_BACKUP	same as purge	Set 0 to upgrade without a prior backup

If validation fails, the backup is **kept** and the script prints a restore command.

14.5 4. Restore from backup

Use when workspace/ was corrupted, or you need to roll back **data** (not just images).

```
cd ~/open-fdd
RESTORE_WORKSPACE=1 ./scripts/openfdd_site_update.sh
```

This stops containers, extracts the backup, applies an optional historian cap, recreates containers, validates, and purges the backup on success.

14.5.1 Historian size cap on restore

Large sites may not need full feather history on a small edge disk. Default: **keep the newest ~200 GiB** of feather_store shard files; drop oldest shard-*.feather files until under cap.

Variable	Default	Meaning
RESTORE_FEATHER_MAX_GIB	200	Max historian size after restore
RESTORE_FEATHER_MAX_GIB=0	—	Restore all feather data (no cap)

Example — full historian restore:

```
RESTORE_WORKSPACE=1 RESTORE_FEATHER_MAX_GIB=0 ./scripts/
openfdd_site_update.sh
```

Runtime historian caps (ongoing poll) use OFDD_FEATHER_MAX_GIB in bridge env — see [Configuration reference](#)({{ "/appendix/configuration/" | relative_url }}).

14.6 5. Fetch scripts without bootstrap

If scripts are missing:

```
mkdir -p ~/open-fdd/scripts
BASE=https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts
for f in openfdd_site_lib.sh openfdd_site_backup.sh
openfdd_site_update.sh openfdd_check_ghcr_platform.sh; do
    curl -fsSL -o ~/open-fdd/scripts/"$f" "$BASE/$f"
done
chmod +x ~/open-fdd/scripts/openfdd_site_*.sh
```

14.7 Related

- [Updating the stack](#)({{ "/quick-start/updating/" | relative_url }}) — operator checklist
- [Backup and restore](#)({{ "/ops/backup-restore/" | relative_url }}) — Ansible / bensserver paths
- [\[Health check\]](#)({{ "/quick-start/health-check/" | relative_url }})
- [REST API reference](#)({{ "/appendix/bridge_api/" | relative_url }})

15 Raspberry Pi edge bootstrap

16 Raspberry Pi edge bootstrap

Open-FDD edge deploy on **Raspberry Pi 4/5 (64-bit)** uses the same bootstrap as x86 Linux. Docker must be installed; the host must be **aarch64 / arm64** (not 32-bit Raspberry Pi OS).

Bootstrap and update scripts **auto-detect** linux/arm64 on Pi hosts. No --platform flag is required when GHCR publishes ARM64 manifests for your tag (3.1.6+).

16.1 One-liner

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \  
https://github.com/bbartling/open-fdd/raw/refs/heads/master/  
scripts/openfdd_edge_bootstrap.sh  
bash /tmp/openfdd_edge_bootstrap.sh --start
```

16.2 Error: no matching manifest for linux/arm64/v8

This means **GHCR does not yet have an ARM64 build** for that image tag (not a Docker install problem).

Error response from daemon: no matching manifest for linux/arm64/v8 in the manifest list entries

16.2.1 Confirm

```
cd ~/open-fdd
./scripts/openfdd_check_ghcr_platform.sh
# explicit: ./scripts/openfdd_check_ghcr_platform.sh --platform
linux/arm64
```

Or manually:

```
docker buildx imagetools inspect ghcr.io/bbartling/openfdd-
bridge:latest | grep -E 'linux/arm64|linux/amd64'
```

You need **linux/arm64** on all three: openfdd-bridge, openfdd-commission, openfdd-mcp-rag.

16.2.2 Option A — Native ARM64 images (recommended)

Open-FDD publishes **multi-arch** images (amd64 + arm64) from the GitHub **Publish Docker images to GHCR** workflow. After a release that includes ARM64:

```
bash /tmp/openfdd_edge_bootstrap.sh --start
```

16.2.3 Option B — QEMU emulation (lab only, slow)

Use only while waiting for ARM64 publish:

```
sudo apt update
sudo apt install -y qemu-user-static binfmt-support
docker run --privileged --rm tonistiigi/binfmt --install amd64

cd ~/open-fdd
OPENFDD_DOCKER_PLATFORM=linux/amd64 docker compose pull
OPENFDD_DOCKER_PLATFORM=linux/amd64 docker compose up -d

docker compose ps
curl -sf http://127.0.0.1:8765/health | jq .
```

Expect slower startup and higher CPU use. Not recommended for production BACnet polling.

16.2.4 Option C — Build on the Pi

```
git clone https://github.com/bbartling/open-fdd.git
cd open-fdd
./scripts/docker_build.sh
# point compose at local tags – see scripts/docker_build.sh header
```

16.3 OpenClaw / AI agent on Pi

1. Bootstrap must finish with `docker compose ps` showing bridge, commission, mcp-rag **Up**.
2. Agent needs **terminal/shell** access on the Pi (not chat-only) to run docker and curl.
3. Use the [README OpenClaw bootstrap prompt](#) — it checks `uname -m` and GHCR pull before declaring success.
4. Login: integrator / password in `~/open-fdd/workspace/auth.env.local` (never paste passwords into chat).

16.4 Validate

```
docker compose ps
curl -sf http://127.0.0.1:8765/health | jq .
grep BACNET_BIND ~/open-fdd/workspace/bacnet/commissioning/
commission.env
```

Dashboard: `http://<pi-ip>:8765/` (or Caddy on `:80` when configured).

16.5 Related

- [Edge site lifecycle](#)(({ { "/quick-start/site-lifecycle/" | relative_url } })
- [Run with Docker](({ { "/quick-start/docker/" | relative_url } })

17 Local development

18 Local development

18.1 Clone and auth

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local
# Edit integrator/operator passwords and OFDD_AUTH_SECRET (32+
chars)
```

18.2 Start stack

```
./scripts/build_and_test.sh           # dashboard build + pytest
./scripts/docker_build.sh
./scripts/openfdd_stack.sh up
./scripts/stack_health_check.sh
```

URL	Service
http://127.0.0.1:8765	Bridge (direct)
http://127.0.0.1:8765/docs	OpenAPI

18.3 BACnet bench overlay (optional)

```
docker compose -f docker/compose.dev.yml -f docker/
compose.bench.yml up -d
# commission.env – set BACNET_BIND to your OT interface
```

18.4 Production-like local stack (Caddy, LAN scans)

Same shape as an Acme edge deploy: **prod React UI, Caddy on :80**, bridge on loopback **8765**, auth enabled. Use for OWASP ZAP or nmap from another machine on the LAN.

```
./scripts/pentest_production_stack.sh start    # build prod UI +
Caddy + auth.pentest.local
./scripts/pentest_production_stack.sh status   # bridge,
commission, caddy, mcp-rag
./scripts/pentest_production_stack.sh verify   # LAN IP, /health,
ZAP target URL
./scripts/pentest_production_stack.sh stop     # when done
```

Credentials: workspace/auth.pentest.local (generated on first start). ZAP target is usually http://<lan-ip>/ (benserver example: http://192.168.204.18/). If LAN clients cannot reach :80, run `sudo ./scripts/open_lan_port.sh 80`.

See also [Health checks \(developer\)](#)(({ { "/developer/health-check/" | relative_url } }) for probe details and [ZAP baseline expectations](({ { "/security/zap-baseline/" | relative_url } }) for accepted Medium/Low findings.

18.5 Docs preview

```
cd docs && bundle install && bundle exec jekyll serve  
# http://127.0.0.1:4000/open-fdd/
```

19 Build Docker images

20 Build Docker images

20.1 Local build

```
./scripts/docker_build.sh  
# Optional tar for air-gap:  
./scripts/docker_build.sh --save
```

Images: openfdd-bridge, openfdd-commission, openfdd-mcp-rag.

20.2 Publish to GHCR (maintainers)

GitHub Actions workflow **Publish Docker images to GHCR** — manual dispatch or git tag vX.Y.Z. Publishes **pyproject.toml version** (e.g. 3.1.3), minor alias (3.1), and **latest**, then prunes old GHCR versions.

Maintainer checklist:

1. Green CI on master
2. Run **Publish Docker images to GHCR** workflow (or push tag vX.Y.Z)
3. Verify ghcr.io/bbartling/openfdd-**{bridge,commission,mcp-rag}**: 3.1.3 and :latest
4. Edge hosts: OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_ghcr.sh --limit <host>

Details live in .github/workflows/ and scripts/docker_build.sh — not duplicated here.

21 Developer Guide

22 Developer Guide

Clone, build, test, and contribute to Open-FDD.

Topic	Page
Local setup	Local development ({{ "/" developer/local-setup/"
Docker builds	Build Docker images ({{ "/" developer/docker-build/"
Health checks	Health checks (developer) ({{ "/"developer/health- check/"
Tests & CI	Tests and CI ({{ "/"developer/ tests-ci/"
Security testing	[Release & LAN scan cycle] ({{ "/"developer/security- testing/"
PRs	Contributing ({{ "/"developer/ contributing/"

AI-assisted contributors: see repository AGENTS.md for workspace layout, skill routing, and agent policy (one paragraph in public docs; details stay internal).

23 Tests and CI

24 Tests and CI

24.1 Local tests

```
./scripts/build_and_test.sh # UI + workspace bridge
pytest
python3 -m pytest open_fdd/tests -q # PyPI package tests
cd workspace/dashboard && npm test # dashboard unit tests (if
configured)
```

24.2 CI (GitHub Actions)

Workflow	Trigger	Checks
ci.yml	PR / push	pytest, bridge security audit, dashboard build, Jekyll docs
docs-pages.yml	push master	GitHub Pages site
publish-open-fdd.yml	tag v* / open-fdd-v*	PyPI wheel (Trusted Publishing)
docker-publish.yml	tag v* / manual	GHCR bridge, commission, mcp-rag

24.3 Docs build (same as CI)

```
cd docs && bundle exec jekyll build -d /tmp/open-fdd-site
```

Fix any template or link errors before opening a PR that touches docs/.

24.4 Security-related CI

ci.yml includes bridge security tests (tests/workspace_bridge/test_security.py). That complements — but does not replace — LAN [ZAP + Nmap scans]({{ "/security/zap-baseline/" | relative_url }}) on each patch revision. See [Security testing cycle](#)({{ "/developer/security-testing/" | relative_url }}).

25 Contributing

26 Contributing

26.1 Expectations

- Open an issue or discuss large changes before big refactors.
- Keep PRs focused; include test or docs updates when behavior changes.

- Do not commit secrets (auth.env.local, Ansible secrets/*.env.local, commissioning CSVs with real site data).
- Follow existing code style in workspace/api/ and workspace/dashboard/.

26.2 Pull request checklist

☐

Tests pass locally (./scripts/build_and_test.sh or scoped pytest)

☐

Docs updated if user-facing behavior changed

☐

docs/ Jekyll build succeeds

☐

No credentials in diff

☐

Auth/Caddy/header changes: re-run [pentest verify + LAN ZAP]({{ "/developer/security-testing/" | relative_url }}) and update [ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}) if expectations changed

26.3 Repository layout (short)

Path	Contents
workspace/api/	Operator Bridge (FastAPI)
workspace/dashboard/	React SPA
open_fdd/	PyPI package (arrow_runtime, playground)
docker/	Compose files and image contexts
infra/ansible/	Edge deploy playbooks
docs/	This site (Jekyll + Just the Docs)

26.4 Docs links (GitHub Pages)

The site is a **project page** at <https://bbartling.github.io/open-fdd/> (baseurl: /open-fdd in docs/_config.yml).

- **Do not** use {% raw %}/path/{% endraw %} in Markdown body text — it omits baseurl and 404s on GitHub Pages.
- **Do** use [label]({{ "/section/page/" | relative_url }}) for internal links.
- After `bundle exec jekyll build`, run `python3 scripts/check_docs_internal_links.py --site _site` from the repo root.

26.5 Security issues

Do not file public issues for vulnerabilities. Contact the maintainer or use GitHub private security advisories.

26.6 Agent-assisted development

Cursor/Claude contributors: read AGENTS.md and skills/ for automation boundaries. Public docs intentionally avoid agent playbook detail.

27 Health checks (developer)

28 Health checks (developer)

Extended validation for engineers with a **full git checkout** — CI, compose overlays, and pre-release gates.

IT operators on edge hosts should use [Quick Start — health check]({{ "/quick-start/health-check/" | relative_url }}) and Deployment validation({{ "/ops/deployment-validation/" | relative_url }}) instead.

28.1 Clone and local stack

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local #
loopback dev only
./scripts/docker_build.sh # or: ./
scripts/openfdd_stack.sh prod
./scripts/openfdd_stack.sh up
```

GHCR production pull (no local build):

```
OPENFDD_IMAGE_TAG=2026.06.07-edge ./scripts/openfdd_stack.sh prod
```

28.2 Stack health script

```
./scripts/stack_health_check.sh
OPENFDD_BASE_URL=http://192.168.204.18:8765 ./scripts/
stack_health_check.sh
./scripts/stack_health_check.sh --require-ollama
```

Probes: containers up, /health, /api/model/health, sites, tree, SPARQL graph (with integrator auth when configured).

28.3 Compose overlays

Overlay	Use
docker/compose.dev.yml	Local dev bind-mount
docker/compose.bench.yml	Host-network BACnet lab
docker/compose.prod.yml	GHCR images instead of local build
docker/compose.edge.yml	IT edge template (copy to ~/open-fdd/ docker-compose.yml)

```
docker compose -f docker/compose.dev.yml -f docker/  
compose.bench.yml ps  
docker compose -f docker/compose.dev.yml config --images
```

28.4 BACnet poll pipeline

```
tail -1 workspace/bacnet/polls/samples.csv  
cat workspace/data/bacnet_ingest_state.json  
docker compose logs --since 10m commission | grep -i poll  
docker compose logs --since 10m bridge | grep -i ingest
```

28.5 pytest (bridge security / BACnet)

```
PYTHONPATH=workspace/api pytest tests/workspace_bridge/  
test_security.py -q
```

28.6 Pentest production stack (LAN ZAP)

```
./scripts/pentest_production_stack.sh start  
./scripts/pentest_production_stack.sh verify
```

Then from a **LAN workstation**, run the packaged scan:

- Windows: scripts/security/Run-OpenFddSecurityScan.ps1
- macOS/Linux: scripts/security/run_openfdd_security_scan.sh

See [scripts/security/README.md](#), [Security — ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}), and the full [security testing cycle](#)({{ "/developer/security-testing/" | relative_url }}).
Host-side notes: [workspace/deploy/PENTEST.md](#).

28.7 Post-deploy insurance (inventory hosts)

For automated checks against named inventory hosts, see [infra/ansible/scripts/post_deploy_check.sh](#) (maintainer tooling — not required for IT docker-pull sites).

29 Security testing cycle

30 Security testing cycle

How Open-FDD validates each **patch revision** before it ships to the live Acme HVAC bench — local automation, AI-assisted PR review, LAN passive scans, and edge post-deploy checks.

Related security docs: [ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}), [TLS and certificates](#)({{ "/security/tls-and-certs/" | relative_url }}), [LAN hardening](#)({{ "/security/lan-hardening/" | relative_url }}), [Linux host hardening](#)({{ "/security/linux-host-hardening/" | relative_url }}), [Tenable remediation]({{ "/security/tenable-remediation/" | relative_url }}), [Authenticated scanning \(roadmap\)](#)({{ "/security/authenticated-scanning/" | relative_url }}).

30.1 Who owns what

Concern	Owner	Where
HTTP security headers (CSP, X-Frame, COOP, ...)	Bridge middleware	workspace/api/openfdd_bridge/security_headers.py
TLS termination + HSTS	Caddy	workspace/deploy/caddy/ , Ansible Caddyfile.j2
OT LAN TLS certificates	Site integrator / maintainer	./scripts/setup_caddy_certs.sh → workspace/deploy/caddy/certs/

Concern	Owner	Where
Pentest credentials	Generated per stack	Ansible /etc/ openfdd/caddy/ on edge workspace/ auth.pentest.local (never commit)
LAN ZAP + Nmap scans	Developer workstation on test LAN	scripts/security/

See [TLS and certificate ownership]({{ "/security/tls-and-certs/" | relative_url }}) for cert lifecycle.

30.2 Per-revision workflow

Typical cycle on a **control machine** (local dev) through a **lab edge host** (live bench):

flowchart LR

```

A[Local dev + pytest] --> B[Pentest stack verify]
B --> C[LAN ZAP + Nmap]
C --> D[PR + GitHub CI]
D --> E[CodeRabbit AI review]
E --> F[Merge + tag open-fdd-vX.Y.Z]
F --> G[GHCR + PyPI publish]
G --> H[Acme upgrade + post_deploy_check --full]
H --> I[Bump next patch locally]

```

30.2.1 1. Local functional tests

```

./scripts/build_and_test.sh
PYTHONPATH=workspace/api pytest tests/workspace_bridge/
test_security.py -q
infra/ansible/scripts/bench_5007_arrow_battery.sh # when FDD/
Arrow paths change

```

30.2.2 2. Pentest production stack (on Open-FDD host)

```

./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify

```

Confirms Caddy :80 fronts the bridge, auth is enabled, BACnet writes disabled, and security headers are singular (no Caddy duplicates). Details: [Health checks — pentest]({{ "/developer/health-check/" | relative_url }})#pentest-production-stack-lan-zap).

30.2.3 3. Host + exposure checks (on edge VM)

After IT Nessus scans or before Acme deploy sign-off:

```
./scripts/security/check_host_security.sh
./scripts/security/check_openfdd_exposure.sh --edge-ip <lan-ip>
```

Host remediation (dry-run first): `./scripts/security/remediate_ubuntu_host.sh`

30.2.4 4. LAN security scan (from laptop)

Use the packaged scripts — do not hand-roll one-off docker/nmap commands unless debugging:

Platform	Command
Windows	<code>.\scripts\security\Run-OpenFddSecurityScan.ps1</code>
macOS / Linux	<code>./scripts/security/run_openfdd_security_scan.sh</code>

Setup (Docker Desktop, Nmap): [scripts/security/README.md](#).

Target URL is usually `http://<lan-ip>/` (Caddy), not `:8765`.

Review `openfdd-security-report/90-quick-findings-summary.txt` and compare against [ZAP baseline expectations]({{ "/security/zap-baseline/" | relative_url }}).

30.2.5 5. Pull request + automated review

- Open PR against master
- **GitHub Actions** (`ci.yml`): pytest, bridge security audit, dashboard build, Jekyll docs
- **CodeRabbit** (or similar AI PR review): style, security footguns, missing tests/docs

Do not merge with failing CI or unresolved duplicate-header regressions.

30.2.6 6. Release

```
git tag open-fdd-v3.0.4 # example
git push origin open-fdd-v3.0.4
```

Tag triggers GHCR `:latest` and PyPI publish workflows.

30.2.7 7. Lab edge validation

After pulling new images on a live bench VM:

```

OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
    infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full

```

Re-run LAN ZAP + Nmap against the edge LAN IP if Caddy or auth behavior changed. Site-specific example: [GL36 lab note] ({{ "/examples/acme-gl36-lab/" | relative_url }}).

30.2.8 8. Next cycle

Bump patch version locally (pyproject.toml, open_fdd/__init__.py) on master for the next development round — do not tag until the next release is ready.

30.3 What each scan layer covers

Layer	Tool	Depth
Unit / integration	pytest, test_security.py	Header middleware, auth gates
Stack smoke	pentest_production_stack.sh verify	Live headers, health, auth file
Passive web	ZAP baseline	Public routes, CSP, cookies, CORS
Port exposure	Nmap scoped	One host, known service ports
Authenticated API/dashboard	ZAP full + login	Roadmap — [Authenticated scanning]({{ "/security/authenticated-scanning/"

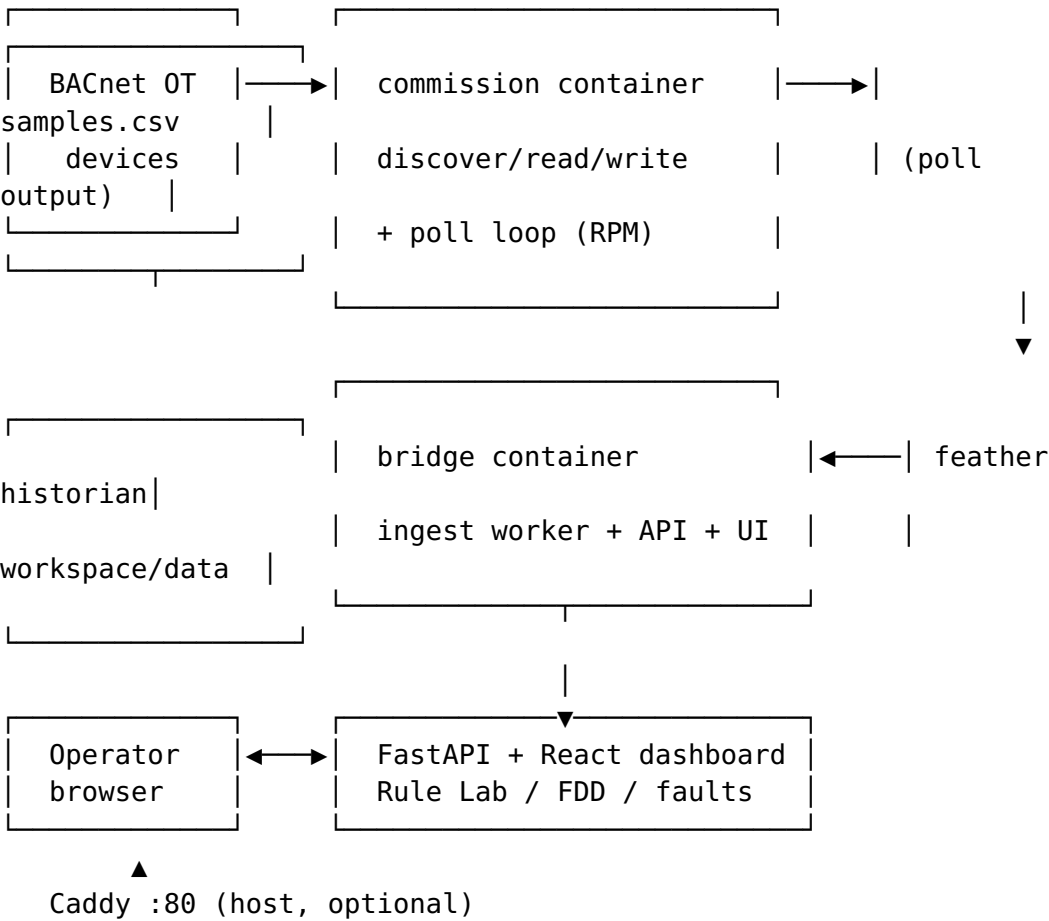
Baseline ZAP does **not** exercise integrator login or protected /api/* routes deeply.

30.4 Open-source contributors

The scripts/security/ folder is part of the public repo so any fork can run the same LAN smoke against their bench. PRs that change Caddy, auth, or security_headers.py should update [ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}) and re-run the scan scripts.

31 System overview

32 System overview



32.1 Docker edge stack (production)

Three GHCR images — see `Containers({{ "/architecture/containers/" | relative_url }})`:

Container	BACnet OT	Operator LAN
bridge	No (HTTP only)	Yes, via Caddy :80 or loopback :8765
commission	Yes (host network, :47808)	No (internal :8767)
mcp-rag	No	No (internal :8090)

BACnet **polling** is not a separate container — it runs inside **commission**. The bridge only **ingests** poll CSV into feather.

32.2 Components

Layer	Responsibility
Operator Bridge	Auth, REST/WebSocket API, compiled dashboard, Rule Lab, historian ingest
BACnet commission	Who-Is, read/write (gated), scheduled poll loop → <code>samples.csv</code>
Historian	Feather files per point; local-first retention
FDD runner	<code>workspace/data/rules_py/*.py</code> on timer or <code>POST /api/rules/batch</code>
Model	Brick TTL + <code>model.json</code> bindings (<code>fdd_input</code> , equipment tree)
MCP RAG	Optional doc retrieval for agent tools
Caddy	Host reverse proxy <code>:80</code> → bridge (typical edge)
Ollama	Optional host LLM for building insight (not required for FDD)

32.3 Deployment

IT operators: [\[Quick Start — Docker\]](#)(`{{ "/quick-start/docker/" | relative_url }}`). Modes and Caddy: [Deployment modes](#)(`{{ "/architecture/deployment-modes/" | relative_url }}`).

32.4 Optional PyPI-only path

`pip install open-fdd ships Arrow runtime + playground lint helpers` **without** the Operator Bridge UI. Use for offline rule tests or CI. See [\[Appendix — Python package\]](#)(`{{ "/appendix/python-package/" | relative_url }}`).

33 PyArrow & DataFusion SQL rules

34 PyArrow & DataFusion SQL rules

Open-FDD evaluates faults on a **single PyArrow historian table** (telemetry). You author rules in **one of two backends**:

Backend	Rule Lab field	Best for
PyArrow	<code>apply_faults_arrow(table, cfg, context)</code>	Rolling windows, flatline, rate-of-change, hunting, schedule gates, multi-column HVAC logic
DataFusion SQL	<code>backend: datafusion_sql + sql:</code>	Simple thresholds, spreads, boolean CASE WHEN, SQL-readable rules for operators

Both backends share the same **confirmation window** (`min_true_rows`, `poll_interval_s`, `min_elapsed_minutes`) and produce the same boolean fault mask for batch FDD.

Install SQL support (optional):

```
pip install 'open-fdd[datafusion]'
```

34.1 Mini tutorial — same rule, both backends

Goal: Flag when outside-air temperature (`oa-t`) is above **85 °F**.

Historian column: `oa-t` (from BRICK `fdd_input` / point binding). Poll interval: **60 s**. Confirmation: **5 consecutive true rows** (~5 minutes).

34.1.1 PyArrow

Save in Rule Lab or workspace/data/rules_py/oa_t_high.py:

```
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    col = str((cfg or {}).get("value_column") or "oa-t")
    high = float((cfg or {}).get("high", 85.0))
    vals = pc.cast(table[col], "float64")
    return pc.greater(vals, high)
```

Rule metadata:

```
{
  "id": "oa-t-high-arrow",
  "backend": "arrow",
  "config": { "high": 85.0, "value_column": "oa-t" },
  "fault_code": "BLD-B"
}
```

34.1.2 DataFusion SQL

Same logic as SQL — register with backend: datafusion_sql:

```
SELECT
  *,
  "oa-t" > 85.0 AS fault
FROM telemetry
```

Rule metadata:

```
{
  "id": "oa-t-high-sql",
  "backend": "datafusion_sql",
  "fault_column": "fault",
  "sql": "SELECT *, \"oa-t\" > 85.0 AS fault FROM telemetry"
}
```

34.1.3 Prove parity

In **Rule Lab** → **Compare backends**, run the PyArrow and SQL rules on the same lookback window. Masks must match row-for-row (see bench rules bench-oa-temp-high-arrow / bench-oa-temp-high-sql).

CLI / CI:

```
pytest open_fdd/tests/validation/test_datafusion_arrow_parity.py -
```

q

34.2 When to use which backend

34.2.1 Prefer DataFusion SQL

- One or two column comparisons ($>$, $<$, BETWEEN, $\text{ABS}(a - b)$)
- Operator-facing rules that should read like SQL
- Stateless row-wise logic (no rolling window)
- Proving a rule before a future Rust port

34.2.2 Prefer PyArrow

- Rolling min/max/sum, flatline detection, rate-of-change
- PID hunting / excessive command reversals
- Schedule or occupancy gating combined with sensor logic
- Reuse of cookbook helpers (`sensor_bounds_mask`, `arrow_rolling_min`, etc.)
- Anything not expressible in a single `SELECT ... FROM telemetry`

34.2.3 Do not use SQL for

- Rolling windows, hunting detectors, file I/O, joins to external tables, or ML features — use PyArrow modules instead.

SQL is linted server-side: single statement, `FROM telemetry` only, boolean `fault` column required. See `[Arrow rule contract]({{ "/rule-authoring/arrow-rule-contract/" | relative_url }})`.

34.3 Workflow

1. Bind points in **Model** → assign rule in **Rule Lab**.
 2. Set `fault_code` from `Fault codes` (`{{ "/fault-codes/" | relative_url }}`).
 3. Configure confirmation in rule config — `Fault confirmation` (`{{ "/rule-cookbook/fault-confirmation/" | relative_url }}`).
 4. Run `POST /api/rules/batch` or wait for the scheduled FDD loop.
 5. View results on the dashboard or in `fdd_results.json`.
-

34.4 More recipes

Topic	Page
Full Arrow helper library	[Python recipes (Arrow)] (<code>{{ "/rule-cookbook/python-recipes-arrow/" }}</code>)

Topic	Page
Additional SQL patterns	DataFusion SQL recipes ({ { "/rule-cookbook/datafusion-sql-recipes/"
GL36 / AHU patterns	GL36 & sensor patterns ({ { "/rule-cookbook/expression-cookbook/"
Contract & types	Rule authoring ({ { "/rule-authoring/"

35 Deployment modes

36 Deployment modes

How Open-FDD is typically run: local development, lab LAN trials, and production edge on a trusted OT/IT network. Open-FDD is designed for **building LAN or OT edge** deployment — not direct exposure to the public internet.

36.1 Mode summary

Mode	Stack	BACnet	Front door	Auth
Local dev	Compose on control machine	Optional bench simulator	127.0.0.1:8765	OFDD_AUTH_DISABLED=1 optional on loopback
Lab LAN	Three GHCR containers	Real or simulator VLAN	Caddy :80 or direct :8765	Required unless explicit insecure lab flags
Edge production	Three GHCR containers + host Caddy	OT NIC via commission	Caddy :80 or :443 → loopback bridge	Required — workspace/auth.env.local

36.2 v3 container stack (all edge modes)

Image	Role
openfdd-bridge	

Image	Role
	Operator Bridge API, dashboard, feather historian ingest
openfdd-commission	BACnet discover/read/write + poll loop
openfdd-mcp-rag	Doc-search sidecar for agent tools

openfdd-bacnet-poll is retired. BACnet polling runs inside **commission** only. Do not run the legacy poll container or openfdd-bacnet-poll systemd unit alongside Docker commission — that double-polls the OT network.

See [Containers](#)({{ "/architecture/containers/" | relative_url }}) for ports, networks, and persistence.

36.3 Caddy HTTP (typical edge)

- Caddy listens on building LAN :80
- Reverse-proxies to bridge at 127.0.0.1:8765
- Bridge stays loopback-only; operators bookmark `http://<edge-ip>/`

Bootstrap and compose: [Quick Start — Docker]({{ "/quick-start/docker/" | relative_url }}).

36.4 Caddy TLS (OT LAN)

For HTTPS on the edge LAN (self-signed or site CA):

1. Generate certs: `./scripts/setup_caddy_certs.sh` (control machine) or site PKI
2. Start with `OFDD_CADDY_MODE=tls`
3. Trust the CA on operator PCs

Details: [TLS and certificates](#)({{ "/security/tls-and-certs/" | relative_url }}). Security headers are set by the **Operator Bridge**; Caddy adds **HSTS** only in TLS mode.

36.5 BACnet data path

OT devices —UDP 47808→ commission (poll loop) → samples.csv
 ↓
 ▼

bridge (ingest) → feather

historian

The bridge thread named `bacnet_poll_worker` is **ingest only** — it does not bind BACnet.

36.6 What not to do on production edges

- Port-forward `:8765` to the public internet without TLS and strong auth
- Run `docker compose down -v` or delete workspace/ on a live site
- Deploy the retired `openfdd-bacnet-poll` image alongside commission

36.7 Related

Topic	Page
First deploy	[Quick Start — Docker]({{ "/quick-start/docker/"
Upgrade	<u>Updating the stack</u> ({{ "/quick-start/updating/"
LAN security	<u>LAN hardening</u> ({{ "/security/lan-hardening/"
Lab example site	[Examples — GL36 lab note]({{ "/examples/acme-gl36-lab/"

37 Rule authoring

38 Rule authoring

Production FDD on the edge uses **PyArrow tables** from the feather historian. Optional **DataFusion SQL** covers simple stateless rules. Both backends share confirmation settings and batch execution.

Page	Purpose
[PyArrow & DataFusion SQL]({{ "/relative_url }}"	

Page	Purpose
rule-cookbook/dual-backend-rules/"	
[Arrow rule contract]({{ "/rule-authoring/arrow-rule-contract/"	relative_url }}
[Data types & units]({{ "/rule-authoring/data-types-and-units/"	relative_url }}
[Rust readiness]({{ "/rule-authoring/rust-readiness/"	relative_url }}

38.1 Packaging

```

pip install open-fdd # PyArrow runtime (required)
pip install 'open-fdd[datafusion]' # optional SQL backend

```

38.2 YAML files in the repo

Location	Role
open_fdd/faults/catalog/*.yaml	Fault-code metadata (not executable)
open_fdd/default_rules/**/*.yaml	Starter metadata — implement as rules_py Arrow/SQL modules
workspace/data/rules_py/*.py	Executable production rules

38.3 Fault codes

Assign letter codes (VAV-C, AHU-A, ...) from the `Fault codes({{ "/fault-codes/" | relative_url }}` reference — not legacy numeric VAV-03 strings.

39 Python recipes (full Arrow library)

40 Python recipes — full Arrow library

Copy-paste **apply_faults_arrow** modules for Rule Lab. PyArrow + pyarrow.compute only.

- Set **fault_code** in Rule Lab metadata (letter codes or Grade-A AHU-ECON-001)
- Map historian column names in the data model / rule bindings (ma-t, oa-t, stat_zn-t, ...)

Backend choice: [PyArrow & DataFusion SQL]({{ "/rule-cookbook/dual-backend-rules/" | relative_url }}) · Patterns: [GL36 & sensor patterns](#)({{ "/rule-cookbook/expression-cookbook/" | relative_url }})

40.1 Shared helpers (paste once per rule file)

```
import pyarrow as pa
import pyarrow.compute as pc

from open_fdd.arrow_runtime.cookbook import mixing_envelope_mask
from open_fdd.arrow_runtime.windows import (
    arrow_rolling_min,
    arrow_rolling_max,
    arrow_abs_diff,
)

def _norm_cmd(col: pa.ChunkedArray) -> pa.ChunkedArray:
    c = pc.cast(col, pa.float64())
    return pc.if_else(pc.greater(c, 1.0), pc.divide(c, 100.0), c)

def _col(table, name: str) -> pa.ChunkedArray:
    return pc.cast(table[name], pa.float64())
```

```
def _hypot_tol(a: float, b: float) -> float:
    return (a * a + b * b) ** 0.5
```

40.2 GL36 AHU Rules A-M

40.2.1 Rule A — duct static low at full fan speed (AHU-A)

```
FAULT_CODE = "AHU-A"
SP, SP_SP, FAN = "duct-static-pressure", "duct-static-pressure-
sp", "supply-fan-speed-command"
SP_MARGIN, DRV_HI, DRV_NEAR = 0.12, 0.93, 0.06
```

```
def apply_faults_arrow(table, cfg, context=None):
    sp, sp_sp = _col(table, SP), _col(table, SP_SP)
    fan = _norm_cmd(table[FAN])
    return pc.and_(
        pc.less(sp, pc.subtract(sp_sp, SP_MARGIN)),
        pc.greater_equal(fan, DRV_HI - DRV_NEAR),
    )
```

40.2.2 Rules B & C — blended air outside OAT/RAT band (AHU-D)

Uses `mixing_envelope_mask` (covers below-band and above-band).

```
FAULT_CODE = "AHU-D"

def apply_faults_arrow(table, cfg, context=None):
    return mixing_envelope_mask(
        table,
        {**cfg, "mixing_tol": 1.15, "normalize_cmd_percent": 1},
        mat_col="ma-t",
        oat_col="oa-t",
        rat_col="ra-t",
        fan_col="supply-fan-speed-command",
    )
```

40.2.3 Rule D — discharge cold when heating commanded (AHU-B)

```
FAULT_CODE = "AHU-B"
BLEND_TOL, SAT_TOL, FAN_DT = 1.15, 1.15, 0.55
```

```

def apply_faults_arrow(table, cfg, context=None):
    mat, sat = _col(table, "ma-t"), _col(table, "sa-t")
    valve = _norm_cmd(table["heating-valve-command"])
    fan = _norm_cmd(table["supply-fan-speed-command"])
    rhs = pc.add(pc.subtract(mat, BLEND_TOL), FAN_DT)
    cold = pc.less_equal(pc.add(sat, SAT_TOL), rhs)
    return pc.and_(cold, pc.greater(valve, 0.01), pc.greater(fan,
0.01))

```

40.2.4 Rule E — SAT too low with full heating (AHU-C)

```

FAULT_CODE = "AHU-C"
SUPPLY_ERR = 1.0

```

```

def apply_faults_arrow(table, cfg, context=None):
    sat = _col(table, "sa-t")
    sp = _col(table, "sa-t-sp") if "sa-t-sp" in
table.column_names else _col(table, "supply-air-temperature-
setpoint")
    valve = _norm_cmd(table["heating-valve-command"])
    fan = _norm_cmd(table["supply-fan-speed-command"])
    return pc.and_(
        pc.less(sat, pc.subtract(sp, SUPPLY_ERR)),
        pc.greater(valve, 0.9),
        pc.greater(fan, 0.01),
    )

```

40.2.5 Rule F — SAT/MAT mismatch in economizer mode (AHU-E)

```

FAULT_CODE = "AHU-E"
FAN_DT, BLEND_TOL, SAT_TOL, ECON_MIN = 0.55, 1.15, 1.15, 0.12
TOL = _hypot_tol(SAT_TOL, BLEND_TOL)

```

```

def apply_faults_arrow(table, cfg, context=None):
    mat, sat = _col(table, "ma-t"), _col(table, "sa-t")
    damper = _norm_cmd(table["oa-damper-command"])
    cool = _norm_cmd(table["cooling-valve-command"])
    mismatch = pc.greater(pc.abs(pc.subtract(pc.subtract(sat,
FAN_DT), mat)), TOL)
    econ = pc.and_(pc.greater(damper, ECON_MIN), pc.less(cool,
0.1))
    return pc.and_(mismatch, econ)

```

40.2.6 Rule G — ambient too warm for free cooling (AHU-E)

```
FAULT_CODE = "AHU-E"
OAT_TOL, FAN_DT, SAT_TOL, ECON_MIN = 1.15, 0.55, 1.15, 0.12

def apply_faults_arrow(table, cfg, context=None):
    oat = _col(table, "oa-t")
    sp = _col(table, "sa-t-sp")
    damper = _norm_cmd(table["oa-damper-command"])
    cool = _norm_cmd(table["cooling-valve-command"])
    warm = pc.greater(oat, pc.add(pc.subtract(sp, FAN_DT),
SAT_TOL - OAT_TOL))
    econ = pc.and_(pc.greater(damper, ECON_MIN), pc.less(cool,
0.1))
    return pc.and_(warm, econ)
```

40.2.7 Rule H — OAT/MAT mismatch (econ + mech cooling) (AHU-E)

```
FAULT_CODE = "AHU-E"
OAT_TOL, BLEND_TOL = 1.15, 1.15
TOL = _hypot_tol(OAT_TOL, BLEND_TOL)

def apply_faults_arrow(table, cfg, context=None):
    oat, mat = _col(table, "oa-t"), _col(table, "ma-t")
    cool = _norm_cmd(table["cooling-valve-command"])
    damper = _norm_cmd(table["oa-damper-command"])
    return pc.and_(
        pc.greater(pc.abs(pc.subtract(mat, oat)), TOL),
        pc.greater(cool, 0.01),
        pc.greater(damper, 0.9),
    )
```

40.2.8 Rule I — OAT/MAT mismatch (economizer only) (AHU-E)

```
FAULT_CODE = "AHU-E"
TOL = _hypot_tol(1.15, 1.15)

def apply_faults_arrow(table, cfg, context=None):
    oat, mat = _col(table, "oa-t"), _col(table, "ma-t")
    damper = _norm_cmd(table["oa-damper-command"])
    return
pc.and_(pc.greater(pc.abs(pc.subtract(mat, oat)), TOL),
pc.greater(damper, 0.9))
```

40.2.9 Rule J — discharge above blended in cooling (AHU-B)

```
FAULT_CODE = "AHU-B"  
FAN_DT, BLEND_TOL, SAT_TOL, ECON_MIN = 0.55, 1.15, 1.15, 0.12  
TOL = _hypot_tol(SAT_TOL, BLEND_TOL)
```

```
def apply_faults_arrow(table, cfg, context=None):  
    sat, mat = _col(table, "sa-t"), _col(table, "ma-t")  
    cool = _norm_cmd(table["cooling-valve-command"])  
    damper = _norm_cmd(table["oa-damper-command"])  
    hot = pc.greater(sat, pc.add(mat, pc.add(TOL, FAN_DT)))  
    mode_a = pc.and_(pc.greater(damper, 0.9), pc.greater(cool,  
0.0))  
    mode_b = pc.and_(pc.less_equal(damper, ECON_MIN),  
pc.greater(cool, 0.9))  
    return pc.and_(hot, pc.or_(mode_a, mode_b))
```

40.2.10 Rule K — discharge above setpoint in full cooling (AHU-C)

```
FAULT_CODE = "AHU-C"  
SAT_TOL, ECON_MIN = 1.15, 0.12
```

```
def apply_faults_arrow(table, cfg, context=None):  
    sat = _col(table, "sa-t")  
    sp = _col(table, "sa-t-sp")  
    cool = _norm_cmd(table["cooling-valve-command"])  
    damper = _norm_cmd(table["oa-damper-command"])  
    high = pc.greater(sat, pc.add(sp, SAT_TOL))  
    full = pc.or_(  
        pc.and_(pc.greater(damper, 0.9), pc.greater(cool, 0.9)),  
        pc.and_(pc.less_equal(damper, ECON_MIN), pc.greater(cool,  
0.9)),  
    )  
    return pc.and_(high, full)
```

40.2.11 Rule L — cooling coil ΔT when inactive (CH-c)

```
FAULT_CODE = "CH-C"  
ENTER_TOL, LEAVE_TOL, ECON_MIN = 1.15, 1.15, 0.12  
TOL = _hypot_tol(ENTER_TOL, LEAVE_TOL)
```

```
def apply_faults_arrow(table, cfg, context=None):  
    ent = _col(table, "cooling-coil-entering-air-temperature")  
    leave = _col(table, "cooling-coil-leaving-air-temperature")
```

```

        heat = _norm_cmd(table["heating-valve-command"])
        cool = _norm_cmd(table["cooling-valve-command"])
        damper = _norm_cmd(table["oa-damper-command"])
        drop = pc.greater(pc.subtract(ent, leave), TOL)
        mode_a = pc.and_(pc.greater(heat, 0), pc.equal(cool, 0),
pc.less_equal(damper, ECON_MIN))
        mode_b = pc.and_(pc.equal(heat, 0), pc.equal(cool, 0),
pc.greater(damper, ECON_MIN))
        return pc.and_(drop, pc.or_(mode_a, mode_b))

```

40.2.12 Rule M — heating coil ΔT when inactive (AHU-B)

```

FAULT_CODE = "AHU-B"
ENTER_TOL, LEAVE_TOL, FAN_DT, ECON_MIN = 1.15, 1.15, 0.55, 0.12
TOL = _hypot_tol(ENTER_TOL, LEAVE_TOL) + FAN_DT

```

```

def apply_faults_arrow(table, cfg, context=None):
    ent = _col(table, "heating-coil-entering-air-temperature")
    leave = _col(table, "heating-coil-leaving-air-temperature")
    heat = _norm_cmd(table["heating-valve-command"])
    cool = _norm_cmd(table["cooling-valve-command"])
    damper = _norm_cmd(table["oa-damper-command"])
    rise = pc.greater(pc.subtract(leave, ent), TOL)
    m1 = pc.and_(pc.equal(heat, 0), pc.equal(cool, 0),
pc.greater(damper, ECON_MIN))
    m2 = pc.and_(pc.equal(heat, 0), pc.greater(cool, 0),
pc.greater(damper, 0.9))
    m3 = pc.and_(pc.equal(heat, 0), pc.greater(cool, 0),
pc.less_equal(damper, ECON_MIN))
    return pc.and_(rise, pc.or_(m1, m2, m3))

```

40.3 Starter pack (VAV / AHU baseline)

40.3.1 01_vav_zone_temp_bounds_occupied (VAV-C)

```

from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask,
_unoccupied_mask

```

```

FAULT_CODE = "VAV-C"

```

```

def apply_faults_arrow(table, cfg, context=None):
    occupied = pc.invert(_unoccupied_mask(table, cfg))
    oob = sensor_bounds_mask(table, "zone_temp", cfg)
    return pc.and_(oob, occupied)

```

40.3.2 02_vav_zone_temp_flatline_occupied (VAV-C)

```
from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask,  
_unoccupied_mask
```

```
FAULT_CODE = "VAV-C"
```

```
def apply_faults_arrow(table, cfg, context=None):  
    occupied = pc.invert(_unoccupied_mask(table, cfg))  
    flat = sensor_flatline_mask(table, "zone_temp", cfg)  
    return pc.and_(flat, occupied)
```

40.3.3 03_vav_damper_command_extreme_flatline (VAV-D)

```
from open_fdd.arrow_runtime.cookbook import flatline_1h_mask
```

```
FAULT_CODE = "VAV-D"
```

```
DAMPER = "damper-position-command"
```

```
def apply_faults_arrow(table, cfg, context=None):  
    stuck = flatline_1h_mask(table, {**cfg, "column": DAMPER})  
    wide = pc.greater(_norm_cmd(table[DAMPER]), 0.975)  
    return pc.and_(stuck, wide)
```

40.3.4 04_ahu_runtime_outside_schedule (BLD-C)

Use run-hours analytics script + schedule compare; boolean mask:

```
from open_fdd.arrow_runtime.cookbook import _fan_on_mask,  
_unoccupied_mask
```

```
FAULT_CODE = "BLD-C"
```

```
def apply_faults_arrow(table, cfg, context=None):  
    return pc.and_( _fan_on_mask(table, cfg),  
_unoccupied_mask(table, cfg))
```

40.3.5 05 / 06 / 07 — see Rules A, sensor_bounds_mask on sa-t / sensor_flatline_mask on sa-t

40.4 Economizer starters

40.4.1 ahu_econ_100oa_temp_tracking_fault (AHU-E)

```
FAULT_CODE = "AHU-E"
DAMPER_FULL, FAN_ON, TRACK_TOL, FAN_HEAT = 0.95, 0.5, 2.0, 0.5

def apply_faults_arrow(table, cfg, context=None):
    oat, mat, sat = _col(table, "oa-t"), _col(table, "ma-t"),
    _col(table, "sa-t")
    damper = _norm_cmd(table["oa-damper-command"])
    fan = _col(table, "supply-fan-status")
    econ = pc.and_(pc.greater(fan, FAN_ON),
pc.greater_equal(damper, DAMPER_FULL))
    mat_bad = pc.greater(pc.abs(pc.subtract(mat, oat)), TRACK_TOL)
    sat_bad = pc.greater(pc.abs(pc.subtract(pc.subtract(sat,
FAN_HEAT), mat))), TRACK_TOL)
    return pc.and_(econ, pc.or_(mat_bad, sat_bad))
```

40.4.2 ahu_mech_cooling_when_free_cooling_available (AHU-E)

```
FAULT_CODE = "AHU-E"
MARGIN, DAMPER_LOW, COOL_ON = 2.0, 0.50, 0.10

def apply_faults_arrow(table, cfg, context=None):
    oat = _col(table, "oa-t")
    sp = _col(table, "sa-t-sp")
    damper = _norm_cmd(table["oa-damper-command"])
    cool = _norm_cmd(table["cooling-valve-command"])
    free = pc.less_equal(oat, pc.subtract(sp, MARGIN))
    return pc.and_(free, pc.less(damper, DAMPER_LOW),
pc.greater_equal(cool, COOL_ON))
```

40.4.3 ahu_oa_damper_excess_open_extreme_ambient (AHU-E)

```
FAULT_CODE = "AHU-E"
DAMPER_MIN, HI_OAT, LO_OAT = 0.50, 70.0, 40.0

def apply_faults_arrow(table, cfg, context=None):
    oat = _col(table, "oa-t")
    damper = _norm_cmd(table["oa-damper-command"])
    extreme = pc.or_(pc.greater_equal(oat, HI_OAT),
pc.less_equal(oat, LO_OAT))
    return pc.and_(pc.greater_equal(damper, DAMPER_MIN), extreme)
```

40.5 VAV zones

40.5.1 zone_reheat_warm_ambient (VAV-A)

```
FAULT_CODE = "VAV-A"  
T_CUTOFF, REHEAT_MIN = 78.0, 0.52
```

```
def apply_faults_arrow(table, cfg, context=None):  
    oat = _col(table, "oa-t")  
    reheat = _norm_cmd(table["reheat-valve-command"])  
    return pc.and_(pc.greater(oat, T_CUTOFF), pc.greater(reheat,  
REHEAT_MIN))
```

40.5.2 zone_damper_valve_full_open (VAV-D)

```
FAULT_CODE = "VAV-D"  
FULL_OPEN, ROLL = 97.5, 105
```

```
def apply_faults_arrow(table, cfg, context=None):  
    d = _norm_cmd(table["damper-position-command"])  
    hi = pc.greater(d, FULL_OPEN)  
    rmin = arrow_rolling_min(d, ROLL)  
    return pc.and_(hi, pc.greater(rmin, FULL_OPEN))
```

40.5.3 Zone / IAQ bounds

Use `sensor_bounds_mask(table, "zone_temp", cfg)` or `sensor_bounds_mask` with CO₂ profile — see [sensor validation]({{ "/rule-cookbook/expression-cookbook/" | relative_url }}#sensor-validation-bounds-flatline-rate-of-change).

40.6 Central plant

40.6.1 dp_below_sp_pump_max (CH-E)

```
FAULT_CODE = "CH-E"  
DP_MARGIN, PMP_HI, PMP_NEAR = 2.2, 0.93, 0.06
```

```
def apply_faults_arrow(table, cfg, context=None):  
    dp = _col(table, "differential-pressure")  
    sp = _col(table, "differential-pressure-sp")  
    pump = _norm_cmd(table["pump-speed-command"])  
    return pc.and_(  
        pc.less(dp, pc.subtract(sp, DP_MARGIN)),
```

```

        pc.greater_equal(pump, PMP_HI - PMP_NEAR),
    )

```

40.6.2 flow_high_pump_max (CH-B)

```

FAULT_CODE = "CH-B"
FLOW_HI, PMP_HI, PMP_NEAR = 1100.0, 0.93, 0.06

```

```

def apply_faults_arrow(table, cfg, context=None):
    flow = _col(table, "water-flow")
    pump = _norm_cmd(table["pump-speed-command"])
    return pc.and_(pc.greater(flow, FLOW_HI),
pc.greater_equal(pump, PMP_HI - PMP_NEAR))

```

40.6.3 plant_supply_temp_deadband (CH-D)

```

FAULT_CODE = "CH-D"
SP_BAND = 2.2

```

```

def apply_faults_arrow(table, cfg, context=None):
    temp = _col(table, "chilled-water-supply-temperature")
    sp = _col(table, "chilled-water-supply-temperature-sp")
    pump = _norm_cmd(table["pump-speed-command"])
    low = pc.less(temp, pc.subtract(sp, SP_BAND))
    high = pc.greater(temp, pc.add(sp, SP_BAND))
    return pc.and_(pc.greater(pump, 0.01), pc.or_(low, high))

```

40.6.4 chiller_excessive_runtime (CH-F)

```

from open_fdd.arrow_runtime.cookbook import arrow_rolling_mean

```

```

FAULT_CODE = "CH-F"
WINDOW, MAX_ON = 276, 264 # 5-min data ≈ 23 h window, 22 h max on

```

```

def apply_faults_arrow(table, cfg, context=None):
    on = pc.greater(_col(table, "chiller-status"),
0.5).cast(pa.float64())
    roll = arrow_rolling_mean(on, WINDOW)
    count = pc.multiply(roll, float(WINDOW))
    return pc.greater(count, float(MAX_ON))

```

40.7 Heat pumps

40.7.1 hp_discharge_cold_when_heating (HP-D)

```
FAULT_CODE = "HP-D"  
MIN_SAT, ZONE_COLD, FAN_ON = 85.0, 69.0, 0.01
```

```
def apply_faults_arrow(table, cfg, context=None):  
    sat = _col(table, "sa-t")  
    zone = _col(table, "stat_zn-t")  
    fan = _col(table, "supply-fan-status")  
    return pc.and_(  
        pc.greater(fan, FAN_ON),  
        pc.less(zone, ZONE_COLD),  
        pc.less(sat, MIN_SAT),  
    )
```

40.8 Opportunistic / ventilation

40.8.1 econ_active_warm_ambient (AHU-E)

```
FAULT_CODE = "AHU-E"  
T_CUTOFF, DPR_MIN = 63.0, 0.42
```

```
def apply_faults_arrow(table, cfg, context=None):  
    oat = _col(table, "oa-t")  
    damper = _norm_cmd(table["oa-damper-command"])  
    return pc.and_(pc.greater(oat, T_CUTOFF), pc.greater(damper,  
DPR_MIN))
```

40.8.2 mech_cool_when_econ_available (AHU-E)

```
FAULT_CODE = "AHU-E"  
T_CUTOFF, DPR_MAX = 63.0, 0.32
```

```
def apply_faults_arrow(table, cfg, context=None):  
    oat = _col(table, "oa-t")  
    damper = _norm_cmd(table["oa-damper-command"])  
    cool = _norm_cmd(table["cooling-valve-command"])  
    return pc.and_(pc.less(oat, T_CUTOFF), pc.less(damper,  
DPR_MAX), pc.greater(cool, 0.01))
```

40.8.3 low_oa_fraction_estimated (VAV-B)

```
FAULT_CODE = "VAV-B"  
OA_MIN, GAP = 21.0, 2.2
```

```
def apply_faults_arrow(table, cfg, context=None):  
    mat, rat, oat = _col(table, "ma-t"), _col(table, "ra-t"),  
    _col(table, "oa-t")  
    fan = _norm_cmd(table["supply-fan-speed-command"])  
    denom = pc.subtract(oat, rat)  
    safe = pc.greater(pc.abs(denom), GAP)  
    num = pc.subtract(mat, rat)  
    oa_pct = pc.multiply(pc.divide(num, denom), 100.0)  
    low = pc.less(oa_pct, OA_MIN)  
    return pc.and_(pc.greater(fan, 0.01), safe, low)
```

40.8.4 preheat_excess_temp (AHU-B)

```
FAULT_CODE = "AHU-B"  
EXCESS = 2.2
```

```
def apply_faults_arrow(table, cfg, context=None):  
    pre = _col(table, "preheat-coil-leaving-air-temperature")  
    sp = _col(table, "sa-t-sp")  
    oat = _col(table, "oa-t")  
    valve = _norm_cmd(table["preheat-valve-command"])  
    hot_oa = pc.and_(  
        pc.greater(oat, sp),  
        pc.greater(pc.subtract(pre, oat), EXCESS),  
    )  
    cold_oa = pc.and_(  
        pc.less(oat, sp),  
        pc.greater(pc.subtract(pre, sp), EXCESS),  
    )  
    return pc.and_(pc.greater(valve, 0.01), pc.or_(hot_oa,  
cold_oa))
```

40.9 Weather station

40.9.1 weather_temp_spike (BLD-B)

```
from open_fdd.arrow_runtime.cookbook import rate_of_change_mask  
  
FAULT_CODE = "BLD-B"
```

```
def apply_faults_arrow(table, cfg, context=None):
    return rate_of_change_mask(
        table,
        {**cfg, "max_per_sample": 16.0, "column": "oa-t"},
        col="oa-t",
    )
```

40.9.2 weather_gust_lt_wind (BLD-B)

```
FAULT_CODE = "BLD-B"
```

```
def apply_faults_arrow(table, cfg, context=None):
    gust = _col(table, "wind-gust-speed")
    wind = _col(table, "wind-speed")
    return pc.and_(
        pc.is_valid(gust),
        pc.is_valid(wind),
        pc.less(gust, wind),
    )
```

40.9.3 RH bounds

```
from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask
```

```
FAULT_CODE = "DC-C"
```

```
def apply_faults_arrow(table, cfg, context=None):
    merged = {**cfg, "value_kind": "rh"}
    return sensor_bounds_mask(table, "relative_humidity", merged)
```

40.10 Schedule + weather gating (occupied hours)

```
from open_fdd.arrow_runtime.cookbook import _unoccupied_mask,
_fan_on_mask
```

```
FAULT_CODE = "BLD-C"
```

```
def apply_faults_arrow(table, cfg, context=None):
    """Fan running when schedule says unoccupied."""
    return pc.and_( _fan_on_mask(table, cfg),
        _unoccupied_mask(table, cfg))
```

Pass occupied_start_hour, occupied_end_hour, tz_offset_hours in Rule Lab cfg.

40.11 FC4 — PID hunting

Open-FDD implements GL36-style FC4 as an Arrow-native **PID / control hunting** detector — sample-window rolling step counts on the historian poll interval.

Ports gist check_hunting intent to Arrow. Two modes:

Mode	cfg["hunting_mode"]	Use on
Per-loop command	command (default)	VAV damper, reheat, pump VFD, RTU cooling %
AHU operating-state bitmap	ahu_os	RTU/AHU with economizer + fan + heat + cool columns in wide frame

Fault codes: **VAV-F** (VAV outputs), **CH-G** (plant pump), **RTU-E** (RTU cooling capacity), **AHU-G** (multi-signal OS / legacy FC4 alias).

```
"""pid_hunting_fc4.py – deploy via setup_gl36_fdd.py or Rule Lab."""

import pyarrow as pa

from open_fdd.arrow_runtime.cookbook import
pid_hunting_ahu_os_mask, pid_hunting_command_mask

FAULT_CODE = "VAV-F" # or AHU-G, CH-G, RTU-E per binding

def apply_faults_arrow(table, cfg, context=None):
    mode = str(cfg.get("hunting_mode") or
"command").strip().lower()
    if mode in ("ahu", "ahu_os", "os"):
        return pid_hunting_ahu_os_mask(table, cfg)
    col = str(cfg.get("value_column") or cfg.get("column") or
"").strip()
    if col and col not in table.column_names:
```

```

        return pa.array([False] * table.num_rows, type=pa.bool_)
    return pid_hunting_command_mask(table, cfg, col=col or None)

```

Tuning (Acme 60 s poll, 1 h window):

```

cfg = {
    "poll_interval_s": 60,
    "hunting_window_hours": 1.0,
    "delta_os_max": 10,          # max command steps or OS bitmap
    changes per window
    "min_command_delta": 0.03,   # 3 % minimum step to count as a
    reversal
    "min_active_command": 0.05,
    # loop must be modulating above 5 %
}

```

Legacy interlink: LEGACY_CODE_MAP["FC4"] → **AHU-G**; per-loop VAV/
plant rules use family-specific codes above.

40.12 Next steps

1. Copy a module into workspace/data/rules_py/<site>_<rule>.py
2. Bind feather columns on the Rule Lab row
3. Set fault_code metadata
4. Quick-test on 3-24 h feather window
5. Ship via setup_gl36_fdd.py or Rule Lab save

See also: [GL36 & sensor patterns](#)(({ { "/rule-cookbook/expression-cookbook/" | relative_url })} · [Fault codes](#)(({ { "/fault-codes/" | relative_url })} · [Rule Lab](#)(({ { "/operator-bridge/rule-lab/" | relative_url })}))

41 Containers

42 Containers

Open-FDD v3 edge sites run **three** GHCR containers plus optional **host** services (Caddy, Ollama). There is no separate BACnet poll container — **field BACnet polling lives in commission.**

Quick answer: Who binds BACnet UDP 47808? **openfdd-commission.** Bridge only ingests samples.csv into the feather historian.

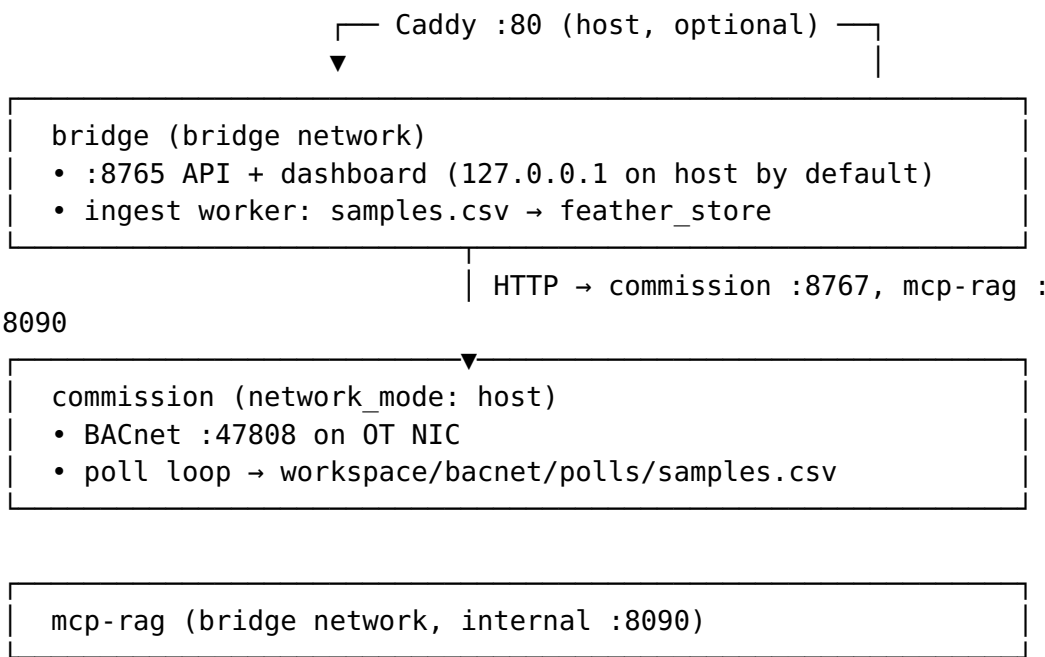
42.1 GHCR images

Image	Compose service	Role
ghcr.io/ bbartling/ openfdd- bridge	bridge	Operator API, React dashboard, feather ingest
ghcr.io/ bbartling/ openfdd- commission	commission	BACnet discover / read / write + poll loop
ghcr.io/ bbartling/ openfdd-mcp- rag	mcp-rag	Doc-search sidecar for agent tools

Tag: **latest** on [GitHub Packages](#). Retired: openfdd-bacnet-poll (poll runs inside **commission**). GHCR retains only the **three newest** versions per image.

Template: docker/compose.edge.yml → copy to ~/open-fdd/docker-compose.yml.

42.2 Stack diagram



42.3 Responsibilities

Concern	Container
BACnet Who-Is / read / write	commission
BACnet scheduled RPM reads	commission (bacnet_poll_loop)
CSV → feather historian	bridge (feather ingest worker — bacnet_poll_worker module name is historical)
Operator login, Rule Lab, faults UI	bridge
Agent doc search	mcp-rag

The bridge thread named bacnet_poll_worker is **ingest only** — it does not bind BACnet.

42.4 Network and ports (typical edge)

Endpoint	Reachable from	Notes
Caddy :80	Building LAN	Reverse proxy to bridge
Bridge :8765	Loopback (default)	Bind 127.0.0.1 in compose
Commission : 8767	Bridge / loopback	HTTP API for BACnet ops
Commission : 47808	OT BACnet LAN	UDP; requires network_mode: host
MCP :8090	Bridge container network	Not exposed to LAN by default

42.5 Long-lived deployment

All compose services set restart: unless-stopped. After host reboot:

```
sudo systemctl enable docker
cd ~/open-fdd && docker compose ps
```

If containers are down: `docker compose up -d` (idempotent).

See `Run with Docker images({ { "/quick-start/docker/" | relative_url }}`).

42.6 Persistent data

State lives on the **host** under `workspace/` (bind mount). Containers are replaceable; **backup workspace/** before image upgrades.

Path	Contents
<code>workspace/data/feather_store/</code>	Historian
<code>workspace/data/*.json</code>	Model, rules, FDD results
<code>workspace/bacnet/commissioning/</code>	<code>commission.env</code> , <code>points.csv</code>
<code>workspace/bacnet/polls/samples.csv</code>	Poll output
<code>workspace/auth.env.local</code>	Login secrets
<code>workspace/api/static/app/</code>	Dashboard bundle (if updated separately)

42.7 Optional host services

Service	Role
Caddy	TLS or plain HTTP :80 → 127.0.0.1:8765
Ollama	Optional LLM on :11434 (not in the three-image stack)
FDD loop timer	Optional <code>openfdd-fdd-loop.timer</code> on host (docker exec batch)

43 Fault confirmation / minimum duration

44 Fault confirmation

A raw rule condition can flicker true for one sample. **Fault confirmation** delays the final fault mask until the condition stays true for N consecutive rows or N minutes.

44.1 Config fields

Field	Type	Meaning
min_true_rows	int	Consecutive true samples required
min_elapsed_minutes	float	Elapsed time from streak start (uses timestamp)
poll_interval_s	float	Fallback when timestamps missing
timestamp_column	str	Default timestamp

44.2 Example

At **60-second polling**, min_true_rows: 5 means the condition must persist about **5 minutes** before fault becomes true.

PyArrow:

```
def apply_faults_arrow(table, cfg, context=None):  
    return pc.greater(table["duct-t"], 80.0)
```

Rule config:

```
{"min_true_rows": 5, "poll_interval_s": 60}
```

DataFusion SQL (same config — confirmation runs after SQL):

```
SELECT  
    *,  
    "duct-t" > 80.0 AS fault  
FROM telemetry
```

The runtime applies confirmation in `open_fdd.arrow_runtime.confirmation` for both backends.

44.3 Paired smoke default (5 minutes)

The hardcoded paired FDD harness (`open_fdd/validation/paired_fdd_contract.py`) sets **5-minute confirmation** on every smoke rule:

```
        {"min_elapsed_minutes": 5, "min_true_rows": 5, "poll_interval_s":  
60}
```

This suppresses false positives during bench 5007 / Acme OAT spread toggles. See [Paired FDD smoke](#).

45 Data flow

46 Data flow

46.1 Telemetry ingest

Open-FDD supports three commissioning drivers. Each appends long-format samples to workspace CSVs and writes **feather shards** tagged by source (`bacnet`, `modbus`, `json_api`).

46.1.1 BACnet

1. **Commission poll loop** (inside the commission container) reads enabled BACnet points from `points.csv` on a schedule (host network, BACnet :47808).
2. RPM results append to `workspace/bacnet/polls/samples.csv`.
3. **Bridge ingest worker** loads new rows into `feather_store/bacnet/`.

BACnet devices → commission (poll loop) → `samples.csv` → bridge (ingest) → `feather_store/bacnet`

46.1.2 Modbus TCP

1. Operator configures registers on the **Modbus** tab (or `read_and_store API`).
2. Poll worker reads holding/input registers via Modbus TCP.

3. Samples append to workspace/modbus/polls/samples.csv → feather_store/modbus/.

46.1.3 JSON API (HTTP/HTTPS)

1. Operator configures REST endpoints on the **JSON API** tab — GET/POST, Bearer or Basic auth, \${ENV:VAR} placeholders from json_api.env.local, optional TLS verify off for self-signed OT gateways.
2. Poll worker issues HTTP requests (one GET can fan out to multiple JSON paths, e.g. OpenWeather web-oat-t / web-rh / web-weather-desc).
3. Samples append to workspace/json_api/polls/samples.csv → feather_store/json_api/.
4. FDD batch **merges** json_api columns with BACnet/Modbus on the same site (nearest timestamp, 30 min tolerance) for cross-source rules.

OT REST / weather API → bridge JSON API worker → samples.csv → feather_store/json_api

↘ merged

into FDD frame with bacnet/modbus

Showcase: [OpenWeatherMap bundle]({{ "/drivers/json-api/" | relative_url }})#openweathermap-showcase-recommended-demo).
Details: Driver framework({{ "/drivers/" | relative_url }}), [BACnet polling]({{ "/bacnet/polling/" | relative_url }}), JSON API driver({{ "/drivers/json-api/" | relative_url }}), Containers({{ "/architecture/containers/" | relative_url }}).

46.2 Rule evaluation

1. Rules are Python files in workspace/data/rules_py/ with metadata in rules_store.json.
2. **Bindings** map rules to points via the Brick model (fdd_input tags).
3. Batch runner (POST /api/rules/batch or host timer) produces fdd_results.json.
4. Dashboard **faults** view and check-engine status read aggregated results.

46.3 Model sync

- Commissioning imports update model.json / TTL graph.
- POST /api/model/sync-ttl refreshes SPARQL-backed tree used by the UI.

46.4 Retention

Feather files grow on disk; retention is configurable (workspace/data.env.local). Image upgrades must **preserve** workspace/data/ — backup before docker compose up -d --force-recreate. See [Updating the stack](#)({{ "/quick-start/updating/" | relative_url }}).

47 Architecture

48 Architecture

Open-FDD v3 edge: **three Docker containers** (bridge, commission, mcp-rag), BACnet polling in **commission**, feather ingest in **bridge**, optional host Caddy and Ollama.

Page	Topic
System overview ({{ "/architecture/overview/" relative_url }})	
Deployment modes ({{ "/architecture/deployment-modes/" relative_url }})	
Data flow ({{ "/architecture/data-flow/" relative_url }})	
Containers ({{ "/architecture/containers/" relative_url }})	
Driver framework ({{ "/drivers/" relative_url }})	

Operators: [Quick Start — Docker]({{ "/quick-start/docker/" | relative_url }}) — no git clone on the edge host.

BACnet polling: [Polling](#)({{ "/bacnet/polling/" | relative_url }}) — commission container only.

49 DataFusion SQL recipes

50 DataFusion SQL recipes

Additional SQL patterns beyond the [PyArrow & DataFusion tutorial]({{ "/rule-cookbook/dual-backend-rules/" | relative_url }}). Requires `pip install 'open-fdd[datafusion]'`. SQL runs **server-side only**.

Not for: rolling windows, PID hunting, flatline, or schedule logic — use [Python recipes (Arrow)]({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}).

50.1 Temperature high

```
SELECT
    *,
    "duct-t" > 80.0 AS fault
FROM telemetry
```

Quote hyphenated historian columns.

50.2 Humidity high

```
SELECT
    *,
    "oa-h" > 70.0 AS fault
FROM telemetry
```

50.3 Occupied + temperature

```
SELECT
    *,
    CASE
        WHEN occupied = true AND "stat_zn-t" > 76.0 THEN true
        ELSE false
    END AS fault
FROM telemetry
```

50.4 With fault confirmation

Add to rule config:

```
{"min_true_rows": 5, "poll_interval_s": 60}
```

PyArrow remains better for rolling windows, custom helpers, and ML prep. See [PyArrow recipes](#).

51 Rule Cookbook

52 Rule Cookbook

Arrow-native fault rules for the Operator Bridge historian.
Production rules live in `workspace/data/rules_py/` and run via `POST /api/rules/batch`.

52.1 Start here

Page	Content
[PyArrow & DataFusion SQL] ({{ "/rule-cookbook/dual-backend-rules/" relative_url }})	
Rule authoring ({{ "/rule-authoring/" relative_url }})	
Fault confirmation ({{ "/rule-cookbook/fault-confirmation/" relative_url }})	
[Python recipes (Arrow)] ({{ "/rule-cookbook/python-recipes-arrow/" relative_url }})	
DataFusion SQL recipes ({{ "/rule-cookbook/datafusion-sql-recipes/" relative_url }})	
GL36 & sensor patterns ({{ "/rule-cookbook/expression-cookbook/" relative_url }})	
Windowing & debugging ({{ "/rule-cookbook/windowing-debugging/" relative_url }})	
	relative_url }}

Page	Content
	[Central plants]({{ "/"rule-cookbook/central-plants/"

52.2 Rule Lab

Author and test in the dashboard **Rule Lab** tab (/rule-lab): lint, preview on historian window, compare PyArrow vs SQL masks.

API: [Appendix — REST API]({{ "/"appendix/bridge_api/" | relative_url }}) (Rules & FDD, Rule Lab sections).

53 GL36 & sensor patterns

54 GL36 & sensor patterns

Fault-code mapping and sensor-validation patterns for **PyArrow** rules. For backend choice and a side-by-side tutorial, see [PyArrow & DataFusion SQL]({{ "/"rule-cookbook/dual-backend-rules/" | relative_url }}).

Executable modules live in workspace/data/rules_py/. Copy-paste sources: [Python recipes (Arrow)]({{ "/"rule-cookbook/python-recipes-arrow/" | relative_url }}).

54.1 Rule structure

1. **Historian columns** — Feather names from BACnet poll (oa-t, sa-t, stat_zn-t, ...) or BRICK labels via model export.
2. **Module constants** — Thresholds at top of file or read from cfg for site tuning.
3. **apply_faults_arrow** — Returns a **boolean PyArrow array** (True = fault sample).
4. **fault_code** — Set in Rule Lab; must exist in GET /api/faults/catalog.

```
import pyarrow.compute as pc
from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

def apply_faults_arrow(table, cfg, context=None):
    return sensor_bounds_mask(table, "zone_temp", cfg)
```

54.1.1 Command scaling (0-1 vs 0-100 %)

```
def _norm_cmd(col):  
    return pc.if_else(pc.greater(col, 1.0), pc.divide(col,  
100.0), col)
```

54.1.2 Occupied hours (schedule gating)

Use `open_fdd.arrow_runtime.cookbook._unoccupied_mask` or compare local hour from timestamp. Fan on when unoccupied → **BLD-C**.

54.2 Sensor validation (bounds, flatline, rate of change)

Use `open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES` for defaults, or tune per site in Rule Lab `cfg`.

54.2.1 Bounds (out of range)

Sensor kind	Min	Max	Flatline tol	Max Δ / hour	Max Δ / 15 min	Fault code
Zone temp (°F)	55	90	0.10	4.0	2.0	VAV-C
Supply air temp	50	110	0.15	8.0	3.0	AHU-C, RTU-C
Return air temp	55	95	0.10	3.0	1.5	AHU-D
Mixed air temp	40	110	0.15	6.0	2.5	AHU-D
Outdoor air temp	-40	130	0.10	12.0	6.0	BLD-B
Duct static (inH ₂ O)	-0.5	3.0	0.02	0.5	0.25	AHU-A
Relative humidity (%)	0	100	1.0	15.0	8.0	DC-C
Chilled water (°F)	40	90	0.10	4.0	2.0	CH-D
Hot water (°F)	70	200	0.15	6.0	3.0	CH-D
	50	110	0.15	5.0	2.5	CH-A

Sensor kind	Min	Max	Flatline tol	Max Δ / hour	Max Δ / 15 min	Fault code
Condenser water (°F)						
CO ₂ (ppm, occupied)	400	1000	5.0	200	80	VAV-B
Discharge air temp	45	120	0.15	10.0	4.0	VAV-E, HP-D

When OAT/RAT/MAT are all present, prefer **mixing envelope** over RAT bounds alone.

54.2.2 Flatline (stuck sensor)

Default **12 samples $\approx$ 1 hour** at 5-minute poll.

```

from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask

def apply_faults_arrow(table, cfg, context=None):
    return sensor_flatline_mask(table, "outdoor_air_temp", cfg)
# BLD-B

```

54.2.3 Rate of change (spike)

```

from open_fdd.arrow_runtime.cookbook import rate_of_change_mask
from open_fdd.arrow_runtime.sensor_catalog import cfg_from_profile

def apply_faults_arrow(table, cfg, context=None):
    merged = cfg_from_profile("outdoor_air_temp", cfg)
    merged["samples_per_hour"] = 12
    return rate_of_change_mask(table, merged, col="oa-t")

```

54.2.4 Mixing envelope (MAT vs OAT/RAT)

```

from open_fdd.arrow_runtime.cookbook import mixing_envelope_mask

def apply_faults_arrow(table, cfg, context=None):
    return mixing_envelope_mask(
        table,
        {**cfg, "mixing_tol": 1.15},
        mat_col="ma-t",
        oat_col="oa-t",
        rat_col="ra-t",
        fan_col="supply-fan-speed-command",
    )

```

54.3 GL36-inspired AHU rules

ASHRAE Guideline 36-style patterns. Assign `fault_code` in Rule Lab.

Rule	Summary	Code	Module
A — duct static low @ full fan	SP below setpoint at high VFD	AHU-A	[Rule A]({ { "/rule-cookbook/python-recipes-arrow/"
B — blend below band	MAT below OAT/RAT envelope	AHU-D	[Rules B & C]({ { "/rule-cookbook/python-recipes-arrow/"
C — blend above band	MAT above OAT/RAT envelope	AHU-D	[Rules B & C]({ { "/rule-cookbook/python-recipes-arrow/"
D — discharge cold when heating	SAT low vs MAT, heat valve open	AHU-B	[Rule D]({ { "/rule-cookbook/python-recipes-arrow/"
E — SAT low, full heating	SAT below SP, valve > 90%	AHU-C	[Rule E]({ { "/rule-cookbook/python-recipes-arrow/"
F — SAT/MAT mismatch econ	Econ mode, SAT $\neq$ MAT	AHU-E	[Rule F]({ { "/rule-cookbook/python-recipes-arrow/"
G — ambient warm free cool	OAT > SAT SP, econ open, cool off	AHU-E	[Rule G]({ { "/rule-cookbook/python-recipes-arrow/"
H — OAT/MAT mismatch econ+mech	Mech + econ, MAT $\neq$ OAT	AHU-E	[Rule H]({ { "/rule-cookbook/python-recipes-arrow/"
I — OAT/MAT mismatch econ-only	Econ only, MAT $\neq$ OAT	AHU-E	[Rule I]({ { "/rule-cookbook/python-recipes-arrow/"
J — discharge above	SAT > MAT in cooling	AHU-B	[Rule J]({ { "/rule-cookbook/

Rule	Summary	Code	Module
blend cooling K — discharge above SP full cool	SAT > SP, full cooling	AHU-C	python-recipes- arrow/" [Rule K]({ { "/ rule-cookbook/ python-recipes- arrow/"
L — cooling coil ΔT when off	CHW drop when valves closed	CH-C	[Rule L]({ { "/ rule-cookbook/ python-recipes- arrow/"
M — heating coil ΔT when off	HW rise when valves closed	AHU-B	[Rule M]({ { "/ rule-cookbook/ python-recipes- arrow/"
FC4 — PID hunting	Excessive command reversals	AHU-G / VAV-F / CH-G / RTU-E	[FC4]({ { "/rule- cookbook/ python-recipes- arrow/"

54.4 Baseline deploy bundle

Pattern	Arrow approach	Fault code
Zone temp bounds (occupied)	sensor_bounds_mask + occupied mask	VAV-C
Zone temp flatline (occupied)	sensor_flatline_mask + occupied	VAV-C
Damper command flatline	flatline_1h_mask on damper cmd	VAV-D
Runtime outside schedule	after_hours_fan_satisfied_mask	BLD-C
Duct static not maintained	[Rule A]({ { "/rule-cookbook/ python-recipes-arrow/"	relative_url } } #rule-a-duct- static-low-at-full- fan-speed-ahu-a)
Internal temp bounds / flatline	sensor_bounds_mask / sensor_flatline_mask on SAT	AHU-C

Economizer starters: [economizer section]({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}#economizer-starters).

VAV, plant, heat pump: [VAV]({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}#vav-zones) · Central plant({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}#central-plant) · Heat pumps({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}#heat-pumps).

54.5 Data quality vs equipment faults

Symptom	Pattern	Code
Flatline 1 h	flatline_1h	VAV-C, AHU-C, BLD-B
OOB sample	oob_rolling	sensor-specific
Spike between polls	rate_of_change	BLD-B
No new samples	stale_points	BLD-D
MAT $\notin$ [OAT, RAT]	mixing_envelope	AHU-D

Gate sensor faults with **fan on**, **valve open**, or **occupied** where appropriate — [Sensor & data quality]({{ "/fault-codes/sensor-quality/" | relative_url }}).

54.6 Metric (°C) equivalents

Set `cfg["temp_unit"] = "metric"` or scale constants: $55\text{ °F} \approx 12.8\text{ °C}$, $90\text{ °F} \approx 32.2\text{ °C}$, $2.0\text{ °F tol} \approx 1.1\text{ °C}$.

54.7 Binding to the fault catalog

1. Pick a **letter code** from Fault codes({{ "/fault-codes/" | relative_url }}).
2. Set **fault_code** in Rule Lab.
3. Batch FDD aggregates into GET `/api/faults/status`.

Next: [Python recipes (Arrow)]({{ "/rule-cookbook/python-recipes-arrow/" | relative_url }}) · Rule Lab({{ "/operator-bridge/rule-lab/" | relative_url }})

55 GL36 algorithm stubs

56 GL36 algorithm stubs (doc-only)

Draft **supervisory** PyArrow patterns for the future Algorithms tab. These mirror Niagara GL36 blocks documented in:

README_TRIM_RESPOND.md

FDD rules return **fault masks**. Algorithms return **setpoints or request integers** — the stubs below use `apply_algorithm_arrow` as the planned entriypoint. For early testing in Rule Lab, you can adapt the logic into advisory FDD rules (boolean “sequence unhealthy” flags).

56.1 Shared lookback helper

```
import pyarrow.compute as pc

LOOKBACK_HOURS = 6

def _kit_lookback_stats(table, *, hours=None):
    h = hours if hours is not None else LOOKBACK_HOURS
    ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
    tmin = pc.min(ts).as_py()
    tmax = pc.max(ts).as_py()
    span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
    print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")
```

56.2 VAV zone cooling requests (0-3)

Simplified from GL36 VAV box request generator — zone temp vs cooling setpoint bands.

```
"""GL36 VAV cooling requests – doc stub."""

import pyarrow.compute as pc

ZONE_TEMP = "zn-t"
ZONE_SP = "zn-t-sp"
ZONE_DEMAND = "zn-cool-loop"
```

```

HIGH_DIFF_F = 5.0
MED_DIFF_F = 3.0
LOOP_ON = 95.0
LOOP_OFF = 85.0

def apply_algorithm_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    tz = pc.cast(table[ZONE_TEMP], "float64")
    sp = pc.cast(table[ZONE_SP], "float64")
    diff = pc.subtract(tz, sp)
    demand = pc.cast(table[ZONE_DEMAND], "float64")
    # Last row request level for kit demo (full port needs timer
state)

    last_diff = diff[-1].as_py() if table.num_rows else 0.0
    last_demand = demand[-1].as_py() if table.num_rows else 0.0
    if last_diff >= HIGH_DIFF_F:
        req = 3
    elif last_diff >= MED_DIFF_F:
        req = 2
    elif last_demand > LOOP_ON:
        req = 1
    else:
        req = 0
    print(f"cooling_requests={req} dT={last_diff:.1f}F
loop={last_demand:.0f}%")
    return {"cooling_requests": req}

```

56.3 AHU duct static trim & respond (advisory fault)

Detect **duct static stuck high** while VAV pressure requests are low — trim & respond may not be trimming.

```

"""AHU duct static trim advisory – doc stub."""

```

```

import pyarrow.compute as pc

```

```

DUCT_SP = "duct-static"
DUCT_SP_MAX = 1.50
DUCT_SP_TRIM_TARGET = 0.80
VAV_PRESS_REQ_SUM = "vav-press-req-sum"
LOOKBACK_HOURS = 12

```

```

def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
    sp = pc.cast(table[DUCT_SP], "float64")
    reqs = pc.cast(table[VAV_PRESS_REQ_SUM], "float64")
    stuck_high = pc.and_(pc.greater(sp, DUCT_SP_TRIM_TARGET),

```

```
pc.less(reqs, 1.0))
    return pc.and_(stuck_high, pc.greater(sp, DUCT_SP_MAX * 0.9))
```

56.4 Chiller plant enable (valve request counting)

From GL36 §5.18.15.2 — count AHUs with CHW valve above threshold.

```
"""Chiller plant enable advisory – doc stub."""

import pyarrow.compute as pc

AHU_VALVE_COLS = ["ahu1-clg-vlv", "ahu2-clg-vlv", "ahu3-clg-vlv"]
REQ_THRESH = 95.0
DIS_THRESH = 10.0
MIN_AHUS_REQ = 2

def apply_algorithm_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    requesting = 0
    for col in AHU_VALVE_COLS:
        if col not in table.column_names:
            continue
        v = pc.cast(table[col], "float64")[-1].as_py()
        if v >= REQ_THRESH:
            requesting += 1
    enable = requesting >= MIN_AHUS_REQ
    print(f"chiller_enable={enable} requesting_ahus={requesting}/
{MIN_AHUS_REQ}")
    return {"chiller_enable": enable, "requesting_ahus":
requesting}
```

56.5 Hot water plant HWST trim & respond (advisory)

Flags **HWST trimmed too low** while heating requests remain high.

```
"""HWST trim advisory – doc stub."""

import pyarrow.compute as pc

HWST = "hw-supply-t"
HWST_SP = "hw-supply-t-sp"
HEAT_REQ_SUM = "hw-reset-req-sum"
LOW_HWST_F = 120.0
```

```
LOOKBACK_HOURS = 6
```

```
def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
    temp = pc.cast(table[HWST], "float64")
    reqs = pc.cast(table[HEAT_REQ_SUM], "float64")
    return pc.and_(pc.less(temp, LOW_HWST_F), pc.greater(reqs,
2.0))
```

56.6 Chilled water plant (DP + CHWST reset advisory)

Flags **CHWST at design cold** with low cooling requests — possible stuck 100% plant loop.

```
"""CHW plant reset advisory - doc stub."""

import pyarrow.compute as pc

CHWST = "chw-supply-t"
CHWST_DESIGN_COLD_F = 44.0
COOL_REQ_SUM = "chw-reset-req-sum"
LOOKBACK_HOURS = 6

def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
    temp = pc.cast(table[CHWST], "float64")
    reqs = pc.cast(table[COOL_REQ_SUM], "float64")
    return pc.and_(pc.less(temp, CHWST_DESIGN_COLD_F + 1.0),
pc.less(reqs, 1.0))
```

56.7 Testing later

1. Bind historian columns in **Model & assignments** (fdd_input / brick types).
2. Copy a stub into Rule Lab as an FDD advisory rule, or wait for the **Algorithms** tab kit export.
3. Run python run_test.py from a downloaded kit and confirm _kit_lookback_stats prints expected start / stop / span.

Reference implementation (Niagara Java):
[README_TRIM_RESPOND.md](#).

57 Windowing & debugging

58 Windowing & debugging

58.1 Lookback vs rolling window

	Historian lookback	Rolling window
Set by	Batch lookback_hours, FDD loop, Quick test UI	WINDOW_SAMPLES constant in rule.py
Typical	1 h scheduled; 24 h Update all	12 samples $\approx$ 1 h @ 5 min poll
Validate with	<code>_kit_lookback_stats(table) — see [Lookback window helper] ({ { “/rule-cookbook/lookback-window/”</code>	<code>relative_url }}</code>

58.2 Rolling windows (Arrow)

Use `open_fdd.arrow_runtime.windows` for sample-based rolling min/max and consecutive-true streaks:

```
from open_fdd.arrow_runtime.windows import arrow_rolling_min,
arrow_rolling_max, arrow_consecutive_true
```

Cookbook masks (`flatline_1h_mask`, `spread_1h_mask`, `oob_mask`) default to ~12 samples (~1 h at 5 min poll).

Bench rules use constants: `WINDOW_SAMPLES`, `FLATLINE_TOLERANCE`, `OAT_LOW` / `OAT_HIGH`, etc. Legacy `cfg` keys still work for uploaded rules that read `cfg.get(...)`.

58.3 Rule Lab console

Quick-test and batch responses include backend: `arrow`, `ms`, and flagged row counts. Arrow summary events appear in the event console on fault hits.

59 Central plants (CHW / CTW / BLR)

60 Central plant Arrow recipes

Chiller plant, condenser water, cooling tower, boiler, and hot-water distribution faults use the same Arrow-native primitives as air-side rules.

60.1 Starter fault codes

Code	Primitive	Recipe
CHW-DT-001	low_delta_t_mask	Low chilled-water ΔT syndrome
CHW-DP-001	reset_stuck_mask	CHW differential pressure reset stuck high

60.2 Required point roles (CHW)

- chw_supply_temp, chw_return_temp — ΔT rules
- chw_dp, chw_dp_setpoint — reset stuck rules
- pump_command, pump_feedback — command/feedback mismatch

60.3 Synthetic test pattern

```
import pyarrow as pa
from open_fdd.arrow_runtime.primitives import low_delta_t_mask

table = pa.table({
    "chw_supply_temp": [44.0] * 10,
    "chw_return_temp": [44.5] * 10,
})
mask = low_delta_t_mask(
    table,
    {"min_delta_t": 4.0},
    supply_col="chw_supply_temp",
    return_col="chw_return_temp",
```

```
)
assert mask.to_pylist()[-1] is True
```

See [Fault codes — chiller plant]({{ "/fault-codes/" | relative_url }}) (expanded in 3.0.19).

61 Dashboard overview

62 Dashboard overview

62.1 Primary areas

Area	Purpose
Home / Building insight	Check-engine status; fault cards show equipment, rule, point, historian column, BACnet
Trends	Point history from feather store
Faults	Rule hits grouped by equipment family
Rule Lab	Edit, lint, and test Python FDD rules (PyArrow kit zip)
Algorithms	Supervisory GL36 trim & respond (coming soon)
Model & assignments	Equipment tree, FDD rule pins, commissioning JSON
BACnet	Discovery, reads, commission workflows
Settings / health	Stack status, auth mode

62.2 Live updates

Dashboard tiles use REST polling and WS /ws/dashboard for selected live views (ticket auth via POST /api/auth/ws-ticket).

62.3 Roles

Read-only operators see trends and faults; integrators access Rule Lab and model import. See [Auth and roles](#)({{ "/operator-bridge/auth-roles/" | relative_url }}).

63 Auth and roles

64 Auth and roles

64.1 Production (LAN / edge)

- Set `OFDD_AUTH_SECRET` and role passwords in `workspace/auth.env.local`.
- Bridge on non-loopback addresses **requires** auth unless explicit insecure dev flags are set (lab only).

Role	Typical access
viewer	Read trends, faults, model tree
operator	Run batch FDD, acknowledge views
integrator	Rule Lab, bindings, model import
commission	BACnet writes (plus write guard)
agent	Agent tools / app-edit gates
admin	Full configuration

Login: `POST /api/auth/login` → Bearer JWT on API calls.

64.2 Localhost dev

`OFDD_BRIDGE_HOST=127.0.0.1` with `OFDD_AUTH_DISABLED=1` is acceptable on a single machine. **Never** use auth-disabled mode on a building LAN.

Full hardening: `Security({ { "/security/" | relative_url } })`.

65 Rule Lab

66 Rule Lab

Rule Lab authors **Arrow-native** Python FDD rules against the feather historian.

66.1 Equipment-scoped test and export

Use **Equipment-scoped test & export** on Rule Lab:

1. Select site, equipment, and sensor (point).
2. **Test selected rule** or **Test all for equipment** against recent feather data.
3. **Download equipment kit** — zip with manifest.json, equipment.json, points.json, per-rule samples, expanded helper source, and commissioning export subset.

API: GET /api/rules/export-equipment-kit?equipment_id=... · expanded helpers: GET /api/playground/rules/{id}/source-expanded.

66.2 Export all rules

Integrators can download **Export all rules** — one zip with per-rule kits, manifest.json, catalog snapshots, and model TTL. API: GET /api/rules/export-all-kit. Bench rules use **module constants** at the top of rule.py — no browser config panel, no config.json in the dev kit zip.

```
"""Bench OA-T out of bounds (Arrow)."""

import pyarrow.compute as pc

VALUE_COLUMN = "oa-t"
OAT_LOW = 68.0
OAT_HIGH = 88.0
LOOKBACK_HOURS = 1

def _kit_lookback_stats(table, *, hours=None):
    h = hours if hours is not None else LOOKBACK_HOURS
    ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
    tmin = pc.min(ts).as_py()
    tmax = pc.max(ts).as_py()
    span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
    print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")

def _kit_value_stats(table):
    vals = pc.cast(table[VALUE_COLUMN], "float64")
    print(
        f"column={VALUE_COLUMN} min={pc.min(vals).as_py():.2f} "
        f"max={pc.max(vals).as_py():.2f}
mean={pc.mean(vals).as_py():.2f}"
    )
```

```

def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    _kit_value_stats(table)
    vals = pc.cast(table[VALUE_COLUMN], "float64")
    return pc.or_(pc.less(vals, OAT_LOW), pc.greater(vals,
OAT_HIGH))

```

66.3 Workflow

1. **Download kit** — PyArrow zip (see below)
2. **Edit constants** locally (VALUE_COLUMN, limits, LOOKBACK_HOURS, WINDOW_SAMPLES)
3. **Run** `pip install -r requirements.txt` then `python run_test.py`
4. **Upload** rule.py on Rule Lab (integrator)

Upload validation (Phase B): AST parse, forbidden imports (**no pandas/numpy**), `apply_faults_arrow(table, cfg, context=None)` signature for fault rules. Script-mode rules must use **table** (PyArrow), not legacy **df** DataFrames — lint errors explain this for AI agents.

The bridge runs rules on **PyArrow tables** via `open_fdd.arrow_runtime`, and persists .py sources under `workspace/data/rules_py/`.

- **Quick test** — `POST /api/playground/test-rule` (default 3 h lookback)
- **Batch** — `POST /api/rules/batch / openfdd-fdd-loop timer (1 h default lookback)`
- **Update all records** — 24 h lookback, 6 h chunks (`use_chunks: true`)
- **Templates** — `GET /api/playground/arrow-templates`

Pin points to rules via **Model & assignments** (/model) commissioning JSON or BACnet tree right-click. Export shows Rule Lab **names** on each point (`fdd_rules_linked`); import uses rule **ids** (`fdd_rule_ids`).

Bench seed: `python3 scripts/setup_bench_afdd.py` — imports `bench_import_model.json` and bench cookbook rules with matching id/name pairs.

66.4 Dev kit zip (download)

Download kit on Rule Lab exports `openfdd-rule-kit-<rule_id>.zip` with:

File	Purpose
rule.py	Arrow rule + constants at top; optional <code>_kit_lookback_stats / _kit_value_stats</code> helpers
data.py	<code>SITE_ID, LOOKBACK_HOURS, load_table()</code> reading <code>sample.feather</code>
sample.feather	Full historian window for the kit lookback (all rows, not a CSV preview)
run_test.py	Runs <code>rule.apply_faults_arrow(table, {}, context=...)</code> and prints <code>flagged=N</code>
requirements.txt	<code>open-fdd>=3.0.1</code> and <code>pyarrow</code>
README.md	Install and run instructions
column_map.json	BRICK logical keys → feather column names
kit_meta.json	Export metadata (site, columns, row count, lookback hours)

Not included: `config.json` (retired — tune thresholds as Python constants).

Local test:

```
unzip openfdd-rule-kit-*.zip
cd openfdd-rule-kit-*
pip install -r requirements.txt
python run_test.py
```

Expected console output:

```
site=demo lookback_h=3 rows=...
lookback=3h rows=... start=... stop=... span=2.98h
column=oa-t min=... max=... mean=...
flagged=...
```

66.5 Lookback windows

Rules that need multi-hour history should call `_kit_lookback_stats(table)` (see [\[Lookback window helper\]](#)(`{{ "/rule-cookbook/lookback-window/" | relative_url }}`)). Pass `hours=1, 3, 6, 12, or 24` to validate timestamp span in the console.

Rolling-window rules (`flatline`, `spread`) use `WINDOW_SAMPLES` (~12 samples ≈ 1 h at 5 min poll) — see [Windowing & debugging](#)(`{{ "/rule-cookbook/windowing-debugging/" | relative_url }}`).

66.6 Algorithms (supervisory — coming soon)

GL36 trim & respond and plant reset sequences will live on the **Algorithms**(`{{ "/operator-bridge/algorithms/" | relative_url }}`) tab with the same zip kit shape but `apply_algorithm_arrow` outputs. Reference: README_TRIM_RESPOND.

66.7 Related

- Model workflow(`{{ "/operator-bridge/model-workflow/" | relative_url }}`)
- Arrow recipes(`{{ "/rule-cookbook/arrow-recipes/" | relative_url }}`)
- GL36 algorithm stubs(`{{ "/rule-cookbook/gl36-algorithm-stubs/" | relative_url }}`) (doc-only)

67 JSON API driver

68 JSON API driver

Poll **HTTP or HTTPS** REST endpoints on the OT LAN (or public APIs such as OpenWeatherMap) and store extracted JSON values in the feather historian (`source=json_api`).

Use this driver for gateways, micro-PLCs, IoT hubs, vendor APIs, and **web weather** when you want to cross-check shaky local outdoor-air sensors against a reference.

68.1 Supported features

Feature	Support
Methods	GET, POST
Transport	http:// and https://
TLS certificate verify	On by default; disable for self-signed OT gateways
Authentication	None, Bearer token , HTTP Basic , or query-string API keys via env
Env placeholders	

Feature	Support
	<code>\${ENV:VAR}</code> or <code>\${VAR}</code> in URL, headers, body, bearer token
Custom headers	Optional headers map (merged with auth headers)
Value extraction	Dot-path into JSON body (title, main.temp, weather.0.description)
Multi-extract poll	One HTTP request per URL group; fan-out to multiple json_path rows
Polling	1 / 5 / 15 / 30 min or 1 hour (shared with BACnet/Modbus)
Historian	workspace/json_api/polls/samples.csv → feather_store/json_api/
FDD merge	Scheduled FDD merges bacnet + modbus + json_api columns by nearest timestamp

68.2 Secrets: json_api.env.local

Copy workspace/json_api.env.example → workspace/json_api.env.local (gitignored). The bridge loads this file at startup and when run_local.sh starts the stack.

Use placeholders in commissioning URLs so API keys never land in git:

```
https://api.openweathermap.org/data/2.5/weather?q=${ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=${ENV:OPENWEATHER_UNITS}
```

Check which vars are configured: GET /api/json-api/env/status.

68.3 OpenWeatherMap showcase (recommended demo)

This is the flagship example of what the generic JSON API driver can do — **one URL, three historian columns**, env-based API key, browser tree like BACnet/Modbus.

68.3.1 1. Configure env

```
cp workspace/json_api.env.example workspace/json_api.env.local
# Edit: OPENWEATHER_API_KEY, OPENWEATHER_CITY, OPENWEATHER_UNITS
(imperial = °F)
```

Restart the bridge (or `./scripts/run_local.sh restart`) so env vars load.

68.3.2 2. Register from the UI

1. Sign in as **integrator** → **JSON API** tab.
2. Open the **OpenWeatherMap showcase** panel.
3. Use the **OpenWeatherMap** preset → **Register** (default poll 30 min; choose interval in the form).

Three endpoints appear under `api.openweathermap.org` in the commissioning tree:

Label	JSON path	Historian column	Units
web-oat-t	main.temp	web-oat-t	degF / degC
web-rh	main.humidity	web-rh	%
web-weather-desc	weather.0.description	web-weather-desc	text

Poll worker deduplicates: **one GET** serves all three extractions per cycle.

68.3.3 3. Headless / script

```
# Standalone fetch (like the original playground tester)
python3 scripts/openweather_json_api_example.py

# Register via REST (integrator login)
python3 scripts/openweather_json_api_example.py --register --base
http://127.0.0.1:8000
```

68.3.4 4. FDD: local OA-T vs web weather

With BACnet oa-t and JSON API web-oat-t in the merged site frame, scheduled FDD compares local outdoor air to the weather service.

Item	Detail
Rule module	oat_vs_web_spread_1h.py
Fault code	BLD-B (outdoor air sensor fault)

Item	Detail
Default threshold	<code> oa-t - web-oat-t > 8 °F</code> (<code>max_spread_f</code> in rule cfg)
Acme rule id	<code>acme-oat-vs-web-spread</code> (via <code>setup_gl36_fdd.py</code>)
Local OAT source	RTU 1100-unknown-2 (Outdoor Air Temperature Local) — same value as boiler/AHU networked OAT
Web column	<code>web-oat-t</code> from OpenWeather bundle

Model alias: local BACnet historian column must be `oa-t`:

```
python3 scripts/acme_patch_oat_column.py --point-id 1100-unknown-2 --host "$ACME_HOST" --token "$TOKEN"
```

Or set `external_id` / `fdd_input` to `oa-t` on the chosen `Outside_Air_Temperature_Sensor` in the model.

Units: keep `OPENWEATHER_UNITS=imperial` when BACnet OAT is °F.

Enable the rule in Rule Lab (or `setup_gl36_fdd.py` push) and run an FDD batch. Missing either column → rule returns all-false (no crash).

68.4 Commissioning (UI)

The **JSON API** tab follows [Home Assistant sensor.rest](#):

1. Sign in → **RESTful sensor (JSON API)** tab.
2. Pick a **preset** (JSONPlaceholder, DummyJSON, httpbin, Open-Meteo, OpenAQ, OpenWeather) or enter a **resource URL**.
3. Add one or more **sensors** — each row is a `value_json_path` + historian **label** (multi-value from one GET, like HA `json_attributes`).
4. Click **Test resource** — probes without saving; shows extracted values and raw JSON.
5. Click **Register & poll** — writes `endpoints.csv`, enables polling at the selected interval.
6. Poll choices match BACnet/Modbus: **1 / 5 / 15 / 30 min** or **1 hour**.

68.4.1 API for agents

Endpoint	Purpose
GET <code>/api/json-api/presets</code>	Full preset catalog + poll intervals (human + AI readable)

Endpoint	Purpose
POST /api/json-api/test	Probe resource + multi-sensor extraction
POST /api/json-api/register-bundle	Register one resource with many sensors
POST /api/json-api/presets/{id}/register	One-click preset registration

68.4.2 Bench preset

Use **JSONPlaceholder** — **todo title** → **Test resource** to verify outbound HTTPS without field hardware.

68.5 Authentication examples

68.5.1 Query-string API key via env (OpenWeather)

```
{
  "url": "https://api.openweathermap.org/data/2.5/weather?q=${ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=${ENV:OPENWEATHER_UNITS}",
  "method": "GET",
  "json_path": "main.temp",
  "label": "web-oat-t",
  "auth_type": "none"
}
```

68.5.2 Bearer token (API key)

```
{
  "url": "https://192.168.10.50/api/v1/status",
  "method": "GET",
  "json_path": "sensors.zone_temp",
  "label": "zone-temp",
  "auth_type": "bearer",
  "bearer_token": "${ENV:GATEWAY_API_KEY}",
  "verify_tls": false
}
```

68.5.3 HTTP Basic

```
{
  "url": "http://192.168.10.50/data.json",
  "method": "GET",
  "json_path": "value",
  "label": "sensor-value",
}
```

```

    "auth_type": "basic",
    "basic_user": "operator",
    "basic_password": "changeme"
  }

```

68.5.4 POST with JSON body

```

{
  "url": "https://gateway.local/api/read",
  "method": "POST",
  "json_path": "result.pv",
  "label": "pv",
  "body": { "point": "AHU-1_SAT", "command": "read" },
  "auth_type": "bearer",
  "bearer_token": "..."
}

```

Credentials in endpoints.csv are redacted in API responses (***).
Prefer \${ENV:...} for production keys.

68.6 REST API

Method	Path	Role	Description
GET	/api/json-api/env/status	operator+	Which env vars are configured (no secret values)
POST	/api/json-api/presets/openweather	integrator+	Register 3-point OpenWeather bundle + optional poll
GET	/api/json-api/driver/tree	operator+	Commissioning tree grouped by host
GET	/api/json-api/poll/status	operator+	Poll worker status
POST	/api/json-api/poll/once	integrator+	Run one poll cycle
POST	/api/json-api/request	operator+	One-shot HTTP request (no store)
POST	/api/json-api/read_and_store	integrator+	Request + CSV + feather ingest
POST	/api/json-api/refresh	operator+	Re-fetch one endpoint
PATCH	/api/json-api/endpoint/poll	integrator+	Enable/disable poll interval
DELETE		integrator+	Remove endpoint

Method	Path	Role	Description
	/api/json-api/ endpoint/ {point_id}		

Full route list: [API routes]({{ "/appendix/bridge_api/" | relative_url }}).

68.7 Typical OT URLs

Device type	Example URL	Notes
Web weather	https:// api.openweathermap.org/data/ 2.5/weather?...	Env-based appid; 3 historian columns
Gateway status	http://192.168.1.50/api/ status	Often plain HTTP on LAN
HTTPS BMS API	https://10.0.0.12/rest/ points/zone1	May need verify_tls: false
Local mock	https:// jsonplaceholder.typicode.com/ todos/1	Bench / CI only

68.8 Limitations

- Response body must be **JSON** (not XML or plain text).
- Numeric paths (e.g. main.temp) are stored as strings in poll CSV; FDD rules cast with `pc.cast(..., "float64")`.
- Outbound requests originate from the **bridge** container/host — ensure routing and firewall rules allow the edge box to reach the target (OT subnet or public internet for weather).

68.9 Disable polling (save API quota)

Stop scheduled calls without deleting endpoints: JSON API tree → select OpenWeather points → **Stop polling**, or integrator API:

```
curl -X PATCH http://127.0.0.1:8765/api/json-api/endpoint/poll \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"point_id":"ja-api-openweathermap-org-get-web-oat-
t","enabled":false,"poll_interval_s":0}'
```

API keys live in `json_api.env.local` (gitignored) — never committed; audit logs do not record expanded URLs with `appid=`. See [Logging and audit](#)(`{{ "/ops/logging/" | relative_url }}`).

68.10 Smoke test

```
python3 scripts/smoke_multi_driver_stack.py
```

Validates JSON API ingest alongside BACnet and Modbus, feather isolation, BRICK scope, and FDD batch.

69 Algorithms

70 Algorithms (coming soon)

The **Algorithms** dashboard tab (`/algorithms`) is a placeholder for **supervisory Python sequences** — the mirror image of Rule Lab FDD:

	FDD (Rule Lab)	Algorithms (planned)
Purpose	Detect faults	Compute setpoints, request levels, plant enables
Entrypoint	<code>apply_faults_arrow(table, cfg, context)</code>	<code>apply_algorithm_arrow(table, cfg, context)</code> (planned)
Output	Boolean fault mask	Numeric outputs / structured dict
Historian	Same feather_store PyArrow tables	Same

Until the tab ships, draft patterns live in this doc and in [GL36 algorithm stubs](#)(`{{ "/rule-cookbook/gl36-algorithm-stubs/" | relative_url }}`).

70.1 GL36 Trim & Respond reference

Open-FDD algorithms will follow the same ASHRAE Guideline 36 trim & respond methodology already implemented for Niagara in:

README_TRIM_RESPOND.md

That guide covers:

- VAV zone request generators (cooling + static pressure)

- AHU duct static pressure reset
- AHU supply air temperature reset
- Central plant chiller / boiler enable and HWST / CHWST trim & respond

Porting those blocks to PyArrow is in progress; the dashboard tab shows **COMING SOON** until upload, batch, and BACnet write paths are wired.

70.2 Planned workflow (same kit shape as Rule Lab)

1. Download **algorithm kit** zip from the Algorithms tab (future)
2. Edit **constants** at the top of algorithm.py (no config.json)
3. Run `python run_test.py` locally against `sample.feather`
4. Upload algorithm.py when satisfied

Kit files will match the Rule Lab zip layout (see [Rule Lab — dev kit zip]({{ "/operator-bridge/rule-lab/" | relative_url }}#dev-kit-zip-download)).

70.3 Lookback windows

Rules and algorithms that need multi-hour history should use the shared **lookback stats helper** documented in [Lookback window helper]({{ "/rule-cookbook/lookback-window/" | relative_url }}). Pass `hours=1, 3, 6, 12, or 24` to print row count, timestamp start/stop, and span for console validation.

Scheduled batch defaults: **1 h** lookback on the FDD loop; **Update all records** in Rule Lab uses **24 h** with 6 h chunks.

70.4 AHU supervisory checks (doc-only stubs)

These are **not** uploaded rules yet — copy-paste references for future algorithm or FDD work.

70.4.1 Duct static pressure too high

Flags sustained high duct static (possible stuck setpoint, blocked filter, or trim/respond not trimming).

```
"""AHU duct static high – doc stub (Arrow constants)."""
```

```
import pyarrow.compute as pc
```

```

VALUE_COLUMN = "duct-static"
HIGH_INWC = 1.20
LOOKBACK_HOURS = 3
MIN_SAMPLES_ABOVE = 6

def _kit_lookback_stats(table, *, hours=None):
    h = hours if hours is not None else LOOKBACK_HOURS
    ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
    tmin = pc.min(ts).as_py()
    tmax = pc.max(ts).as_py()
    span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
    print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")

def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    vals = pc.cast(table[VALUE_COLUMN], "float64")
    high = pc.greater(vals, HIGH_INWC)
    # Require several consecutive highs (~30 min at 5 min poll)
    from open_fdd.arrow_runtime.windows import
arrow_consecutive_true

    streak = arrow_consecutive_true(high, MIN_SAMPLES_ABOVE)
    return streak

```

70.4.2 Supply air temperature too cold (cooling coil over-delivering)

```

"""AHU SAT too cold vs setpoint – doc stub."""

import pyarrow.compute as pc

SAT_COLUMN = "sa-t"
SP_COLUMN = "sa-t-sp"
COLD_DELTA_F = 5.0
LOOKBACK_HOURS = 1

def _kit_lookback_stats(table, *, hours=None):
    h = hours if hours is not None else LOOKBACK_HOURS
    ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
    tmin = pc.min(ts).as_py()
    tmax = pc.max(ts).as_py()
    span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
    print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")

```

```
def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    sat = pc.cast(table[SAT_COLUMN], "float64")
    sp = pc.cast(table[SP_COLUMN], "float64")
    return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

70.5 Plant supervisory checks (doc-only)

See [GL36 algorithm stubs](#)({{ "/rule-cookbook/gl36-algorithm-stubs/" | relative_url }}) for chiller plant enable, HWST trim & respond, and CHW DP/CHWST reset patterns aligned with the Niagara README.

70.6 Related

- [Rule Lab](#)({{ "/operator-bridge/rule-lab/" | relative_url }}) — live FDD authoring
- [Model workflow](#)({{ "/operator-bridge/model-workflow/" | relative_url }}) — point bindings
- [Windowing & debugging](#)({{ "/rule-cookbook/windowing-debugging/" | relative_url }}) — rolling windows

71 Model workflow

72 Model workflow

72.1 Brick model

- **Equipment tree** — SPARQL over synced TTL (workspace/data/data_model.ttl)
- **Point registry** — BACnet inventory → series_id for historian bindings
- **Health score** — orphan points, missing fdd_input, stale telemetry

72.2 Typical integrator flow

1. Import or sync BACnet point registry.

- 2. Map equipment classes (AHU, VAV, ...) in the model UI — set brick_type on points and feeds on equipment (AHU→VAV).
- 3. Tag points with fdd_input roles (zone temp, SAT, damper cmd, ...).
- 4. Bind Rule Lab rules to points, equipment, or BRICK classes (bindings in rules_store.json).
- 5. Run POST /api/model/sync-ttl after bulk imports.
- 6. Validate on **Trend plot** with FDD overlays (?fdd=1) — faults render on a right-hand 0/1 axis per device scope.

AI Agent and commissioning JSON import can bulk-apply steps 2-4; integrator should review model health before production FDD. See [FDD and assignments](#)({{ "/operator-bridge/fdd-assignments/" | relative_url }}) and [Ollama and analytics](#)({{ "/operator-bridge/ollama-analytics/" | relative_url }}).

Export: GET /api/model/export (TTL/JSON). Commissioning bundle: GET /api/model/commissioning-export — includes per-point fdd_rules_linked (Rule Lab names) plus fdd_rules[] catalog. Use **Import / export** tab preview table or **Point → FDD rule pins** before editing raw JSON. API detail in [Appendix](#)({{ "/appendix/bridge_api/" | relative_url }}).

73 Driver framework

74 Driver framework

Open-FDD edge commissioning uses a **shared driver pattern** for OT data sources. Each driver polls field devices on a schedule, stores long-format samples in workspace CSVs, and ingests into the **feather historian** under a distinct source tag for FDD and plots.

74.1 Drivers

Driver	Protocol	Feather source	Commissioning tab
BACnet ({{ "/bacnet/" relative_url }})	relative_url }}	MS/TP / IP (: 47808)	bacnet
Modbus	Modbus TCP	modbus	Modbus
[JSON API] ({{ "/drivers/json-api/" relative_url }})	relative_url }}	HTTP/HTTPS REST (+ [OpenWeather showcase]	relative_url }} #openweathermap-showcase-

Driver	Protocol	Feather source	Commissioning tab
		({{ { "/drivers/json-api/"	recommended-demo))

All three drivers share the same operator workflow:

1. **Discover or configure** — add devices/endpoints to the driver registry.
2. **Request once** — on-demand read to verify connectivity and JSON path / register map.
3. **Request & store** — append to poll CSV and write a feather shard.
4. **Enable polling** — 1 / 5 / 10 / 15 minute intervals via tree right-click or bulk toolbar.
5. **Poll all now** — force one worker cycle without waiting for the interval.

74.2 Data layout

```
workspace/
  bacnet/commissioning/  points.csv, discovered inventory
  bacnet/polls/          samples.csv
  modbus/commissioning/  registers.csv
  modbus/polls/          samples.csv
  json_api/commissioning/ endpoints.csv
  json_api/polls/        samples.csv
  data/feather_store/
    bacnet/{site_id}/    shard-*.feather
    modbus/{site_id}/
    json_api/{site_id}/
```

74.3 Historian and FDD

- **Plot tab** — numeric columns from bacnet and modbus sources (GET /api/timeseries/plot?source=...).
- **String JSON fields** — stored in feather under json_api; use the JSON API tree or feather directly (plot is numeric-only).
- **FDD batch** — merges bacnet + modbus + json_api historian columns by nearest timestamp; rules use BRICK-bound external_id names (e.g. compare BACnet oa-t vs JSON API web-oat-t — see [JSON API]({{ { "/drivers/json-api/" | relative_url }}})).
- **BRICK model** — bind points in **Model & assignments** commissioning JSON; sync TTL for SPARQL scope.

74.4 Rule Lab (Arrow upload/download)

1. **Download kit** — zip with rule.py (constants at top), data.py, sample.feather (~3h), run_test.py, requirements.txt.
2. Local test: pip install -r requirements.txt then edit constants in rule.py and python run_test.py.
3. **Upload rule.py** — Arrow-only; must define apply_faults_arrow(table, cfg, context=None).
4. Browser shows **read-only** source; use Quick test + Update all records.

API: GET /api/rules/export-kit, POST /api/rules/upload.

74.5 Smoke validation

From repo root on a running stack (http://127.0.0.1:8765):

```
python3 scripts/smoke_multi_driver_stack.py
python3 scripts/validate_modbus_temp_e2e.py --no-server
```

74.6 Related docs

- [Data flow](#)({{ "/architecture/data-flow/" | relative_url }}) — poll → CSV → feather → FDD
- [\[API routes\]](#)({{ "/appendix/bridge_api/" | relative_url }}) — REST reference
- [\[BACnet polling\]](#)({{ "/bacnet/polling/" | relative_url }}) — commission container loop

75 Operator Bridge

76 Operator Bridge

The Operator Bridge is the FastAPI service and React dashboard operators use daily.

Page	Topic
Dashboard overview ({{ "/operator-bridge/dashboard/" relative_url }})	relative_url }}
Auth and roles ({{ "/operator-bridge/auth-roles/" relative_url }}	relative_url }}
	relative_url }}

Page	Topic
Rule Lab ({{ "/operator-bridge/rule-lab/"	
Algorithms ({{ "/operator-bridge/algorithms/"	relative_url }}
Model workflow ({{ "/operator-bridge/model-workflow/"	relative_url }}
FDD and assignments ({{ "/operator-bridge/fdd-assignments/"	relative_url }}
Ollama and analytics ({{ "/operator-bridge/ollama-analytics/"	relative_url }}

REST details: [Appendix — API routes]({{ "/appendix/bridge_api/" | relative_url }}). Env files: [Workspace env files](#)({{ "/appendix/workspace-env-files/" | relative_url }}).

77 Ollama and analytics

78 Ollama and analytics

78.1 What runs without login

The **Building status** home page (/) and GET /openfdd-agent/building-insight are **public** (same policy as fault status). Anyone on the LAN can read the briefing sentence and underlying numeric levers; only the **AI Agent** chat and stack revision endpoints require sign-in.

78.2 Ollama setup

1. Copy workspace/ollama.env.example → workspace/ollama.env.local.
2. Pick a model for available RAM (comments in the example file):

RAM tier	Suggested model
4 GB Pi	qwen3:0.6b
8 GB	qwen3:1.7b

RAM tier	Suggested model
16 GB	qwen3:4b
32 GB	qwen3:8b
64 GB	qwen3:14b

1. Set OFDD_OLLAMA_BASE_URL (default `http://127.0.0.1:11434`).
2. Optional: OFDD_OLLAMA_GPU_MODE (`cpu` / `auto` / `gpu`),
OFDD_OLLAMA_TIMEOUT_S for slow CPUs.
3. Bootstrap helper: `./scripts/bootstrap_ollama.sh`.

Bridge reads env via `run_local.sh` / `compose`. Model resolution:
`workspace/api/openfdd_bridge/ollama_profiles.py`.

78.3 Automatic analytics (deterministic, always on)

These pandas calculations run on a refresh interval and feed the home dashboard **whether or not Ollama is up**:

Lever	Module	Refresh env	What it computes
Zone temperature snapshot	<code>zone_temp_analytics.py</code>	OFDD_ZONE_TEMP_INTERVAL_S (default 3600s)	Site-wide and per AHU zone temperature averages, spread out-of-band counts using BRICK feeds (AHU→VAV) which are modeled; falls back to all Zone_Air_Temperature points
Device poll health	<code>device_poll_health.py</code>	stack health cycle	BACnet device health seen / stale poll detection
Operational brief	<code>building_insight.py</code> → <code>get_operational_brief</code>	same as insight	Compact numerical status for integrators
Root-cause hints	<code>root_cause_hints.py</code>	on insight refresh	Multi-zone fault chains using brick:feeds from

Shared lookback window: OFDD_ANALYTICS_LOOKBACK_DAYS (default 14) in `operational_analytics.py`.

API routes (auth optional unless noted):

- GET `/openfdd-agent/building-insight` — sentence + zone snapshot + fault sentences

- GET /openfdd-agent/zone-temp-analytics
- GET /openfdd-agent/device-poll-health
- GET /openfdd-agent/ollama/health — integrator diagnostics

78.4 When Ollama is used

`building_insight.get_building_insight():`

1. Refreshes zone + device snapshots (pandas).
2. Calls `ollama_client.health()`.
3. If Ollama is reachable and `should_use_ollama_for_insight()` passes, sends a **compact context** (status, zone levers, feeds chains) to the configured model for a one-sentence briefing (source: "ollama").
4. On failure, uses `_fallback_sentence()` (source: "deterministic").

Context assembly: `building_insight._compact_context()`. Prompt template references BRICK feeds and `root_cause_hints`.

78.5 AI Agent (authenticated)

/agent page uses POST /openfdd-agent/chat with tool access (`agent_tools.py`): BRICK graph, scope bundles, rule binding patches, model health, etc. This is separate from the public home briefing.

78.6 Code map

```
workspace/api/openfdd_bridge/
  building_insight.py      # cached home briefing + Ollama
sentence
  zone_temp_analytics.py  # zone temp levers (BRICK feeds)
  device_poll_health.py   # BACnet poll staleness
  operational_analytics.py # lookback window + methodology dict
  root_cause_hints.py     # multi-fault BRICK chain hints
  ollama_client.py        # HTTP client to Ollama
  ollama_profiles.py      # RAM tiers / model names
```

Dashboard consumers: `BuildingStatusPage.tsx`,
`StackStatusStrip.tsx` (WebSocket /ws/dashboard).

79 FDD and assignments

80 FDD and assignments

80.1 What the product actually does

Open-FDD separates **data modeling** (BRICK equipment, points, feeds relationships) from **rule assignment** (which FDD rules apply to which points, equipment, or BRICK classes). Both are editable in **Model & assignments** and via commissioning JSON import.

80.1.1 BRICK modeling (integrator + AI-assisted)

Concern	Where it lives	Notes
Equipment tree	ModelService + TTL sync	AHU, VAV, sensors; equipment_type and brick_type on points
brick:feeds	model_feeds.py, ttl_service.py	AHU→VAV→zone relationships; used by zone analytics and root-cause hints
Point registry	BACnet inventory → model points	fdd_input roles (oa-t, stat_zn-t, ...), series_id / historian column
Commissioning import	POST /api/model/commissioning-import	Bulk JSON: sites, equipment, points, optional fdd_rules

AI Agent (agent_tools.py) can read slim_brick_graph, scope_bundle, patch rule bindings, and upsert points — but it does **not** auto-generate a full site model without operator review. Expected workflow:

1. BACnet discover/poll populates point registry.
2. Integrator or agent tags brick_type, fdd_input, and feeds on equipment.
3. POST /api/model/sync-ttl writes workspace/data/data_model.ttl.

4. Model health (model_health.py) flags orphan points and missing fdd_input.

80.1.2 Rule assignment (three binding kinds)

Saved rules in rules_store.json carry bindings:

```
{
  "point_ids": ["5007-analog-input-1173"],
  "equipment_ids": ["bench-1"],
  "brick_types": ["Outside_Air_Temperature_Sensor"]
}
```

Binding kind	Effect
point_ids	Rule runs for that historian point / series
equipment_ids	Rule runs for all points on equipment (mass bind)
brick_types	Rule runs for any point with that BRICK class on site

Assign via:

- **Model & assignments** UI — equipment/point chips, rule pin menu on trend plot
- **Rule Lab** — bindings panel
- **Commissioning import** — fdd_rules[].bindings and per-point fdd_rule_ids
- **AI Agent** — patch_rule_binding tool

Runtime evaluation: fdd_runner.py (batch FDD) and plot_readings.evaluate_fault_plots() (trend overlays). Plot scope filters rules to those bound to **selected telemetry** (point, equipment, or BRICK class); unbound rules still evaluate site-wide.

Brick bindings (3.0.20+): when a rule binds brick_types (or equipment_ids / point_ids), the batch runner resolves **every** matching historian column via historian_columns_for_rule, runs the Arrow rule per column with value_column set, and OR-combines fault masks. One rule config therefore applies consistently to all zone temps, discharge temps, etc.

80.1.3 CLASS vs device

- **BRICK class** (brick_type on points, or brick_types on rule bindings) — template-style assignment (“all Zone_Air_Temperature_Sensor”).
- **Device / equipment** (equipment_id, BACnet device instance) — instance assignment (“this VAV-12”).
- **Point** (point_ids) — single-sensor assignment.

applies_to on legacy rules is deprecated; use bindings in 3.0+.

80.2 Commissioning JSON shape

Export: GET /api/model/commissioning-export. Import accepts import_ready_json wrappers (see commissioningImport.ts).

80.2.1 Rule names vs ids (Rule Lab alignment)

Rule Lab shows **human names** (Bench OA-T flatline 1h). Saved rules use stable **ids** (bench-oa-t-flatline-1h from setup scripts, or a UUID for uploaded custom rules). Commissioning export includes both so you never have to cross-reference sections by hand:

Field	Direction	Purpose
points[].fdd_rule_ids	import + export	Assignment surface — array of rule ids
points[].fdd_rules_linked	export only	[{id, name}] — Rule Lab names beside each point
fdd_rules[].id	import + export	Stable rule key (matches Rule Lab / rules_store.json)
fdd_rules[].name	import + export	Same label as Rule Lab dropdown
fdd_rules[].source_file	export	Baseline of deployed .py when present (e.g. bench-oa-t-flatline-1h.py)

On import, fdd_rules_linked is stripped (like fdd_rule_ids — not stored in model.json). Use the **Point → FDD rule pins** table on Model & assignments to read names without parsing JSON.

Minimum fields integrators/AI should produce:

- sites, equipment[] with id, equipment_type, optional feeds[]
- points[] with site_id, equipment_id, brick_type, fdd_input, BACnet IDs, optional fdd_rule_ids
- fdd_rules[] with id, name, bindings, enabled (optional source_file on export)

After import, verify in **Trend plot** with ?fdd=1 — fault lanes appear on the right-hand 0/1 axis for bound rules.

80.3 Related

- [Model workflow](#)({{ "/operator-bridge/model-workflow/" | relative_url }})
- [Rule Lab](#)({{ "/operator-bridge/rule-lab/" | relative_url }})
- [\[Trend plot / fault overlays\]](#)({{ "/operator-bridge/dashboard/" | relative_url }}#trend-plot)

81 Network setup

82 Network setup

82.1 Bind address

Commission and poll containers need the **OT interface** IP and UDP **47808**:

```
# workspace/bacnet/commissioning/commission.env
BACNET_BIND=192.168.1.50/24:47808
BACNET_INSTANCE=599999
```

- Use the NIC that can reach controllers (not only the management VLAN unless routed).
- Poll driver uses network_mode: host on Linux edges.

82.2 Discovery range

```
DISCOVER_LOW=1
DISCOVER_HIGH=4194302
DISCOVER_TIMEOUT=30
```

Narrow the range in large sites to avoid broadcast storms.

82.3 Verify

From commission container logs or API: Who-Is → I-Am responses. If zero devices, check firewall, bind IP, and VLAN routing before tuning rules.

83 Discover and read

84 Discover and read

The **commission** container exposes HTTP APIs for discovery and read-property workflows used by the dashboard BACnet tools.

Capability	Notes
Who-Is / I-Am	Device discovery on OT subnet
Read property	Present value, object name, units
RPM	Bulk reads for inventory export
Write property	Gated — see Write safety (<code>{{ "/bacnet/write-safety/"</code>

Exported inventory feeds `points_discovered.csv` and the Brick point registry.

84.1 Operator workflow

1. Confirm `network_setup({{ "/bacnet/network-setup/" | relative_url }}`.
2. Run discover from dashboard or API.
3. Review discovered objects; enable points for poll profiles.
4. Sync model bindings for FDD inputs.

85 Polling

86 Polling

BACnet field reads run in the openfdd-commission container.
The Operator Bridge does not bind UDP 47808.

Retired: openfdd-bacnet-poll image and openfdd-bacnet-poll systemd unit. Poll loop lives in **commission** only — do not run both.

BACnet polling has two stages:

1. **Field reads** — commission poll loop → workspace/bacnet/polls/samples.csv
2. **Historian ingest** — bridge background worker → workspace/data/feather_store/

86.1 Commission poll loop (default)

The **commission** container runs the poll loop at startup. It reads enabled rows from points.csv and appends BACnet RPM results to samples.csv.

Setting	Location
BACnet bind	workspace/bacnet/commissioning/commission.env
Enabled points	workspace/bacnet/commissioning/points.csv
Poll output	workspace/bacnet/polls/samples.csv
Historian	workspace/data/feather_store/

Check activity:

```
tail -1 workspace/bacnet/polls/samples.csv  
docker compose logs --since 5m commission | grep -i poll
```

86.2 Bridge ingest worker

Inside **bridge**, a background thread watches samples.csv and ingests new rows into feather. Disable only for debugging:
OFDD_DISABLE_POLL_WORKER=1.

86.3 Health

Dashboard stack health (GET /health/stack) reports bacnet_poll status from the commission agent — not a separate container.

Stale points trigger data-quality fault patterns — see [Sensor quality faults]({{ "/fault-codes/sensor-quality/" | relative_url }}).

86.4 Poll strategy (edge)

Open-FDD intentionally keeps BACnet field traffic **minimal**:

- **One enabled row per object** in points.csv — poll only what the model/FDD needs.
- **Uniform cycle** — every enabled point is RPM-pollled each cycle; the loop sleeps $\max(15, \min(\text{poll_interval_s}))$ (typically **60 s** on Acme).
- Per-point poll_interval_s in the UI is the **commissioned target**; throughput analytics compare configured vs observed samples.
- **Unit conversion** (e.g. Trane °C → °F) uses device_poll_profiles.csv at ingest — not a poll-rate file.

Acme ships **FDD-minimal polling** (~76 points on a typical GL36 lab when 8 rules are enabled), not the full discovery tree (~340+ GL36 objects or 566 discovered objects). Disable unneeded objects in BACnet commissioning rather than raising global poll rates.

AI agents / MCP: Never enable BACnet polling for every discovered object. Poll **only** historian columns bound to **enabled** FDD rules (rules_store.json brick_types, point_ids, and rule config column hints). Use bacnet_toolshed.fdd_minimal_poll or acme_commission_fdd_minimal.sh — not bulk --all enable scripts.

86.5 Async driver and single-stack I/O

Layer	Behavior
Dedicated bacnet-io thread	One asyncio loop + serial lock for all field I/O
RPM batching	Present-value reads chunked (25 objects) to cut APDUs
Poll loop	Sequential per-device RPM when interruptible=True so operator UI can preempt

Layer	Behavior
Operator UI	INTERACTIVE priority — cancels in-flight background work
Override scans	BACKGROUND , 15 min timeout — see Operator override scans ({{ “/bacnet/override-scans/”
Bridge ingest	Async notify after each poll cycle → feather historian

Do **not** run a second BACnet bind on 47808 (no parallel openfdd-bacnet-poll container). See [Containers](#)({{ “/architecture/containers/” | relative_url }}).

86.6 Operator override scans

Hourly P8 supervisory scans rotate one device at a time. Documented in [Operator override scans](#)({{ “/bacnet/override-scans/” | relative_url }}).

87 Operator override scans

88 Operator override scans (P8)

The commission agent rotates **one BACnet device per hour** (default) through a supervisory priority-array scan. Results are persisted for operator review and portfolio rollup — not for automatic writes.

88.1 Schedule

Setting	Default	Meaning
OFDD_OVERRIDE_SCAN_INTERVAL_S	3600	Seconds between single-device scans
OFDD_OPERATOR_OVERRIDE_PRIORITY	8	BACnet priority counted as

Setting	Default	Meaning
		“operator override”
Device list	points_discovered.csv	Unique device_instance rows
Rotation	registry.json cursor	Full plant rotation $\approx$ device_count $\times$ interval

Example: **33 devices $\times$ 1 h $\approx$ 33 h** for a full pass. Dashboard shows **next device** and **last scan** time.

88.2 What each scan does

1. point_discovery on the target device (commandable objects).
2. read_property on each object’s **priority-array** (not RPM — RPM loses priority slots).
3. Merge into workspace/bacnet/overrides/registry.json and append overrides_export.csv.
4. Advance cursor to the next device.

Fault code **operator_override** (P8) surfaces active overrides on the BACnet tree and building status.

88.3 BACnet I/O priority

Override scans run at **BACKGROUND** priority on the single shared BACnet stack (UDP 47808). Operator reads/writes use **INTERACTIVE** and preempt background work. Scheduled RPM polling also uses BACKGROUND and shares the same asyncio lock — polls and override scans take turns; neither starves the other indefinitely.

Override scans allow up to **15 minutes** per device (`_run_bacnet_override_scan` timeout). Bridge `POST /api/bacnet/overrides/scan-once` proxies with a matching 900 s timeout.

88.4 Manual trigger

```
curl -X POST -H "Authorization: Bearer $TOKEN" \
  http://EDGE:8765/api/bacnet/overrides/scan-once
```

Dashboard: BACnet page → **Scan next device now**.

88.5 Health checks

```
curl -H "Authorization: Bearer $TOKEN" http://EDGE:8765/api/bacnet/overrides/status
```

Expect `device_count > 0`, `last_scan_at` within the last few hours on a healthy edge, and `export_row_count` growing over time.

88.6 Data paths

Path	Content
workspace/bacnet/overrides/registry.json	Per-device last scan + override points
workspace/bacnet/overrides/overrides_export.csv	Flat export for spreadsheets

Preserved across Docker upgrades when `workspace/` is on the edge volume — see [Backup & restore]({{ "/ops/backup-restore/" | relative_url }}).

89 Write safety

90 Write safety

BACnet writes can change live plant behavior. Open-FDD disables writes by default. Enable only with operator sign-off, a tested allowlist, and audit logging.

90.1 Defaults

Control	Default
Supervisory writes	Off
Write allowlist	Empty — must enumerate device/object
Audit	Commands logged when writes enabled

90.2 Enable writes (commission role)

1. Set env flag documented in [Security](#)({{ "/security/" | relative_url }}) (OFDD_BACNET_WRITE_ENABLED or site equivalent).
2. Populate write allowlist CSV / config with explicit object IDs.
3. Use **commission** role JWT; every write should include reason/ note where API supports it.
4. Test on bench hardware before production OT.

90.3 Operator rules

- Never write setpoints or overrides on life-safety or fire/smoke interlocks through experimental rules.
- Prefer read-only polling for FDD until rules are validated in Rule Lab.
- Revert overrides through BACnet priority relinquish where supported.

91 BACnet

92 BACnet

Open-FDD integrates with field controllers via BACnet/IP (and bench MSTP overlays in dev compose).

Page	Topic
Network setup ({{ "/bacnet/network-setup/" relative_url }})	relative_url }}
Discover and read ({{ "/bacnet/discover-read/" relative_url }})	relative_url }}
Polling ({{ "/bacnet/polling/" relative_url }})	relative_url }}
Operator override scans ({{ "/bacnet/override-scans/" relative_url }})	relative_url }}
Write safety ({{ "/bacnet/write-safety/" relative_url }})	relative_url }}

93 Convention

94 Fault code convention

94.1 Format (Grade-A, 3.0.18+)

Short stable code: FAMILY-SUBSYSTEM-NNN — e.g. AHU-ECON-001, CHW-DT-001, DATA-STAL-001.

Semantic ID: dotted canonical_id —
e.g. ahu.economizer.not_using_free_cooling.

Legacy letter codes (AHU-E, VAV-C) remain **aliases** in open_fdd/faults/catalog/*.yaml until bridge sync completes.

Field	Example
code	AHU-ECON-001
canonical_id	ahu.economizer.not_using_free_cooling
family	AHU, VAV, CHW, DATA, CRS, ...
rule_doc_path	Link to rule cookbook recipe

Full schema: open_fdd/faults/schema.py. Catalog loader: open_fdd/faults/catalog.py.

94.2 Categories (Grade-A)

Category	Meaning
energy	Waste, reset stuck, plant inefficiency
comfort	Setpoint not met, poor zone performance
reliability	Short cycling, staging, equipment stress
maintenance	Leaking valves, filters, fouling proxies
indoor_air_quality	Ventilation shortfall, humidity
healthcare_risk	Pressure, isolation, critical-space excursions
data_quality	Stale, flatline, OOB, missing historian
controls_integrity	Overrides, chatter, schedule violations

94.3 Severity

Level	Operator response
info	Log; trend for maintenance window
low	Review in routine rounds
medium	Investigate within days
high	Prompt attention; comfort/energy impact likely
critical	Immediate operational risk (especially healthcare_risk)

94.4 Linking rules

In Rule Lab, set `fault_code` on the rule metadata. Batch FDD aggregates hits into `GET /api/faults/status` by family.

94.5 Cookbook mapping

Each catalog entry lists `cookbook_patterns` (e.g. `flatline_1h`, `mixing_envelope`, `rate_of_change`) — see [GL36 & sensor patterns](#)(`{{ "/rule-cookbook/expression-cookbook/" | relative_url }}`).

Executable rules are PyArrow or DataFusion SQL modules in `workspace/data/rules_py/`. YAML under `open_fdd/default_rules/` is catalog metadata only — see [Rule authoring](#)(`{{ "/rule-authoring/" | relative_url }}` `#yaml-files-in-the-repo`).

95 Live site update

96 Live site update

Upgrade GHCR images on a **live edge host** that already has `~/open-fdd/` — no `git pull` on the host.

IT operators: prefer [\[Quick Start — Updating the stack\]](#) (`{{ "/quick-start/updating/" | relative_url }}`) and `openfdd_site_backup.sh` / `openfdd_site_update.sh` on the edge host. This page adds control-machine UI deploy and Ansible paths.

96.1 Layout on the edge host

```
~/open-fdd/  
  docker-compose.yml  
  workspace/          # site state – backup first
```

workspace/ holds BACnet commissioning, poll CSV, feather historian, BRICK model, FDD rules, and auth.env.local. Image upgrades must **preserve** it.

See [Backup and restore](#)(({ { "/ops/backup-restore/" | relative_url } }) before any upgrade.

96.2 Concepts

Term	Meaning
Container	Running instance (bridge, commission, mcp-rag) — recreated on upgrade
Image	GHCR package (tag latest or dated tag)
workspace/	Open-FDD site state on disk — do not delete

96.3 1. Verify new images on GHCR

Three images must exist for every tag:

```
export NEW_TAG="${NEW_TAG:-latest}"  
  
docker manifest inspect ghcr.io/bbartling/openfdd-bridge:${  
{NEW_TAG} >/dev/null &&  
docker manifest inspect ghcr.io/bbartling/openfdd-commission:${  
{NEW_TAG} >/dev/null &&  
docker manifest inspect ghcr.io/bbartling/openfdd-mcp-rag:${  
{NEW_TAG} >/dev/null &&  
echo "All three images exist for ${NEW_TAG}"
```

96.4 2. Update dashboard UI (when release changed React)

Image pull alone does not update the browser UI. The Operator Bridge serves workspace/api/static/app/ from the bind mount **before** image-baked assets.

From control machine (full repo):

```
cd /path/to/open-fdd
./scripts/build_operator_dashboard.sh prod
cd infra/ansible
RUN_POST_CHECK=0 ./deploy.sh ui --limit <inventory_host>
```

Confirm hash on edge:

```
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.' | head -1
```

One command (UI + images + insurance check):

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
```

96.5 3. Pull and recreate (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

Manual equivalent:

```
cd ~/open-fdd
docker compose pull
docker compose up -d --force-recreate
docker compose ps
curl -sf http://127.0.0.1:8765/health
```

96.6 4. Ansible image-only (control machine)

```
OPENFDD_IMAGE_TAG=latest RUN_POST_CHECK=1 ./scripts/
upgrade_edge_ghcr.sh --limit <inventory_host>
```

96.7 5. Validate

→ Deployment validation({ { “/ops/deployment-validation/” | relative_url } })

96.8 Rollback

Restore docker-compose.yml from backup, pull previous tag, docker compose up -d --force-recreate. workspace/ is unchanged. Details: Backup and restore({ { “/ops/backup-restore/” | relative_url } }).

96.9 Related

- [Updating the stack](#)({{ "/quick-start/updating/" | relative_url }}) — edge-host helper scripts
- [Containers](#)({{ "/architecture/containers/" | relative_url }}) — three-image architecture

97 Central collection

98 OpenFDD RCx Central — collection

OpenFDD Edge sites emit **interval summaries** (not raw historian by default). **OpenFDD RCx Central** ingests rollups over Tailscale/VPN, stores them locally, and feeds the Dash overview and validation jobs.

RCx Central is read-only toward Edge/BACnet. It runs on an analyst workstation or Docker Desktop — not on the OT edge by default.

98.1 Interval summary schema

Defined in `portfolio/store/interval_schema.py` as `FaultIntervalSummary`:

- Identity: `site_id`, `building_id`, `equipment_id`, `equipment_family`
- Window: `timestamp_start`, `timestamp_end`
- Fault: `fault_code`, `canonical_id`, `severity`, `confidence`
- Activity: `active_minutes`, `percent_active`, `occurrence_count`
- Evidence: `evidence_json` (summarized, not full trend)
- Tuning: `tuning_version`, `tuning_params_hash`
- Operator: `operator_status` (new | acknowledged | resolved | false_positive | ...)

98.2 Edge export formats

- JSONL (append-only events)
- Parquet / Feather (batch intervals)

98.3 Collect today (rollup API)

```
source infra/ansible/secrets/acme.env.local
python3 scripts/portfolio_collect.py
```

98.4 Tuning proposals

Human-reviewable proposals use `TuningProposal` in the same module. The AI agent **never** auto-writes BACnet points on edge — use `GET /api/building-agent/tuning-brief` and `POST /api/building-agent/apply-tuning` with `apply: false` first. Maintainer workflow: `skills/openfdd-edge-deploy-tune/SKILL.md`.

99 AHU faults

100 AHU faults

Air handling unit codes (sample from catalog).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
AHU-A	Supply fan performance degradation	warning	Spread/runtime vs baseline	Belt slip, dirty filters, VFD limit	Trend speed vs airflow; filter dP
AHU-B	Simultaneous heating and cooling	critical	Heat + cool outputs active together	Sequencing bug, leaking valve	Trend coil commands; valve feedback
AHU-C	SAT sensor fault	warning	Flatline / OOB SAT	Failed sensor, bad calibration	SAT trend; compare to coil discharge
AHU-D	MAT inconsistent with OAT/RAT	warning	MAT outside OAT-RAT envelope	MAT/OAT/RAT sensor errors	Verify damper positions; cross-check OAT
AHU-E	Economizer not economizing	warning	Free cool available, OA minimum	Stuck damper, lockout	OA damper vs OAT trend

Code	Title	Severity	Detection summary	Likely causes	Operator checks
AHU-F	Damper command vs feedback mismatch	warning	Cmd $\neq$ feedback	Stuck actuator, linkage	Compare cmd/feedback trends

Example rule: flatline_1h on SAT $\rightarrow$ tag **AHU-C**. Recipe: [Python recipes]({{ "/rule-cookbook/python-recipes/" | relative_url }}).

101 Standards crosswalk

102 Standards crosswalk

Grade-A fault definitions carry a standards_crosswalk block in open_fdd/faults/catalog/*.yaml.

Field	Source	Use in Open-FDD
ornl_lbnl_taxonomy	ORNL/ LBNL unified HVAC fault taxonomy	Canonical ontology backbone
ashrae_g36_reference_type	ASHRAE Guideline 36	Trim-and-respond, plant requests, supervisory sequences
ashrae_207_reference_type	ASHRAE Standard 207	Economizer FDD test categories
brick_classes	<u>Brick</u> <u>Schema</u>	Equipment/point semantic roles
haystack_tags	Project Haystack (optional)	Interoperability tags

102.1 Example

```
standards_crosswalk:
  ornl_lbnl_taxonomy: ahu/economizer/free_cooling
  ashrae_g36_reference_type: supply_air_temp_reset
  ashrae_207_reference_type: economizer_fault_detection
```

```
brick_classes: [brick:Air_Handler_Unit, brick:Economizer]
haystack_tags: [equip, ahu, econ]
```

```
Export full catalog: python3 -c "from open_fdd.faults.catalog
import catalog_export; import json;
print(json.dumps(catalog_export(), indent=2))"
```

103 Backup and restore

104 Backup and restore

workspace/ on the edge host is the **live site state** — feather historian, BACnet commissioning files, BRICK model, FDD rules, auth, and dashboard static bundle. Container images are replaceable; **backup workspace/ before every upgrade.**

Canonical operator flow: `Edge site lifecycle({ { "/quick-start/site-lifecycle/" | relative_url } })`.

104.1 Quick backup (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Archives default to `~/openfdd-backups/latest` (overwrites each run).

Fast pre-upgrade backup (skips large workspace/bacnet/polls/ CSV history):

```
BACKUP_INCLUDE_POLL_SAMPLES=0 ./scripts/openfdd_site_backup.sh
```

Custom location:

```
BACKUP_ROOT=~/openfdd-backups/manual ./scripts/
openfdd_site_backup.sh
```

104.2 Upgrade + automatic backup purge

After a successful `./scripts/openfdd_site_update.sh`, the backup directory is **removed** by default once health and workspace layout checks pass. Set `PURGE_BACKUP_AFTER_SUCCESS=0` to retain archives.

104.3 Restore from backup

```
cd ~/open-fdd
RESTORE_WORKSPACE=1 ./scripts/openfdd_site_update.sh
```

Default historian cap on restore: **200 GiB** of newest feather shards. Full restore:

```
RESTORE_WORKSPACE=1 RESTORE_FEATHER_MAX_GIB=0 ./scripts/
openfdd_site_update.sh
```

104.4 What to protect

Path	Contents
workspace/data/ feather_store/	Feather historian
workspace/bacnet/ commissioning/	points.csv, commission.env
workspace/bacnet/ polls/samples.csv	Poll output
workspace/data/*.json	Model, rules store, FDD results
workspace/ auth.env.local	Login secrets
workspace/api/static/ app/	Dashboard bundle (if deployed separately)

104.5 Ansible / bensserver

From bensserver for any edge host (backup runs **on the edge**, no workspace transfer over Tailscale):

```
OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_site.sh --limit
acme_vm_bbartling
OPENFDD_IMAGE_TAG=3.1.3 ./scripts/upgrade_edge_site.sh --limit
acme_vm_bbartling --fast-backup
```

104.6 Safe cleanup

Handled automatically by openfdd_site_update.sh (stopped containers, dangling/unused images). Manual:

```
docker image prune -f
```

Never on live sites: docker compose down -v, docker volume prune, docker system prune -a --volumes, or deleting workspace/.

104.7 Related

- [Edge site lifecycle](#)({ { "/quick-start/site-lifecycle/" | relative_url }})
- [Updating the stack](#)({ { "/quick-start/updating/" | relative_url }})
- [Live site update](#)({ { "/ops/live_site_update/" | relative_url }})

105 OpenFDD RCx Central API

106 OpenFDD RCx Central API

Local analyst API for multi-edge RCx workflows over Tailscale/VPN. **Read-only toward OpenFDD Edge** — no BACnet, no commands.

Package path: portfolio/central/api.py — service name **openfdd-rcx-central**.

106.1 Start

```
pip install -r portfolio/requirements.txt
./scripts/run_central_api.sh    # http://127.0.0.1:8060/health
./scripts/run_portfolio_dash.sh # http://127.0.0.1:8050
```

Bind defaults: 0.0.0.0 (OPENFDD_CENTRAL_API_HOST, OPENFDD_RCX_DASH_HOST). LAN: sudo ./scripts/open_rcx_central_lan_ports.sh.

Agent guide: [docs/agent-skills/rcx-central-dash-agent.md](#)

106.2 Edge registry (browser auth)

Method	Path	Purpose
GET	/api/central/edges	Saved Edge instances (secrets masked)
POST	/api/central/edges	Add/update Edge
POST	/api/central/edges/test	Test connection (health + model)
PUT	/api/central/edges/{site_id}	Update Edge
DELETE	/api/central/edges/{site_id}	Remove Edge

106.3 Analytics & RCx

Method	Path	Purpose
GET	/api/central/overview/{site_id}	Dash payload: KPIs, fault pie, building summary, P8
GET	/api/central/mechanical-summary/{site_id}	Equipment counts, BACnet, narrative
GET	/api/central/fdd-analytics/{site_id}	FDD rules + fault descriptions
GET	/api/central/fdd-preset/{site_id}/{preset_id}	Proxy Edge FDD/BRICK preset (Data Model tab parity)
POST	/api/central/rcx/preview	Data readiness + chart catalog
POST	/api/central/rcx/charts/preview	Matplotlib PNG previews (base64)
POST	/api/central/rcx/report	Download DOCX report

106.4 Validation (legacy collect workflow)

Method	Path	Purpose
GET	/api/central/sites	Edge registry + last check-in
POST	/api/central/sites/{site_id}/collect-validate	Collect rollup + validation
POST	/api/central/validation/run	One-off or scheduled plan
GET	/api/central/validation/jobs	Stored jobs
GET	/api/central/validation/jobs/{id}	Job detail

106.5 RCx report body

```
{
  "site_id": "acme",
  "hours": 24,
  "charts": ["fault_hours_by_severity", "ahu_sat_vs_setpoint"],
  "sections": ["executive_summary", "mechanical_summary"],
  "save_to_volume": true
}
```

Reports save under portfolio/data/reports/ (configurable via OPENFDD_RCX_CENTRAL_DATA).

Legacy endpoint: POST /api/central/rcx/report-legacy

107 VAV faults

108 VAV faults

Variable air volume terminal codes (sample).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
VAV-A	Zone heating/cooling overlap	warning	Heat and cool at terminal	Reheat valve stuck, SAT too low	Zone temp vs SAT; valve cmd
VAV-B	Poor zone temperature control	warning	Persistent deviation from setpoint	Oversized box, bad tuning	PI loop trends; occupancy
VAV-C	Zone temperature sensor fault	warning	Flatline / OOB zone temp	Failed zone sensor	Flatline test; compare to neighbor zones
VAV-D	Minimum airflow not met	warning	Flow below minimum cmd	Damper stuck, pressure issue	Airflow vs damper cmd
VAV-E	Discharge temp deviation from SAT	warning	DAT far from SAT setpoint	Coil issue, sensor drift	DAT vs SAT SP

Example rule: oob_rolling on zone temp → **VAV-C** or comfort bounds per site policy.

109 Docker images (GHCR)

110 Docker images (GHCR)

Full **Open-FDD edge/operator stack** — not the PyPI library.

110.1 Images

GHCR image	Service	Role
ghcr.io/bbartling/openfdd-bridge	bridge	API, dashboard, historian
	commission	

GHCR image	Service	Role
ghcr.io/bbartling/openfdd-commission		BACnet discover/read/poll
ghcr.io/bbartling/openfdd-mcp-rag	mcp-rag	MCP + doc-search sidecar

Image name openfdd-mcp-rag is retained for compatibility; the container runs the unified MCP server.

110.2 Tags

Tag	Meaning
3.0.30	Exact release (preferred for production)
3.0	Minor alias on release tags
latest	Latest tagged release only
edge	Not published by default — use pinned versions on OT hosts

```
export OPENFDD_IMAGE_TAG=3.0.30
docker pull ghcr.io/bbartling/openfdd-bridge:${OPENFDD_IMAGE_TAG}
docker pull ghcr.io/bbartling/openfdd-commission:${
{OPENFDD_IMAGE_TAG}
docker pull ghcr.io/bbartling/openfdd-mcp-rag:${OPENFDD_IMAGE_TAG}
```

110.3 Publish

Triggered by git tag vX.Y.Z → workflow **Publish Docker images to GHCR** (tags X.Y.Z, X.Y, latest).

Manual: Actions → **Publish Docker images to GHCR** (workflow_dispatch) — reads version from pyproject.toml and publishes the same tags (3.1.3, 3.1, latest). Optional image_tag input overrides the version.

110.4 vs PyPI

Need	Use
Embed FDD in your Python/cloud job	pip install open-fdd
BACnet + UI + bridge on an edge host	GHCR images

See [Release process](#)({{ "/developer/release-process/" | relative_url }}).

111 RTU faults

112 RTU faults

Packaged rooftop unit codes (sample).

Code	Title	Severity	Detection summary	Likely causes	Operator checks
RTU-A	Compressor short cycling	warning	Rapid on/off cycles	Low charge, oversized unit	Run-time histogram
RTU-B	Economizer / OA damper fault	warning	OA behavior vs conditions	Actuator, sensor	OA damper vs enthalpy/OAT
RTU-C	Supply air temperature fault	warning	SAT flatline or OOB	Sensor failure	SAT trend vs outdoor conditions
RTU-D	Simultaneous heat and cool	critical	Heat + cool stages together	Control board, relay stuck	Stage status trends

Bind rules to packaged unit fdd_input points (SAT, OAT, compressor cmd, ...).

113 Deployment validation

114 Deployment validation

For the **live Acme site** after GHCR updates, use the dedicated harness documented in [Acme live validation](#)({{ "/ops/acme-live-validation/" | relative_url }}) (acme_post_deploy_validate.sh).

Non-destructive checks after deploy or image upgrade. Run from the **edge host** (smoke) or from a **control machine** with the full repo (insurance suite).

114.1 Edge host smoke (5 minutes)

```
cd ~/open-fdd

docker compose ps
curl -sf http://127.0.0.1:8765/health | jq . || curl -sf http://
127.0.0.1:8765/health

tail -1 workspace/bacnet/polls/samples.csv
du -sh workspace/data/feather_store/

docker compose logs --since 20m | grep -Ei "error|exception|
traceback|critical" | tail -20 || true
```

Via Caddy (building LAN):

```
curl -sf http://<edge-ip>/health
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.' | head -1
```

114.2 Browser checklist

1. Open `http://<edge-ip>/` (Caddy :80, not only loopback :8765)
2. Integrator login (`workspace/auth.env.local` on the edge host)
3. **Host stats** — container image tag / git SHA
4. **BACnet** — driver tree shows devices; last poll timestamp recent
5. **Model & assignments** — commissioning export returns JSON
6. **Rule Lab** — quick-test a saved rule (backend: arrow)

114.3 Control-machine insurance check

From a machine with the full open-fdd repo and Ansible inventory:

```
cd infra/ansible/scripts
./post_deploy_check.sh --limit <inventory_host> --full
```

Or by IP:

```
./post_deploy_check.sh --host <edge-ip> --ssh-user <deploy-user>
--full
```

Checks include: Caddy → React dashboard, `/health/stack`, MCP RAG, BACnet tree, BRICK/SPARQL, Docker log scan.

Do not run on live commissioned sites without approval: `acme_operational_verify.sh` with BACnet rediscover — use `--skip-discover` for smoke only. See [Examples — GL36 lab]({{ "/examples/acme-gl36-lab/" | relative_url }}).

114.4 LAN security smoke (optional)

From a workstation on the same LAN as the edge:

- Windows: `scripts/security/Run-OpenFddSecurityScan.ps1`
- macOS/Linux: `scripts/security/run_openfdd_security_scan.sh`

Expected findings: [ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}).

114.5 Arrow / FDD rules (3.0+)

On control machine:

```
./scripts/validate_fdd_backends.sh --docker
```

On edge after upgrade:

```
grep -R "apply_faults_arrow" workspace/data/rules_py | wc -l
```

114.6 Related

- [Health check]({{ "/quick-start/health-check/" | relative_url }})
- Security testing cycle({{ "/developer/security-testing/" | relative_url }}) (maintainers)

115 Sensor and data quality

116 Sensor and data quality

Sensor faults use cookbook patterns `flatline_1h`, `oob_rolling`, `rate_of_change`, and `mixing_envelope`. Full thresholds: GL36 & sensor patterns({{ "/rule-cookbook/expression-cookbook/" | relative_url }})#sensor-validation-bounds-flatline-rate-of-change).

116.1 Catalog codes

Code	Title	Severity	Pattern	Detection summary
BLD-B	Outdoor air sensor fault	warning	flatline, oob, rate_of_change	OAT flatline, OOB, or unrealistic spike
BLD-D	Stale telemetry	critical	stale_points	No new samples within threshold
AHU-C	SAT sensor fault	warning	flatline_1h, oob_rolling	Supply air temp stuck or out of band
AHU-D	MAT inconsistent with OAT/RAT	warning	mixing_envelope	Mixed air outside OAT-RAT envelope
VAV-C	Zone temperature sensor fault	warning	flatline_1h, oob_rolling	Zone temp stuck or 55–90 °F band breach
RTU-C	Supply air temperature fault	warning	flatline_1h, oob_rolling	Packaged unit SAT fault
CH-D	CHW supply temperature sensor fault	warning	flatline_1h, oob_rolling	Plant CHW sensor fault
DC-C	Datacom sensor anomaly	warning	flatline_1h	CRAH/room sensor flatline or RH OOB
HP-D	Discharge/suction temperature sensor fault	warning	flatline_1h, oob_rolling	Heat pump refrigerant/air temp sensor

116.2 Default bounds (imperial, occupied)

Sensor	Min	Max	Flatline tol	Max Δ /hr
Zone temp	55 °F	90 °F	0.10 °F	4 °F

Sensor	Min	Max	Flatline tol	Max Δ /hr
Supply air temp	50 °F	110 °F	0.15 °F	8 °F
Return air temp	55 °F	95 °F	0.10 °F	3 °F
Duct static	−0.5	3.0 inH ₂ O	0.02	0.5
RH	0 %	100 %	1.0 %	15 %
CHW	40 °F	90 °F	0.10 °F	4 °F
HW	70 °F	200 °F	0.15 °F	6 °F
Condenser water	50 °F	110 °F	0.15 °F	5 °F
CO ₂ (occupied)	400 ppm	1000 ppm	5 ppm	200 ppm

Return-air checks use **narrow bands**; when MAT/OAT/RAT are available, prefer **mixing envelope** over RAT bounds alone.

116.3 Communication quality

- Monitor poll cycle timestamps in dashboard.
- Use `stale_points` for dropout detection (**BLD-D**).
- Do not confuse **comm loss** with **equipment fault** — verify BACnet device online before dispatching mechanical.

116.4 Operator workflow

1. Confirm **BLD-D** (stale) is resolved before trusting other sensor codes.
2. Cross-check **BLD-B** OAT against weather service or JSON API driver.
3. For **VAV-C** / **AHU-C**, compare neighbor zones / coil behavior before replacing sensor.

117 Logging and audit

118 Logging and audit

Open-FDD uses a **two-tier logging model** familiar to IT and security teams: structured **audit/error JSONL** for forensics and compliance, plus **service stdout** for engineering troubleshooting. Defaults include **age pruning**, **size rotation in MB**, and **Docker log caps** — no operator cron required.

118.1 Log types

Tier	Location	Format	Who reads it
Audit	workspace/logs/audit.jsonl	JSON Lines	Security, MSI, integrator API
Errors	workspace/logs/error.jsonl	JSON Lines	Support, integrator API
Service stdout	Docker: docker compose logs · Dev: workspace/.local-run/*.log	Plain text (uvicorn, workers)	Engineering
Historian	workspace/data/feather_store/	Feather	FDD, plots — separate retention (data.env.local)

Structured audit logs are **not** mixed into uvicorn access logs. That separation matches common practice: SIEM-friendly JSONL for auth and OT commands; unstructured stdout for stack traces and poll worker noise.

118.2 Audit record schema

Each line in audit.jsonl / error.jsonl is one JSON object:

Field	Example	Notes
@timestamp	2026-06-07T19:57:21+00:00	UTC ISO-8601
event_id	UUID	Unique per event
event_type	auth.login.failure	Stable taxonomy (see below)
severity	warning	debug ... critical
outcome	failure	success, failure, denied, ...

Field	Example	Notes
service	openfdd-bridge	
host	hostname	
action	login	Human-readable verb
actor	{username, role}	After successful auth
client	{ip, user_agent}	Respects OFDD_TRUST_X_FORWARDED_FOR behind Caddy
http	method, path, status	When HTTP-triggered
resource	BACnet object, model id, ...	When applicable
detail	Sanitized map	Passwords, tokens, secrets stripped by key name
request_id	UUID	On audited HTTP paths

118.2.1 Auth events (industry-standard login trail)

event_type	When
auth.login.success	Valid credentials
auth.login.failure	Bad password (generic client message)
auth.login.lockout	Rate limit exceeded (default 5 failures / 5 min)

Login rate limits and lockouts are **audited**; responses never reveal whether the username exists.

118.2.2 OT / commissioning events

event_type	When
bacnet.command	Supervisory write (allowed, denied, dry-run)
bacnet.discover	Who-Is / point discovery
model.write	BRICK model mutations
agent.tool	Agent tool calls (prompts hashed/truncated)
api.access	Sensitive POST/PUT/PATCH/DELETE (BACnet, ingest)
error.unhandled	Unhandled exception (also in error.jsonl)

Not audited: routine GETs (dashboard reads, plot data, health). This reduces noise while keeping **auth and field commands** on the trail.

118.2.3 Secret redaction

Never logged: passwords, bearer tokens, `OFDD_AUTH_SECRET`, private keys. Detail objects drop keys containing password, token, secret, authorization. Agent tool arguments use additional hashing for prompts and rule source.

Opt-in full prompt logging (lab only): `OFDD_AUDIT_LOG_PROMPTS=1`.

118.3 Integrator API (read logs in the UI stack)

Method	Path	Role
GET	<code>/api/audit/events?limit=100</code>	integrator
GET	<code>/api/audit/errors?limit=100</code>	integrator
GET	<code>/api/audit/summary</code>	integrator

Use these for post-incident review without shell access. Forward `workspace/logs/*.jsonl` to SIEM (rsyslog, Filebeat, Wazuh) on production edges.

118.4 Retention and rotation (defaults)

Configured in `workspace/data.env.local` (copy from `data.env.example`):

Variable	Default	Behavior
<code>OFDD_LOG_RETENTION_DAYS</code>	90	Drop audit/error JSONL lines older than N days; delete aged <code>.local-run</code> and archived <code>*.log / audit/*.jsonl</code>
<code>OFDD_LOG_MAX_MB</code>	50	When <code>audit.jsonl</code> or <code>error.jsonl</code> exceeds N MB, archive to <code>audit.{UTC}.jsonl</code> and start fresh
<code>OFDD_LOCAL_RUN_LOG_MAX_MB</code>	25	

Variable	Default	Behavior
OFDD_LOG_ROTATE_INTERVAL_HOURS	6	Rotate workspace/.local-run/*.log (bridge, Caddy, FDD loop, ...) Background retention while bridge is running (not only at restart)

Rotation runs at **bridge startup** and on the **scheduled interval**. No logrotate.d entry is required for audit JSONL.

Optional path overrides:

```
OFDD_AUDIT_LOG_PATH=/var/log/openfdd/audit.jsonl
OFDD_ERROR_LOG_PATH=/var/log/openfdd/error.jsonl
```

118.5 Docker edge: json-file caps (not journald for app audit)

Production stacks use **Docker Compose** (docker/compose.edge.yml). Container stdout uses the **json-file driver with size limits**:

```
logging:
  driver: json-file
  options:
    max-size: "25m"
    max-file: "5"
```

That caps each service at roughly **125 MB** of rotated container logs on disk — the same pattern many teams use instead of unbounded docker logs.

Why audit JSONL stays on disk (not journald)?

- Append-only files under workspace/logs/ survive container recreate and bind-mount to the host.
- JSONL lines import cleanly into SIEM without journal field parsing.
- OT incident response often needs **months** of auth/BACnet write history on the edge box.

When journald still applies: native systemd units (Caddy, Ollama on bare metal) — use `journalctl -u <unit>` per your Linux runbook. Bridge audit remains in `workspace/logs/` even when uvicorn stdout goes to Docker or `.local-run/`.

118.6 Local dev (run_local.sh)

File	Service
workspace/.local-run/bridge.log	Bridge / uvicorn
workspace/.local-run/commission.log	BACnet commission agent
workspace/.local-run/caddy.log	Caddy
workspace/.local-run/fdd_loop.log	Scheduled FDD
workspace/.local-run/mcp_rag.log	MCP RAG
workspace/.local-run/ollama.log	Ollama

These are **engineering logs**. Structured audit still lands in `workspace/logs/audit.jsonl` when auth is enabled.

118.7 Quick checks

```
# Last auth failures
grep 'auth.login' workspace/logs/audit.jsonl | tail -5

# Integrator API (after login)
curl -s -H "Authorization: Bearer $TOKEN" http://127.0.0.1:8765/
api/audit/summary | jq .

# Docker edge
docker compose logs bridge --tail 50
```

118.8 Comparison to typical web apps

Practice	Open-FDD
Structured auth audit	Yes — JSONL with actor, IP, outcome
Login rate limit + logout audit	Yes
Secret redaction in logs	Yes — key-based + agent sanitization
Request correlation ID	Yes — on audited mutation paths
Centralized stdout JSON	No — stdout is plain text; audit is JSONL

Practice	Open-FDD
Full HTTP access log	No — mutations and auth only (OT noise reduction)
Log rotation / size caps	Yes — defaults above + Docker max-size
SIEM-ready export	Yes — file tail / Filebeat on workspace/logs/

For BACnet write safety and credential generation, see [SECURITY.md](#) and [Secrets and auth](#)({{ "/security/secrets-auth/" | relative_url }}).

118.9 Related

- [JSON API driver](#)({{ "/drivers/json-api/" | relative_url }}) — OpenWeatherMap showcase, env-based API keys (json_api.env.local)
- [Live site update](#)({{ "/ops/live_site_update/" | relative_url }}) — docker compose logs after upgrades
- [Data flow](#)({{ "/architecture/data-flow/" | relative_url }}) — historian retention (OFDD_FEATHER_*) is separate from audit logs

119 GHCR image retention

120 GHCR image retention

Open-FDD publishes edge images to **GHCR** (ghcr.io/bbartling/openfdd-*). During rapid development, old package versions accumulate. This retention system prunes **noise** while protecting rollback tags for **Acme**, our live validation building.

120.1 Policy (conservative)

Always keep:

- latest and edge
- Latest **5** SemVer release tags per image (configurable)
- Tags listed in scripts/ghcr-retention-protected-tags.txt
- Current Acme deployed tag (workflow input or --current-acme-tag)
- Previous Acme known-good tag (--previous-acme-tag)

Delete candidates (only with `--confirm-delete`):

- Untagged versions older than **7** days
- SHA-only tags older than **30** days
- Feature/dev-style tags older than **30** days
- SemVer releases beyond the latest N window (unless protected)

Images are defined in `docker/images.yaml`:

- `openfdd-bridge`
- `openfdd-commission`
- `openfdd-mcp-rag`
- `openfdd-cloud-exporter`

120.2 Dry-run first

The script defaults to **dry-run**. It prints a keep/delete table and optional JSON/Markdown reports. Nothing is deleted until you pass `--confirm-delete`.

There is **no scheduled automatic deletion** yet — only manual runs locally or via GitHub Actions `workflow_dispatch`.

120.3 Protect Acme rollback tags

Before cleanup, edit `scripts/ghcr-retention-protected-tags.txt`:

```
latest
edge
v3.0.31
v3.0.30
```

Or pass tags at runtime:

```
./scripts/ghcr_prune_packages.sh --all-images --dry-run \  
  --current-acme-tag v3.0.31 \  
  --previous-acme-tag v3.0.30
```

120.4 Run locally

```
gh auth login    # needs read:packages + delete:packages for  
confirm-delete
```

```
./scripts/ghcr_prune_packages.sh --all-images --dry-run  
./scripts/ghcr_prune_packages.sh --image openfdd-bridge --dry-run \  
  --json-out reports/ghcr-prune-plan.json \  
  --confirm-delete
```

```
--markdown-out reports/ghcr-prune-plan.md

# After reviewing the plan:
./scripts/ghcr_prune_packages.sh --all-images --confirm-delete \
--current-acme-tag v3.0.31 --previous-acme-tag v3.0.30
```

120.5 GitHub Actions workflow

Workflow: **Prune old GHCR images** (.github/workflows/ghcr-prune.yml)

1. Actions → **Prune old GHCR images** → **Run workflow**
2. Leave **Dry run only** = true for the first run
3. Download the ghcr-prune-report artifact
4. Re-run with **Dry run only** = false only after reviewing the plan
5. Set **Current Acme deployed tag** and **Previous Acme rollback tag** when known

120.6 Permissions

Deleting package versions requires:

- Local: gh auth login with delete:packages (and read:packages)
- Actions: permissions.packages: write on the workflow (already set)

If deletion fails, the script prints:

```
Deleting GHCR package versions requires package delete
permissions...
```

120.7 Verify images still pull

After pruning:

```
docker pull ghcr.io/bbartling/openfdd-bridge:latest
docker pull ghcr.io/bbartling/openfdd-commission:latest
```

On Acme:

```
export OPENFDD_IMAGE_TAG=v3.0.31
./scripts/upgrade_edge_ghcr.sh --limit acme_vm_bbartling
./scripts/acme_post_deploy_validate.sh --limit acme_vm_bbartling
--quick
```

120.8 Rollback example

```
export OPENFDD_IMAGE_TAG=v3.0.30
./scripts/upgrade_edge_ghcr.sh --limit acme_vm_bbartling
```

120.9 Related issues

Track open Acme/deploy blockers in [GitHub issue #276](#).

121 Fault Codes

122 Fault Codes

Open-FDD uses a **check-engine** model: one building-level status (green / yellow / red) with a catalog of fixed fault codes per equipment family.

Page	Content
Convention ({ { "/fault-codes/convention/"	relative_url }}
AHU faults ({ { "/fault-codes/ahu/"	relative_url }}
VAV faults ({ { "/fault-codes/vav/"	relative_url }}
RTU faults ({ { "/fault-codes/rtu/"	relative_url }}
[Sensor & data quality]({ { "/fault-codes/sensor-quality/"	relative_url }}
Short lookup ({ { "/fault-codes/short-lookup/"	relative_url }}

Live catalog API: GET /api/faults/catalog on the Operator Bridge.

123 Operations

124 Operations

Runbooks for live edge hosts after the initial `Quick Start({{ "/quick-start/" | relative_url }})` deploy.

Page	When to use
Live site update({{ "/ops/live_site_update/" relative_url }})	<code>relative_url }}</code>
Backup and restore({{ "/ops/backup-restore/" relative_url }})	<code>relative_url }}</code>
Logging and audit({{ "/ops/logging/" relative_url }})	<code>relative_url }}</code>
Deployment validation({{ "/ops/deployment-validation/" relative_url }})	<code>relative_url }}</code>
ACME live validation({{ "/operations/acme-live-validation/" relative_url }})	<code>relative_url }}</code>
Bench 5007 long FDD smoke({{ "/operations/bench-5007-long-fdd-smoke/" relative_url }})	<code>relative_url }}</code>
Bench 5007 dual-source smoke({{ "/operations/bench-5007-dual-source-smoke/" relative_url }})	<code>relative_url }}</code>

Site-specific lab notes (example BACnet scope, GL36 rules):
[Examples & lab notes\({{ "/examples/" | relative_url }}\)](#).

For first-time deploy: `[Quick Start — Docker]({{ "/quick-start/docker/" | relative_url }})`.

125 Portfolio

126 Portfolio (central desk)

Multi-site analytics on **benserver** — not part of the edge Docker stack.

Topic	Page
Collect rollups from edge sites	Central collection ({{ "/"portfolio/central-collection/"

Repo layout: portfolio/ (collector, Dash, sites.json). Maintainer tuning API flow: skills/openfdd-edge-deploy-tune/SKILL.md.

127 Acme VAV/AHU rule guidance

128 Acme VAV/AHU FDD rule bundle

Practical Arrow-native rule set for **acme** (vm-bbartling): one RTU/AHU (acme-vm-bbartling-rtu-01, BACnet **1100**) feeding **JCI VAVs** (instances 1-100, imperial °F at wire) and **Trane VAVs** (11000-13000, metric °C converted to °F before feather via device_poll_profiles.csv).

Install / refresh rules:

```
python3 scripts/setup_gl36_fdd.py --site-id acme --building-id vm-bbartling \
--ahu-equipment-id acme-vm-bbartling-rtu-01 --fan-point-id 1100-analog-output-1
```

Validate bundle locally:

```
python3 scripts/acme_validate_fdd_bundle.py
```

128.1 Mixed manufacturers and units

Manufacturer	BACnet instances	Wire units	Open-FDD handling
JCI VAV	1-100	°F	Rules use temp_unit: imperial; no poll conversion
Trane VAV	11000-13000	°C	metric_temp_f in edge_config/acme/vm-bbartling/device_poll_profiles.csv converts to °F before feather
RTU AHU	1100	°F	AHU rules bind to equipment / fan point

All zone-temperature rules share **imperial bounds** after poll conversion. If a new Trane box is added without a profile row, zone OOB/flatline will be wrong until the profile is added and data re-ingested.

Command scaling (damper, reheat, fan): rules normalize **0-1 vs 0-100%** with `pc.if_else(greater(c, 1.0), divide(c, 100), c)`.

Poll interval: Acme BACnet poll loop is **60 s**. Flatline windows use `poll_interval_s: 60` and `flatline_minutes: 60` → **60 samples per hour**, not the legacy 5-min × 12 assumption.

128.2 Rule matrix

Phase	Priority	Rule ID (prefix acme-)	Module	Fault	Equip
0	1	oat-bounds	oat_sensor_bounds.py	BLD-B	Build
0	2	oat-flatline-1h	oat_sensor_flatline.py	BLD-B	Build
0	3	oat-spike	oat_sensor_spike.py	BLD-B	Build
0	3b	oat-vs-web-spread	oat_vs_web_spread_1h.py	BLD-B	Build

Phase	Priority	Rule ID (prefix acme-)	Module	Fault	Equipment
0	4	zn-t-oob-occupied	vav_zone_temp_bounds_occupied.py	VAV-C	VAV
0	5	zn-t-flatline-1h	vav_zone_temp_flatline_occupied.py	VAV-C	VAV
0	6	da-t-flatline-1h	flatline_1h.py	VAV-E	VAV
0	7	sat-flatline-1h	flatline_1h.py	AHU-C	AHU
0	8	sap-flatline-1h	flatline_1h.py	AHU-F	AHU
1	9	vav-damper-stuck	vav_damper_stuck_flatline.py	VAV-D	VAV
1	10	vav-airflow-low	vav_airflow_low.py	VAV-D	VAV
1	11	vav-reheat-warm-ambient	zone_reheat_warm_ambient.py	VAV-A	VAV
1	12	vav-reheat-leak	vav_reheat_dat_vs_sat.py	VAV-A	VAV
2	13	ahu-afterhours-runtime	ahu_afterhours_runtime.py	BLD-C	AHU
2	14	mat-oob-economizer	oob_rolling.py	AHU-D	AHU
2	15	ahu-run-hours	ahu_run_hours.py	—	AHU
—	—	Stale telemetry	Poll health dashboard	BLD-D	All

128.2.1 Phase 2+ (not enabled by default)

Enable only after point coverage on RTU **1100** is verified (OAT, MAT, RAT, SAT, OA damper, cooling cmd):

- **AHU-A** duct static not maintained — needs static SP + fan speed (future rule)
- **AHU-D** mixed air envelope — `mixing_envelope_mask` + fan on
- **AHU-E** economizer trio — 100% OA tracking, mech cooling when free cooling available, OA damper excess open

128.2.2 Phase 3 rollups (future)

- Many simultaneous **VAV-D** airflow-low → suggest AHU static / fan issue
- Many **VAV-B/C** comfort faults → AHU scheduling or SAT problem
- Hot-water plant support when HW supply temp is modeled

128.3 Data model checks

Before enabling more rules:

1. **No duplicate BACnet device instances** on equipment rows (poll health merges by `bacnet_device_id`).
2. **No duplicate point_id** rows with different `external_id` / `fdd` aliases (causes `historian_lag` double-count).
3. **Trane devices** in `device_poll_profiles.csv` with `metric_temp_f`.
4. Run `scripts/acme_validate_fdd_bundle.py` after model edits.

128.4 Tuning notes

- **acme-zn-t-oob-occupied**: if flag rate > ~70%, use tuning brief / bounds auto-tune (needs ≥85% flagged + analytics on Arrow sweep).
- Disable noisy rules via `enabled: false` in `setup_gl36_fdd.py` rather than deleting modules.
- Economizer and reheat-leak rules stay off until OAT/MAT/SAT columns are confirmed on the 7-day feather window.

128.5 JSON API weather + BACnet OAT

- OpenWeather registered on edge JSON API tab (`web-oat-t`, `web-rh`, ~20–30 min poll).
- Local OAT: bind **1100-unknown-2** (RTU outdoor air local) with `external_id: oa-t` — `scripts/acme_patch_oat_column.py`.

- Rule acme-oat-vs-web-spread flags when local vs web spread exceeds 8 °F.

128.6 BACnet override scans

Commission agent rotates **one device/hour** through P8 priority-array reads. Status: GET /api/bacnet/overrides/status. See Operator override scans({ { "/bacnet/override-scans/" | relative_url } }).

128.7 Docker backup / recovery

Edge upgrades via scripts/upgrade_edge_ghcr.sh preserve workspace/data and feather volumes on the VM. After upgrade:

1. curl /health → expect matching openfdd_version
2. ./infra/ansible/scripts/acme_operational_verify.sh
3. Re-push rules if rules_store.json was reset: setup_gl36_fdd.py --host ... --token ...

Feather backup: snapshot workspace/data/feather/ (or site pack export) before major model changes.

129 Cursor agent safeguards

130 Cursor agent safeguards

Long-running shell jobs **crash Cursor** when agents block-wait, poll in loops, or stay attached to child processes.

130.1 What crashes the IDE

Unsafe (agents)	Why
pytest tests/workspace_bridge (~4 min)	Blocks chat turn; heavy output
./scripts/smoke_paired_fdd_harness.sh --short attached	30 min blocking wait
sleep 60 / Await / tail -f poll loops	

Unsafe (agents)	Why
Running harness .py directly from agent	Agent keeps terminal session alive Same as attached smoke

130.2 Safe pattern

Launch in **systemd user unit** (or setsid), then **poll status once** — never wait.

130.2.1 Paired FDD smoke

```
./scripts/run_paired_fdd_smoke_isolated.sh --short --bench-only
./scripts/smoke_paired_fdd_status.sh --mode short
```

130.2.2 Bench 5007 half-hour smoke (FDD + health + RCx)

```
./scripts/smoke_bench_5007_half_hour.sh
./scripts/smoke_bench_5007_half_hour_status.sh
```

130.2.3 workspace_bridge pytest (CI parity)

```
./scripts/run_workspace_bridge_pytest_isolated.sh
./scripts/workspace_bridge_pytest_status.sh
```

130.2.4 Humans in tmux/SSH

```
./scripts/smoke_paired_fdd_harness.sh --short --attached
```

--attached is **refused** when CURSOR_AGENT is set.

130.3 Implementation

File	Role
scripts/lib/cursor_agent_guard.sh	Shared refuse/poll helpers
scripts/run_paired_fdd_smoke_isolated.sh	systemd smoke launcher
scripts/smoke_paired_fdd_status.sh	Read-only smoke status

File	Role
scripts/ run_workspace_bridge_pytest_isolated.sh	systemd pytest launcher
scripts/ workspace_bridge_pytest_status.sh	Read-only pytest exit code

Default: smoke_paired_fdd_harness.sh redirects to the isolated launcher unless --attached.

131 Bench 5007 half-hour smoke

132 Bench 5007 half-hour smoke

30-minute validation for **device 5007** on the bench site (demo):

- **PyArrow vs DataFusion SQL** parity (smoke-paired-zn-t-bacnet-* and Niagara twins)
- **API health** (/api/health, bench poll status)
- **BACnet P8 override scan** — hourly rotation, one device per cycle (/api/bacnet/overrides/status)
- **Frontend** — SPA index + static assets (optional browser console with playwright)
- **Service logs** — bounded tail scan for ERROR/Exception lines
- **RCx DOCX** — smoke results embedded in smoke_validation section

132.1 Cursor / SSH safe launch

Never block-wait from a Cursor agent (crashes IDE on 30 min sleeps).

```
./scripts/run_local.sh start
./scripts/smoke_bench_5007_half_hour.sh
./scripts/smoke_bench_5007_half_hour_status.sh
```

Artifacts:

File	Content
workspace/reports/artifacts/ paired_fdd_smoke_validation.{md,json}	FDD parity cycles

File	Content
workspace/reports/artifacts/ bench_5007_half_hour_health.json	Health probe history
workspace/reports/rcx/ bench_5007_half_hour_smoke_rcx_*.docx	RCx report with smoke section

Regenerate RCx only:

```
python3 scripts/generate_half_hour_smoke_rcx_report.py
```

132.2 Environment

Variable	Default	Meaning
OPENFDD_SMOKE_HEALTH_PROBES	on for --short --bench-only	API/UI/log probes every 5 min
OPENFDD_SMOKE_RCX_REPORT	1	Build DOCX at end
OPENFDD_SMOKE_FRONTEND_BROWSER	off	JS console capture (needs playwright + chromium)

132.3 Override scan expectations

Commission agent runs `bacnet_override_scan_loop` with `OFDD_OVERRIDE_SCAN_INTERVAL_S=3600` (one device per hour). The smoke validates:

- `scan_interval_s` $\approx$ 3600
- `operator_priority` = 8 (P8)
- `device_count` $\geq$ 1 and cursor present
- Override summary flows into RCx `model_health` / analyst override notes via `build_rcx_report_context()`

133 Bench 5007 dual-source smoke

134 Bench 5007 dual-source smoke

Validates BACnet MS/TP device **5007** and Niagara baskStream station **bench9065** on the same physical bench.

134.1 Run

```
./scripts/run_local.sh start
./scripts/smoke_bench_5007_dual_source.sh
```

Environment (optional):

Variable	Default
OPENFDD_BASE_URL	http://127.0.0.1:8765
OPENFDD_AUTH_ENV	workspace/auth.env.local
OPENFDD_SMOKE_SITE_ID	demo
OPENFDD_SMOKE_POLL_SECONDS	60
OPENFDD_SMOKE_DURATION_MINUTES	10

The script wraps `scripts/smoke_benserver_bench.py` and adds DataFusion SQL lab preview when `open-fdd[datafusion]` is installed.

134.2 Paired semantic points

Semantic	BACnet column	Niagara column
oa-t	oa-t	niagara-oa-t
oa-h	oa-h	niagara-oa-h
duct-t	duct-t	niagara-duct-t
stat_zn-t	stat_zn-t	niagara-stat-zn-t

Import `workspace/data/bench_dual_source_model.json` if the model is empty.

135 Bench 5007 long FDD smoke

136 Bench 5007 long FDD smoke

Validates **data-source-agnostic FDD**, **PyArrow vs DataFusion SQL equivalence**, and the **fault confirmation window** on BACnet 5007 + Niagara bench9065 (site demo).

Read-only: polls devices; only Open-FDD evaluation thresholds change. Never prints credentials.

136.1 Commands

Mode	Command
CI synthetic	<code>python scripts/ smoke_bench_5007_long_fdd.py -- synthetic</code>
Dev dry-run (~15 min)	<code>python scripts/ smoke_bench_5007_long_fdd.py --dry- run</code>
Default (2 h)	<code>./scripts/ smoke_bench_5007_long_fdd.sh</code>
Overnight (12 h)	<code>python scripts/ smoke_bench_5007_long_fdd.py -- overnight</code>

136.2 Key environment

Variable	Default
OPENFDD_BASE_URL	http://127.0.0.1:8765
OPENFDD_SMOKE_DURATION_MINUTES	120 (720 if OPENFDD_SMOKE_OVERNIGHT=true)
OPENFDD_SMOKE_POLL_SECONDS	60
OPENFDD_SMOKE_BASELINE_MINUTES	20
OPENFDD_SMOKE_CONFIRMATION_ROWS	10
OPENFDD_SMOKE_CONFIRMATION_MINUTES	10
OPENFDD_SMOKE_PRIMARY_SEMANTIC	duct-t

Variable	Default
OPENFDD_SMOKE_FORCED_THRESHOLD_F	80.0

136.3 Fault confirmation window

At 1-minute polling, `min_true_rows = 10` means the raw condition must stay true for about **10 minutes** before a **confirmed fault** is reported. See [fault-confirmation.md](#).

136.4 Artifacts

- `workspace/reports/bench_5007_long_fdd_<timestamp>.md`
- `workspace/reports/bench_5007_long_fdd_<timestamp>.json`
- `workspace/reports/bench_5007_long_fdd_<timestamp>_events.csv`

136.5 Pass / fail

Verdict	Meaning
PASS	All checks passed; confirmation fully vectorized (not expected today)
WARN	Core FDD uses <code>pyarrow_compute</code> / <code>datafusion_sql</code> ; confirmation engine is <code>python_loop_over_arrow_mask</code> (known tech debt)
FAIL	<code>python_list</code> core crunching, early confirmed fault, backend mismatch, missing data, or secrets in output

Live mode uses `POST /api/bench/long-fdd/evaluate` (validation/smoke only).

136.6 Related

- [Dual-source smoke](#)
- [Fault confirmation](#)

137 ACME live validation

138 ACME live validation

ACME (site_id: acme, Acme Building) is the **live BACnet HVAC validation site** on the OT LAN. Use it for real VAV/AHU behavior, RCx rule tuning, and overnight FDD runs.

Do **not** confuse ACME with **Bench 5007** dual-source validation (site demo, BACnet 5007 + Niagara bench9065). Bench 5007 proves backend equivalence; ACME proves production HVAC rules on field hardware.

Reference model (no secrets): workspace/data/fixtures/acme_data_model.json. Validate locally:

```
python scripts/validate_acme_model_context.py
python scripts/validate_acme_model_context.py --json
```

138.1 Model export / import

1. **Export** — GET /api/model/commissioning-export (dashboard **Import / export** tab or API).
2. **Validate** — python scripts/validate_acme_model_context.py on saved JSON; dashboard dry-run before import.
3. **Import** — POST /api/model/commissioning-import only after review. Preserve point id, fdd_input, metadata.series_id, and existing fdd_rule_ids. Never invent rule IDs.

Someday ACME will be updated from live export again; until the bench harness is stable, treat the committed fixture as the regression baseline.

138.2 Run FDD (read-only)

Goal	Command / API
Post-upgrade smoke	./scripts/acme_post_deploy_validate.sh --limit acme_vm_bbartling --full
Selected rules	Rule Lab preview / POST /api/playground/test-rule
All enabled rules	POST /api/rules/batch → workspace/data/fdd_results.json
Overnight cycles	

Goal	Command / API
	<pre> OPENFDD_LIVE_ACME=1 python3 scripts/ acme_overnight_fdd_validate.py --limit acme_vm_bbartling </pre>

Normal validation is **read-only**: no BACnet writes, no Pooge reset, no setpoint commands. BACnet writes require commission role + explicit enable — see [BACnet write safety]({{ "/security/bacnet-writes/" | relative_url }}).

138.3 Reports

- Post-deploy JSON: reports/acme-live-validate.json (gitignored)
- Overnight FDD: under reports/ from overnight script
- FDD batch: workspace/data/fdd_results.json (local edge only)

Redact tokens and hostnames before sharing.

138.4 Fault confirmation

Rules may fire **raw** faults before **confirmed** faults. Configure `min_true_rows` and `min_elapsed_minutes` in rule cfg (e.g. 10 rows at 1-minute poll $\approx$ 10 minutes). See [Fault confirmation](#)({{ "/rule-cookbook/fault-confirmation/" | relative_url }}).

138.5 Safe rules on ACME

Prefer read-only sensor bounds, flatline, spread (OAT vs weather), damper stuck, and zone temperature rules. Avoid supervisory writes unless explicitly commissioned and guarded.

138.6 Related

- [ACME VAV/AHU rule guidance](#)({{ "/operations/acme_vav_ahu_rule_guidance/" | relative_url }})
- [Deployment validation](#)({{ "/ops/deployment-validation/" | relative_url }})
- [Bench 5007 long FDD smoke](#)({{ "/operations/bench-5007-long-fdd-smoke/" | relative_url }}) — dual-source bench only

139 AI agent context

140 AI agent context

Canonical map for humans and MCP/RAG agents operating Open-FDD on the OT edge. **No secrets** in docs, exports, reports, or chat context.

140.1 1. Open-FDD in one paragraph

Open-FDD is an **edge FDD platform**: BACnet / Niagara / JSON API drivers ingest telemetry into a **BRICK-style JSON model** and **Feather/Arrow historian**; **PyArrow** (primary) and optional **DataFusion SQL** rules produce boolean fault masks with a shared **confirmation window**; the **Operator Bridge** dashboard and **MCP sidecar** expose commissioning, Rule Lab, batch FDD, and doc search. Deploy via **GHCR Docker** (openfdd-bridge, openfdd-commission, openfdd-mcp-rag) on a trusted LAN — not the public internet.

140.2 2. Deployment mental model

Piece	Role
GHCR stack	openfdd-bridge (API + UI), openfdd-commission (BACnet poll), openfdd-mcp-rag (doc RAG)
Host Caddy	:80 / :443 front door to bridge
workspace/	Model, rules, feather store, auth env — persist across image upgrades
Auth	workspace/auth.env.local (gitignored); JWT via POST / api/auth/login
Update	<u>Edge site lifecycle</u> ({{ "/quick-start/site-lifecycle/"
CPU / platform	Auto-detect: x86_64 → linux/amd64, aarch64 → linux/arm64. Check: ./scripts/openfdd_check_ghcr_platform.sh. Override: OPENFDD_DOCKER_PLATFORM or bootstrap --platform. Pi:

Piece	Role
	[Raspberry Pi edge]({{ "/quick-start/raspberry-pi-edge/"

Rebuild MCP index after doc changes: ./scripts/build_mcp_rag_index.sh.

140.3 3. Source / driver modes

Mode	Feather source	When
BACnet direct	bacnet	Production OT — commission container polls :47808
Niagara baskStream	niagara_baskstream	Bench / sites with Niagara HTTP API
JSON API	json_api	Weather, REST sensors, OpenWeather demo
Platform API profile	scaffold	Future portfolio central — not production duplicate poll

Do **not** run legacy BACnet poll alongside Docker **commission** on the same device (double poll). Validation smokes may intentionally dual-source **Bench 5007** only.

140.4 4. Data model contract

Commissioning bundle: GET /api/model/commissioning-export → POST /api/model/commissioning-import (dry-run in UI first).

Field	Rule
sites[], equipment[], points[], fdd_rules[]	Required structure
Point id	Preserve on import — never renumber
fdd_input	Rule column role (oa-t, zn-t, sat, ...)
metadata.series_id	Historian series key

Field	Rule
metadata.external_ref	Stable feather/BACnet reference
fdd_rule_ids	Must reference existing fdd_rules[].id — never invent rule on import
Column map	open_fdd.arrow_runtime.build_column_map_from_model_points(site_id)

Live HVAC reference model: **ACME** (site_id: acme) — fixture at workspace/data/fixtures/acme_data_model.json. Bench dual-source model: site demo / device 5007 — not ACME.

140.5 5. Rule execution contract

Backend	Use when
PyArrow (apply_faults_arrow)	Full HVAC logic, rolling windows, ML prep, cookbook helpers
DataFusion SQL (backend: datafusion_sql)	Simple threshold / CASE / SQL-readable rules; optional pip install open-fdd[datafusion]

Both normalize to **Arrow boolean mask** / ArrowRuleResult. SQL runs **server-side only** against registered telemetry; unsafe SQL is rejected. No browser-side SQL. No pandas row-loop runtime on edge.

140.6 6. Fault confirmation window

- **Raw fault** — rule mask true on one row.
- **Confirmed fault** — min_true_rows consecutive trues and/or min_elapsed_minutes elapsed.
- Example: **10 rows** at **1-minute** polling $\approx$ **10 minutes** confirmed.

See `FaultConfirmation`({ { “/rule-cookbook/fault-confirmation/” | relative_url } }).

140.7 7. Validation modes

Mode	Purpose
Bench 5007	Dual-source BACnet vs Niagara backend equivalence — [long FDD smoke]({ { “/operations/bench-5007-long-fdd-smoke/”

Mode	Purpose
ACME live	Real BACnet HVAC, VAV/AHU rules, RCx — ACME live validation ({{ “/operations/acme-live-validation/”
Synthetic CI	python scripts/ smoke_bench_5007_long_fdd.py -- synthetic --dry-run
Model fixture	python scripts/ validate_acme_model_context.py

140.8 8. Security rules

- **No secrets** in docs, logs, exports, MCP bundle, or validation reports.
- Bridge binds **127.0.0.1** by default; LAN exposure requires explicit operator choice.
- **BACnet writes** disabled/guarded — see [BACnet write safety] ({{ “/security/bacnet-writes/” | relative_url }}).
- Niagara / JSON API service accounts: dedicated, read-only where possible.
- Full tracebacks only when `OFDD_DEBUG_TRACEBACKS=1`.

140.9 9. Where to look

Topic	Doc
Start	Quick Start ({{ “/quick-start/”
Drivers	Driver framework ({{ “/drivers/”
Rules	[Rule Cookbook — PyArrow & DataFusion] ({{ “/rule-cookbook/dual-backend-rules/”
API	REST API reference ({{ “/appendix/bridge_api/”
Operations	Operations ({{ “/ops/”
Architecture	[ADR: Rust-ready Arrow contract]({{ “/adr/adr-rust-ready-arrow-fdd-contract/”
MCP	MCP server ({{ “/ai/mcp-server/”
Contributor policy	<code>AGENTS.md</code> (repo root)

141 LAN hardening

142 LAN hardening

Open-FDD targets **trusted building LAN / OT edge** deployment. The Operator Bridge is not intended for direct public internet exposure.

Full mode matrix: `Deployment modes({{ "/architecture/deployment-modes/" | relative_url }})`.

142.1 Auth and bind

Mode	Bridge bind	Auth
Local dev	127.0.0.1	Optional OFDD_AUTH_DISABLED=1 on loopback
Lab LAN	0.0.0.0	Required unless explicit insecure lab flags
Edge production	Loopback + Caddy on LAN	Required — OFDD_AUTH_SECRET + role passwords

Startup **fails** on non-loopback bind without credentials (unless insecure lab flags are set).

142.2 Network exposure

- Put **Caddy** on :80 / :443; keep bridge :8765 on loopback when possible.
- Do not port-forward the Operator Bridge to the public internet without TLS and strong auth.
- Keep BACnet on OT VLANs; management UI on IT VLAN with firewall rules.
- BACnet **writes** are gated by default — [BACnet write guard] (`{{ "/security/bacnet-writes/" | relative_url }}`).

142.3 Rule Lab

Rule Lab runs **operator-authored Python** in a sandbox. Restrict integrator accounts; review rules before enable on production sites.

142.4 TLS and LAN security scans

Topic	Page
Certificates (self-signed / site CA)	TLS and certificates ({{ "/" security/tls-and-certs/"
ZAP + Nmap bench smoke	[ZAP baseline] ({{ "/"security/ zap-baseline/"
Ubuntu host + Tenable/Nessus	Linux host hardening ({{ "/" security/linux-host-hardening/"
Release test cycle (maintainers)	[Security testing] ({{ "/" developer/security-testing/"

142.5 Ollama on the edge

Ollama has **no built-in auth**. Bind OLLAMA_HOST=127.0.0.1:11434 on the host; never expose :11434 on the OT LAN. Open-FDD bridge reaches Ollama server-side only. Validation: scripts/security/check_openfdd_exposure.sh.

143 ZAP baseline (local bench)

144 ZAP baseline (local bench)

Passive [OWASP ZAP baseline](#) against the **pentest production stack** on benserver is the standard LAN security smoke before Acme edge deploys.

Full per-revision workflow (pytest → PR → GHCR → Acme):
[Developer — security testing cycle]({{ "/"developer/security-testing/" | relative_url }}).

Packaged scan scripts (Windows + Mac/Linux): [scripts/security/README.md](#).

144.1 Run

On the Open-FDD host:

```
./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify
```

From a LAN workstation (after verify passes):

```
# Windows – optional integrator auth probe (copy auth.env.local to
scan PC; never commit)
.\scripts\security\Run-OpenFddSecurityScan.ps1 -TargetUrl "http://
192.168.204.18" `
-AuthEnvFile "C:\path\to\auth.env.local"
```

The script writes 40-release-gate-summary.txt with **SECURITY GATE: PASS/FAIL**: - Fails if production JS bundles contain private LAN IPs or bench Niagara defaults - Warns if auth env is missing (Auth loaded: False) — pass -AuthEnvFile or shell vars OFDD_INTEGRATOR_USER / OFDD_INTEGRATOR_PASSWORD (values are never logged)

```
# macOS / Linux
./scripts/security/run_openfdd_security_scan.sh --url http://
192.168.204.18
```

ZAP target (example): <http://192.168.204.18/> — Caddy on :80 only; bridge :8765 stays loopback. TLS bench: see [TLS and certificates](#)({{ "/security/tls-and-certs/" | relative_url }}).

144.2 Expected results (HTTP bench mode)

Finding	Severity	Status
CSP style-src 'unsafe-inline'	Medium	Accepted — Vite/React inline styles; TODO to nonce/hash
Missing COEP	Low	Accepted — require-corp not enabled (breaks some assets)
Port 80 open (Caddy)	Info	Expected — intentional LAN entry

Finding	Severity	Status
Ports 8765, 5173, 8000, 8080 closed	—	Good
Port 443 filtered	—	Expected in HTTP mode; use <code>OFDD_CADDY_MODE=tls</code> for HTTPS bench
Duplicate Referrer-Policy / X-Frame-Options	Low	Fixed — bridge owns headers; Caddy does not duplicate

144.3 Header ownership

Layer	Sets
Bridge (security_headers.py)	CSP, X-Frame-Options: DENY, frame-ancestors 'none', COOP, CORP, Permissions-Policy, Referrer-Policy, X-Content-Type-Options
Caddy HTTP	Reverse proxy only; strips Server banners
Caddy TLS	Adds Strict-Transport-Security only

144.4 Authenticated scan (later)

Baseline scan hits ~4 public endpoints (/health, public agent insight routes). Deep API/dashboard coverage requires ZAP with integrator login — see [Authenticated scanning \(roadmap\)](#)(`{ { "/security/authenticated-scanning/" | relative_url } }`).

144.5 After fixes

Re-run `pentest_production_stack.sh verify`, then the packaged LAN scan from `scripts/security/`. Confirm response headers show **one** X-Frame-Options: DENY and no duplicate Referrer-Policy.

145 TLS and certificates

146 TLS and certificates

Who generates, stores, and renews TLS material for Open-FDD on OT LANs — and how that ties into ZAP/Nmap bench testing.

Developer release cycle: `Security testing cycle({ { "/developer/security-testing/" | relative_url } })`.

146.1 Ownership

Environment	Who owns certs	Storage
Local / benserver bench	Developer / maintainer	workspace/ deploy/caddy/ certs/ (cert.pem, key.pem)
Ansible edge (Acme, customer sites)	Site integrator	/etc/openfdd/ caddy/ on the VM (playbook- managed or manual)
Public internet	Not in scope	Open-FDD targets OT LAN; use site PKI or self- signed

Open-FDD does **not** ship Let's Encrypt or public CA automation. OT benches use **self-signed** or customer-provided certs.

146.2 Generate bench certs (local)

```
./scripts/setup_caddy_certs.sh  
# or: ./scripts/setup_caddy_certs.sh --cn openfdd.local --out  
workspace/deploy/caddy/certs
```

Then start with TLS mode:

```
OFDD_CADDY_MODE=tls ./scripts/run_local.sh  
# or pentest stack with OFDD_CADDY_MODE=tls
```

Caddy serves :443 and redirects :80 → HTTPS. Caddy adds **Strict-Transport-Security** only; all other security headers remain on the **bridge** ([ZAP baseline]({ { "/security/zap-baseline/" | relative_url } }#header-ownership)).

146.3 ZAP / Nmap with TLS bench

When `OFDD_CADDY_MODE=tls`:

- Nmap: expect **443 open**, 80 may redirect
- ZAP: use `https://<lan-ip>/` (add `-k trust` or import `cert.pem` on the scan workstation)
- PowerShell: `.\Run-OpenFddSecurityScan.ps1 -TargetUrl "https://192.168.204.18"`
- Bash: `./run_openfdd_security_scan.sh --url https://192.168.204.18`

Self-signed certs produce ZAP TLS warnings — expected on OT LAN.

146.4 Renewal

Self-signed certs from `setup_caddy_certs.sh` default to **365 days**. Re-run the script before expiry or bake renewal into site runbooks.

Edge Ansible deploys should document whether the integrator rotates certs manually or via site PKI.

146.5 Enterprise CA on Caddy (edge)

When IT requires **clean Nessus SSL plugins** (no self-signed / untrusted chain), install a site-provided certificate on the edge VM instead of `setup_caddy_certs.sh` output.

Typical layout (Ansible / manual):

```
/etc/openfdd/caddy/cert.pem # full chain (leaf + intermediates)
/etc/openfdd/caddy/key.pem  # private key (0600, root or caddy user)
```

Ansible TLS mode (`caddy_mode: tls`) already points Caddy at those paths — see `infra/ansible/templates/Caddyfile.j2`.

Internal enterprise CA workflow:

1. Generate CSR on the edge host or request cert from site PKI for the operator hostname (e.g. `openfdd.acme.local`).
2. Install `cert.pem` + `key.pem` under `/etc/openfdd/caddy/`.

3. Distribute the **CA root** to admin laptops (trust store) so browsers and ZAP scans validate the chain.
4. Monitor expiry:

```
openssl x509 -in /etc/openfdd/caddy/cert.pem -noout -dates
```

Caddyfile snippet (TLS, enterprise cert):

```
:443 {  
    tls /etc/openfdd/caddy/cert.pem /etc/openfdd/caddy/key.pem  
    header {  
        Strict-Transport-Security "max-age=31536000;  
includeSubDomains"  
    }  
    reverse_proxy 127.0.0.1:8765  
}
```

Lab/self-signed mode still produces expected **Medium** Tenable findings — document as accepted risk or switch to enterprise CA. See [Tenable remediation]({{ "/security/tenable-remediation/" | relative_url }}).

146.6 Caddy internal CA mode (lab)

`./scripts/setup_caddy_certs.sh` generates a **self-signed** pair for bench TLS. Trust `cert.pem` on scan workstations (`-k curl`, ZAP import) or accept browser warnings. Not suitable when IT mandates trusted chains.

146.7 Secrets

- **Never commit** `key.pem` or production cert bundles
- `workspace/deploy/caddy/certs/` is for local bench only unless your `.gitignore` already excludes generated keys (verify before push)

147 Authenticated scanning (roadmap)

148 Authenticated scanning (roadmap)

[ZAP baseline]({{ "/security/zap-baseline/" | relative_url }}) is **unauthenticated** — it checks public pages, response headers, and a handful of routes reachable without login. That is the right default for every docker image / patch revision smoke on a test LAN.

For **deep** coverage of integrator dashboards, /api/* routes, WebSocket sessions, and role-gated writes, plan a separate **authenticated** scan — only on fake/bench stacks with explicit approval.

Release context: `Security testing cycle({{ "/developer/security-testing/" | relative_url }})`.

148.1 Current state (3.0.x)

Capability	Status
ZAP baseline via scripts/security/	Shipped
Pentest creds in workspace/ auth.pentest.local	Shipped (host-local, not in git)
ZAP full / active scan automation	Not shipped — manual only
ZAP authenticated context (login script + session)	Roadmap
CI running ZAP against PR previews	Not planned (LAN bench only)

148.2 Why baseline is not enough

After login, the bridge exposes model editing, BACnet/JSON API configuration, Rule Lab, commissioning export, and write-gated commands. Baseline ZAP never exercises:

- Session cookies / JWT persistence
- CSRF or auth bypass on mutating endpoints

- IDOR on site-scoped resources
- Rate limits and audit logging on failed auth

148.3 Planned approach (contributors welcome)

1. **Bench-only target** — same pentest stack; never point active scans at live OT controllers
2. **Login hook** — ZAP zap-full-scan.py or Automation Framework with:
 - POST /api/auth/login using integrator creds from env (not committed)
 - Regex or JSON extractor for session token / cookie
3. **Scope file** — include /api/model/*, /api/commissioning/*, dashboard static assets; exclude BACnet UDP and unrelated LAN hosts
4. **Report parity** — extend scripts/security/ with opt-in - Authenticated / --zap-full flags and separate report prefix (22-zap-authenticated-*)
5. **Gate** — document accepted findings; fail release only on High/Critical regressions

148.4 Manual authenticated ZAP today

1. Start pentest stack; note creds in workspace/auth.pentest.local
2. Open ZAP desktop or docker with API port exposed
3. Configure **Context** → **Authentication** → script or form-based login to /api/auth/login
4. Run **Spider** then **Active Scan** on http://<lan-ip>/ only
5. Compare with baseline report in openfdd-security-report/

Do **not** run active scans against production Acme without a maintenance window and BACnet write lock.

148.5 Related

- [scripts/security/README.md](#) — baseline setup
- [Secrets and auth](#)({{ "/security/secrets-auth/" | relative_url }}) — env file layout
- [LAN hardening](#)({{ "/security/lan-hardening/" | relative_url }}) — bind and auth modes

149 Secrets and auth

150 Secrets and auth

150.1 Files (gitignored)

File	Contents
workspace/ auth.env.local	JWT secret, role passwords
workspace/ json_api.env.local	REST API keys for JSON API <code>\${ENV:VAR}</code> URLs (e.g. OpenWeather OPENWEATHER_API_KEY)
workspace/ data.env.local	Historian + audit log retention (OFDD_LOG_*, OFDD_FEATHER_*)
workspace/ ollama.env.local	Ollama URL, model, RAM tier, GPU mode
workspace/ mcp.env.local	MCP RAG sidecar
workspace/ caddy.env.local	Reverse proxy TLS/HTTP mode
workspace/ pentest.env.local	LAN security scan stack
infra/ansible/secrets/ <host>.env.local	SSH deploy secrets (maintainers)

Copy from *.example templates only. Full inventory: [Workspace env files](#)(`{{ "/appendix/workspace-env-files/" | relative_url }}`).

150.2 Required variables (production)

Variable	Purpose
OFDD_AUTH_SECRET	HMAC signing (32+ chars)
OFDD_INTEGRATOR_USER / PASSWORD	Rule Lab, bindings
OFDD_OPERATOR_USER / PASSWORD	Operations

150.3 GHCR pulls

Use read-only registry credentials on edge if images are private; rotate tokens periodically.

150.4 Backup

Back up workspace/data/ and auth env in secure operator vault — not in git.

Audit JSONL under workspace/logs/ may contain client IPs and usernames — treat like security logs. See `Logging and audit({{ "/ops/logging/" | relative_url }})`.

151 BACnet write allowlists

152 BACnet write allowlists

Supervisory writes require:

1. **Feature flag** enabled in environment (off by default).
2. **Allowlist** file at workspace/bacnet/write_allowlist.json (copy from `write_allowlist.example.json`). If the flag is on but the allowlist is missing, writes are **denied**.
3. **Integrator** role on API calls.
4. **Audit** log review after commissioning sessions.

Allowlist entries support `device_instance`, `object_identifier`, `property_identifier`, optional `priority_min/priority_max`, `value_min/value_max`, and `allowed_values` for binary/multistate targets.

Lab-only override: `OFDD_BACNET_WRITE_ALLOW_ANY=1` (audited; not for production edges).

See operator guide: `[BACnet write safety]({{ "/bacnet/write-safety/" | relative_url }})`.

Never enable blanket write access for agent or integrator automation without human approval per site.

153 Linux host hardening

154 Linux host hardening

Generic **Ubuntu edge VM** guidance for Open-FDD deployments scanned by IT tools (Tenable/Nessus, Qualys, etc.). This complements application-level [LAN hardening](#)({{ "/security/lan-hardening/" | relative_url }}) and [\[TLS\]](#)({{ "/security/tls-and-certs/" | relative_url }}).

Nessus-specific finding tables: [\[Tenable remediation\]](#)({{ "/security/tenable-remediation/" | relative_url }}).

154.1 Scope

Layer	Owner	Repo support
Ubuntu kernel, OpenSSH, apt packages	Site IT / integrator	scripts/security/remediate_ubuntu_host.sh
Docker engine, compose, Caddy	Integrator	Ansible + check_openfdd_exposure.sh
Open-FDD containers (pip/pyOpenSSL)	Open-FDD maintainers	docker/python-security-constraints.txt, GHCR images
Self-signed TLS on OT LAN	Expected lab finding	Enterprise CA path in [TLS] ({{ "/security/tls-and-certs/"

154.2 Quick operator workflow

On the edge VM (SSH):

```
cd ~/open-fdd
./scripts/security/check_host_security.sh
./scripts/security/check_openfdd_exposure.sh --edge-ip "$
(hostname -I | awk '{print $1}')
```

Remediate host OS (maintenance window):

```
./scripts/security/remediate_ubuntu_host.sh # dry-run
sudo ./scripts/security/remediate_ubuntu_host.sh --apply
# if /var/run/reboot-required exists:
sudo reboot
./scripts/security/check_host_security.sh
```

Pull new Open-FDD images after a release that pins container pip/pyOpenSSL:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit
<inventory_host>
```

154.3 Ubuntu version and EOL

Supported targets: **Ubuntu 22.04 LTS** or **24.04 LTS** (64-bit).
Rebuild hosts still on **16.04** or **18.04** — apt upgrade cannot clear an SEoL finding.

Validate a surprising 16.04 Nessus hit:

```
lsb_release -a
cat /etc/os-release
# Scan credential / stale DNS / container banner mismatch?
```

154.4 Kernel and package updates

```
sudo apt update
sudo apt full-upgrade -y
uname -r
test -f /var/run/reboot-required && echo "reboot required"
sudo reboot # maintenance window
```

Optional status tools:

```
ubuntu-security-status # or: pro status
apt list --upgradable | grep -i security
```

154.5 Host Python (pip, wheel, pyOpenSSL)

Tenable often reports **host** pip, wheel, and pyOpenSSL versions installed by Ubuntu packages or admin pip install. Prefer **apt** on the host:

```
python3 -m pip show pip setuptools wheel pyOpenSSL
sudo apt install --only-upgrade python3-pip python3-openssl
python3-wheel
```

Open-FDD **containers** pin toolchain versions at image build — see `docker/python-security-constraints.txt`. Rescan after `docker compose pull`.

154.6 Ollama — no LAN exposure

Ollama has **no built-in authentication** in typical deployments. Treat LAN exposure as **Critical**.

Rule	Implementation
Bind loopback only	OLLAMA_HOST=127.0.0.1:11434 in systemd (ollama.service) or dev bootstrap
Bridge calls server-side	OFDD_OLLAMA_BASE_URL=http://127.0.0.1:11434 or host.docker.internal:11434 from containers
Never publish :11434 on 0.0.0.0	Compose uses 127.0.0.1:11434:11434 when Docker Ollama is enabled

Validation:

```
ss -tulpn | grep 11434
curl -sf http://127.0.0.1:11434/api/tags | head
curl -sf --max-time 3 http://<edge-lan-ip>:11434/api/tags
# must FAIL from another host
```

Firewall (if management host needs GPU Ollama — rare):

```
# Example: block LAN → 11434; allow loopback only (adjust
interface names)
sudo ufw deny 11434/tcp
sudo ufw allow from 127.0.0.0/8 to any port 11434 proto tcp
```

154.7 OpenSSH Terrapin (CVE-2023-48795)

Apply Ubuntu security updates first (openssh-server, openssh-client). Verify:

```
ssh -V
sudo sshd -T | grep -Ei '^(ciphers|macs|kexalgorithms)'
```

Manual algorithm hardening is **optional** and can lock out clients — document rollback (`sshd_config.bak`) before changes. Do not automate cipher stripping in repo scripts.

154.8 ICMP timestamp (Low)

Nessus may report ICMP timestamp disclosure. Optional hardening at host firewall or upstream network — low priority for isolated OT VLANs.

Example (nftables — site-specific):

```
# Block ICMP timestamp / type 13 – verify with IT before applying
sudo nft add rule inet filter input icmp type timestamp-request
drop
```

154.9 TLS and certificate scans

Self-signed or internal CA certificates produce **Medium** Nessus SSL findings (untrusted, self-signed, expiry). That is **expected** for lab/OT encryption-in-transit. For clean SSL plugin results, install a **site enterprise CA** certificate on Caddy — [TLS and certificates](#)({ { “/security/tls-and-certs/” | relative_url } } #enterprise-ca-production).

154.10 Maintainer security audits (CI parity)

```
./scripts/security/run_maintainer_audits.sh
```

Matches GitHub Actions operator-bridge-security job where possible.

155 Tenable / Nessus remediation

156 Tenable / Nessus remediation

Remediation plan for **IT/Tenable.io Nessus** findings on an Open-FDD **OT/LAN edge** deployment (e.g. Acme GL36 lab VM). Open-FDD runs behind a building firewall on Ubuntu with Docker Compose, Caddy, and optional host Ollama.

Related: [Linux host hardening](#)({{ "/security/linux-host-hardening/" | relative_url }}), [TLS and certificates](#)({{ "/security/tls-and-certs/" | relative_url }}), [LAN hardening](#)({{ "/security/lan-hardening/" | relative_url }}), [Security testing cycle](#)({{ "/developer/security-testing/" | relative_url }}).

156.1 Prioritized action plan

Priority	Action	Owner
P0	Confirm Ubuntu is not 16.04/18.04; rebuild if EOL	IT / integrator
P0	Bind Ollama to 127.0.0.1 only; verify LAN cannot reach :11434	Integrator
P0	apt full-upgrade + reboot for kernel USNs (8254, 8278, 8373)	IT / integrator
P1	Upgrade host pip/wheel/pyOpenSSL via apt or documented pip path	IT / integrator
P1	Pull GHCR images built with docker/python-security-constraints.txt	Integrator
P2	OpenSSH packages current; review Terrapin after patch	IT
P2	TLS: accept self-signed findings or install enterprise CA cert	IT decision
P3	ICMP timestamp: optional firewall rule	Network / IT

156.2 Finding classification table

Nessus finding	Severity	Class	Remediation
	Critical		lsb_release -a — rebuild VM on 24.04 LTS if truly

Nessus finding	Severity	Class	Remediation
Canonical Ubuntu Linux SEoL (16.04.x)		Host OS / false positive?	16.04; if host is 22.04/24.04, check scan target IP, credentials, stale asset record
Ollama Unauthenticated Access	Critical	Deployment config	OLLAMA_HOST=127.0.0.1:11434; no 0.0.0.0:11434; firewall deny; see Ollama
Ubuntu kernel USN-8254-1	Critical	Host OS	sudo apt update && sudo apt full-upgrade -y && sudo reboot
pyOpenSSL 22.0.x < 26.0.0 Buffer Overflow	Critical	Host pip and container	Host: apt upgrade python3-openssl or pip ≥26; Container: GHCR image with constraints file
Ubuntu pip USN-8344-1	High	Host OS + container build	apt full-upgrade; image build upgrades pip≥26
SSL Certificate Cannot Be Trusted	Medium	Accepted risk (lab) / deployment	Self-signed Caddy — use enterprise CA for clean scan
SSL Self-Signed Certificate	Medium	Accepted risk (lab)	Same as above
SSL Certificate Expiry	Medium	Deployment / monitoring	openssl x509 -dates; renew setup_caddy_certs.sh or site PKI
SSH Terrapin (CVE-2023-48795)	Medium	Host OS	Upgrade openssh-server; validate ssh -V
Ubuntu wheel USN-8221-1	Medium	Host OS	apt full-upgrade
Ubuntu kernel USN-8278-1, USN-8373-1	Medium	Host OS	apt full-upgrade + reboot
pyOpenSSL 0.14.x Security Bypass	Medium	Host + container	Same as pyOpenSSL Critical — upgrade to ≥26
ICMP Timestamp Request	Low	Network / optional host	Firewall drop; often accepted on isolated OT VLAN

Class legend: (1) repo-owned fix · (2) Docker base/image · (3) host OS · (4) deployment/config · (5) accepted risk · (6) validate false positive

156.3 Operator commands (edge VM)

156.3.1 Validate (read-only)

```
cd ~/open-fdd
./scripts/security/check_host_security.sh
./scripts/security/check_openfdd_exposure.sh --edge-ip "$
(hostname -I | awk '{print $1}')"
lsb_release -a
uname -r
test -f /var/run/reboot-required && cat /var/run/reboot-
required.pkgs
python3 -m pip show pip setuptools wheel pyOpenSSL 2>/dev/null |
grep -E '^(Name|Version):'
ss -tulpn | grep -E ':80|:443|:8765|:11434'
curl -sf http://127.0.0.1:11434/api/tags | head -3
```

From a **different LAN workstation** (must fail):

```
curl -sf --max-time 3 http://<edge-ip>:11434/api/tags
```

156.3.2 Remediate host (maintenance window)

```
./scripts/security/remediate_ubuntu_host.sh # dry-run
./scripts/security/remediate_ubuntu_host.sh --apply
sudo reboot # if reboot-required
./scripts/security/check_host_security.sh
```

156.3.3 Remediate Open-FDD containers

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit
<inventory_host>
docker compose pull && docker compose up -d
```

156.3.4 Rescan

After reboot and image pull, request IT **Nessus rescan** of the edge IP. Compare plugin output to the table above.

156.4 Ollama unauthenticated access (Critical)

Not an Open-FDD code bug — Ollama's API is unauthenticated by design. Risk is **network exposure**.

Repo defaults (3.0.5+):

- Systemd: Environment=OLLAMA_HOST=127.0.0.1:11434 in infra/ansible/templates/ollama.service.j2
- Compose Ollama (optional profile): 127.0.0.1:11434:11434 only
- Acme-style edge: host Ollama + bridge via host.docker.internal:11434 (container → host loopback path)
- post_deploy_check.sh warns if :11434 listens on 0.0.0.0

Re-apply Ansible Ollama unit after upgrade:

```
ansible-playbook -i inventory.yml ollama_bootstrap.yml --limit  
<host> -e enable_ollama=true
```

156.5 pyOpenSSL and pip in containers

Docker build installs docker/python-security-constraints.txt before application requirements:

- pip>=26.0, setuptools>=75.0, wheel>=0.45
- pyOpenSSL>=26.0.0, cryptography>=44.0.0

CI pip-audit runs against bridge + bacnet requirements with the same constraints.

156.6 TLS — expected vs production

Mode	Nessus SSL plugins	When to use
Caddy HTTP :80	No TLS plugins	Dev / HTTP-only VLAN
Caddy self- signed : 443	Medium: untrusted, self- signed, expiry	OT LAN encryption, lab
Enterprise CA cert on Caddy	Clean SSL plugins (if chain trusted)	IT mandates clean Nessus

See [TLS — enterprise CA]({{ "/security/tls-and-certs/" |
relative_url }})#enterprise-ca-on-caddy-edge).

156.7 Residual findings after remediation

Even with a patched host and loopback-only Ollama, IT may still see:

- **SSL** Medium on self-signed TLS (accepted for lab; document in risk register)
- **ICMP** Low (optional network block)
- **Container** plugins if Nessus scans Docker overlay networks — clarify scan scope with IT

156.8 Notes for IT / security team

1. Scan the **edge VM IP**, not individual container IPs, unless container CVE correlation is required.
2. Open-FDD **bridge :8765** should be **loopback-only**; LAN entry is **Caddy :80/:443**.
3. **BACnet UDP 47808** on commission is expected OT traffic — out of scope for web/plugin remediation.
4. Do not expose Ollama **11434** to the building LAN; agent chat is server-side only.
5. Kernel CVEs require **host reboot** — schedule with operations.
6. For authenticated web/API depth, see roadmap [Authenticated scanning]({{ "/security/authenticated-scanning/" | relative_url }}).

156.9 Repo-owned fixes (this document's release)

Item	Location
Container pip/pyOpenSSL pins	docker/python-security-constraints.txt, docker/Dockerfile
Ollama loopback systemd	infra/ansible/templates/ollama.service.j2
Host/exposure check scripts	scripts/security/check_*.sh, remediate_ubuntu_host.sh
Post-deploy Ollama bind warn	infra/ansible/scripts/post_deploy_check.sh
CI pip-audit + constraints	.github/workflows/ci.yml

157 AI agents

158 AI agents and MCP

- [AI agent context](#)({{ "/ai-agent-context/" | relative_url }}) — canonical map for deploy, drivers, model, rules, validation (**start here**)
- [MCP server](#)({{ "/ai/mcp-server/" | relative_url }}) — FastMCP tools over the operator bridge
- Skills under [skills/openfdd-*-agent/](#) and [skills/openfdd-mcp-server/](#)
- Operator memory: [workspace/MEMORY.md](#) (see [workspace/MEMORY.md.example](#))

Internal runbooks (not primary nav): [\[agent-skills/\]\({{ "/agent-skills/bench-validation-agent/" | relative_url }}\)](#).

159 Security

160 Security

LAN deployment security for the Operator Bridge and BACnet write guard. Open-FDD is for **trusted edge / OT LAN** — not direct public internet exposure.

Page	Topic
LAN hardening ({{ "/security/lan-hardening/" relative_url }})	relative_url }}
ZAP baseline (local bench) ({{ "/security/zap-baseline/" relative_url }})	relative_url }}
TLS and certificates ({{ "/security/tls-and-certs/" relative_url }})	relative_url }}
Linux host hardening ({{ "/security/linux-host-hardening/" relative_url }})	relative_url }}
Tenable / Nessus remediation ({{ "/security/tenable-remediation/" relative_url }})	relative_url }}
Authenticated scanning (roadmap) ({{ "/security/authenticated-scanning/" relative_url }})	relative_url }}

Page	Topic
Secrets and auth ({{ "/" security/secrets-auth/"	relative_url }}
BACnet write allowlists ({{ "/" security/bacnet-writes/"	relative_url }}
[Release & LAN scan cycle] ({{ "/" developer/security-testing/"	relative_url }}
Logging and audit ({{ "/" ops/logging/"	relative_url }}

161 GL36 lab site (Acme)

162 GL36 lab site (Acme)

Lab note only. This documents a maintainer **example edge host** (site acme, building vm-bbartling) used for Guideline 36 supervisory FDD development. Replace inventory host names, IPs, and BACnet ranges for your site.

Reference for **ASHRAE Guideline 36** supervisory monitoring on a live lab edge. Rule code lives in `workspace/data/rules_py/`; commissioning uses **Arrow constants** (no `config.json`).

Trim & respond Niagara reference: [README_TRIM_RESPOND](#).

162.1 BACnet poll scope

Device filter (`infra/ansible/scripts/acme_trim_devices.sh`):

Range	Equipment
1-100	JCI VMA/VAV
1100	RTU-01 (AHU)
1002	Hot water plant
11000-13000	Trane UC210 VAV

Poll interval: **60 s**. Commission from control machine:

```
cd infra/ansible
POLL_INTERVAL=60 ./scripts/acme_commission_gl36.sh
```

Points land in edge_backup/local/acme/vm-bbartling/points.csv for Ansible sync.

162.2 Example FDD rules (Arrow)

162.2.1 Duct static pressure high (AHU)

```
"""AHU duct static high – GL36 trim advisory."""

import pyarrow.compute as pc

VALUE_COLUMN = "duct-static"
HIGH_INWC = 1.20
LOOKBACK_HOURS = 3

def _kit_lookback_stats(table, *, hours=None):
    h = hours if hours is not None else LOOKBACK_HOURS
    ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
    tmin, tmax = pc.min(ts).as_py(), pc.max(ts).as_py()
    span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
    print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")

def apply_faults_arrow(table, cfg, context=None):
    _kit_lookback_stats(table)
    return pc.greater(pc.cast(table[VALUE_COLUMN], "float64"),
HIGH_INWC)
```

162.2.2 SAT too cold vs setpoint

```
"""AHU SAT too cold vs setpoint."""

import pyarrow.compute as pc

SAT_COLUMN = "sa-t"
SP_COLUMN = "sa-t-sp"
COLD_DELTA_F = 5.0

def apply_faults_arrow(table, cfg, context=None):
    sat = pc.cast(table[SAT_COLUMN], "float64")
    sp = pc.cast(table[SP_COLUMN], "float64")
    return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

162.2.3 Site rule stubs in the repo

Rule file	Use
bench_stat_zn-t_flatline_1h.py	Zone temp flatline — bind ZN-T per VAV
acme_duct_static_flatline_1h_gl36_duct_t_r_input.py	RTU duct static flatline
oat_vs_web_spread_1h.py	Local OAT vs OpenWeather JSON API

162.2.4 Local OA-T vs web weather (JSON API)

1. Control machine: workspace/json_api.env.local with OPENWEATHER_API_KEY
2. JSON API tab → **OpenWeatherMap** preset → **Register** (default poll **30 min**)
3. Bind BACnet local OAT in the model (oa-t historian column)
4. Enable oat_vs_web_spread_1h.py in Rule Lab — flags when |oa-t - web-oat-t| > 8 °F

162.3 Deploy / upgrade (Ansible, this lab only)

```
cd infra/ansible
OPENFDD_IMAGE_TAG=latest ./deploy.sh docker --limit
<inventory_host> -e enable_bacnet_poll_driver=true
```

Full UI + image upgrade from control machine:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full
```

Do **not** rsync dev auth.env.local to the edge (deploy_sync_auth: false in private host_vars). Regenerate credentials on the edge host if login fails after upgrade.

162.4 AI-assisted data modeling (lab workflow)

1. BACnet poll running (~340 GL36 points @ 60 s on this lab host).

- 2. **Model & assignments** → export commissioning JSON for LLM-assisted binding.
- 3. Import returned JSON via commissioning import.
- 4. Pin FDD rules on ZN-T, DA-T, duct-static, OAT, etc. — export shows readable names in fdd_rules_linked.

Model seed: workspace/data/acme_gl36_model.json.

163 Examples & lab notes

164 Examples & lab notes

Site-specific commissioning notes, GL36 examples, and maintainer lab workflows. These pages are **not** required for a generic edge deployment.

Page	Description
GL36 lab site (Acme)({{ "/examples/acme-gl36-lab/" relative_url }})	

For production IT operators, start with [Quick Start\({{ "/quick-start/" | relative_url }}\)](#) and [Operations\({{ "/ops/" | relative_url }}\)](#).

165 Paired FDD smoke (bench + Acme)

166 Paired FDD smoke

Hardcoded cross-site validation: **bensserver bench 5007** and **Acme** run the **same schedule** with mid-test fault-parameter toggles. PyArrow and DataFusion SQL rules must agree on flagged counts **and** row-level fault masks (CI golden fixtures).

166.1 What is tested

Site	Rule	Normal phase	Blatant phase
Bench (demo)	stat_zn-t bounds on BACnet 5007 + Niagara bench9065	65-75 °F	99-100 °F
Acme	Local oa-t vs web-oat-t spread	≤ 10 °F	≤ 0.001 °F

Rule IDs are fixed in open_fdd/validation/paired_fdd_contract.py. Each site has **arrow** and **datafusion_sql** variants.

166.2 Fault confirmation (5 minutes)

All paired smoke rules use:

```
        {"min_elapsed_minutes": 5, "min_true_rows": 5, "poll_interval_s": 60}
```

A condition must stay true for ~**5 minutes** before it counts as a confirmed fault. See [Fault confirmation](#).

166.3 Modes (hardcoded)

Mode	Duration	Toggle interval
tryout	6 min	3 min
short	30 min	15 min
standard	2 h	every 15 min
overnight	12 h	every 15 min

Toggles alternate **normal** ↔ **blatant** fault parameters via the rules API (same as an operator or AI agent adjusting thresholds mid-run).

166.4 Run (local bench)

Prereqs: local stack on :8765.

```
OFDD_SKIP_UI_BUILD=1 ./scripts/run_local.sh start
./scripts/smoke_paired_fdd_harness.sh --tryout
```

166.5 Run detached (required for in-depth runs)

Do not run long smokes attached from Cursor — they crash the IDE when agents poll/wait. Use the systemd-isolated launcher (no terminal parent):

```
# benserver bench only (recommended for local UI iteration)
./scripts/run_paired_fdd_smoke_isolated.sh --short --bench-only

# bench + Acme FDD (skips UI bundle parity by default)
OPENFDD_LIVE_ACME=1 ./scripts/run_paired_fdd_smoke_isolated.sh --
short

# Poll status – read-only, safe for agents (never tail -f / wait)
./scripts/smoke_paired_fdd_status.sh --mode short
```

Legacy wrapper (also uses isolated launcher when --detached):

```
OPENFDD_LIVE_ACME=1 ./scripts/smoke_paired_fdd_harness.sh --short
--detached
OPENFDD_LIVE_ACME=1 ./scripts/smoke_paired_fdd_harness.sh --
standard --detached
OPENFDD_LIVE_ACME=1 ./scripts/smoke_paired_fdd_harness.sh --
overnight --detached
```

OPENFDD_LIVE_ACME=1 is required for live Acme paired runs (site parity + remote loop).

Logs: /tmp/paired_fdd_smoke_<mode>_*.log · Status JSON: /tmp/paired_fdd_smoke_<mode>.status.json · Cycle log: /tmp/paired_fdd_smoke_<mode>_cycles.jsonl

Monitor outside Cursor (tmux/SSH). Heartbeat every 5 minutes; status JSON updates each poll cycle.

```
cat /tmp/paired_fdd_smoke_overnight.status.json
grep heartbeat /tmp/paired_fdd_smoke_overnight_*.log | tail -3
```

Bundle for download:

```
./scripts/bundle_paired_fdd_smoke_report.sh --mode overnight
# Windows: scp ben@bensserver:/tmp/paired_fdd_smoke_bundle.zip C:
\Users\ben\Downloads\
```

166.6 Auth refresh (overnight harness)

The harness uses scripts/smoke_paired_fdd_auth.py:

1. Decodes JWT exp (no signature verification) and re-logins before expiry (~2 min skew).

2. On HTTP **401**, retries the request **once** after re-login.
3. Successful retry is an **auth_refresh_event** (not an FDD failure).
4. Unrecoverable 401 stops the site loop and sets `auth_failure` in status JSON.

Status/report fields: `auth_refresh_count`, `auth_401_count`, `auth_first_401_at`, `auth_recovered_count`, `auth_unrecovered_count`, `first_unrecoverable_auth_failure_at`.

Security-negative checks preserved: expired, tampered, missing, or wrong-role tokens still fail. The harness does not disable auth or log tokens/secrets.

166.7 Arrow vs DataFusion SQL parity

Live harness: compares paired batch runs (flagged/rows counts + analytics when present) via `open_fdd/validation/paired_fdd_parity.py`.

CI (no live site): `open_fdd/tests/validation/test_datafusion_arrow_parity.py` runs golden tables through PyArrow + DataFusion and asserts identical fault masks (`true_count`, `false_count`, `null_count`, row-level match).

Live Acme tests remain behind `OPENFDD_LIVE_ACME=1`.

166.8 RCx report smoke

Fixture-driven DOCX validation (no live historian):

```
python3 scripts/smoke_rcx_report.py
python3 -m pytest open_fdd/tests/reports/
test_rcx_docx_fixtures.py -q
```

Live edge RCx: `POST /api/reports/rcx/generate` on a running stack (integrator auth required).

166.9 Niagara massive-site / folder mapping (#315)

UI: **Niagara** tab → **Station tree browse** — pick a folder as building boundary, set `default_points_root`, preview before discover.

Helpers: `open_fdd/validation/niagara_folder_mapping.py` · tests: `open_fdd/tests/validation/test_niagara_folder_mapping.py`

Dry-run poll: use **Poll once** after discovery; check Activity log and driver tree counts.

166.10 Reports

- reports/paired_fdd_smoke_validation.md / .json
- reports/site_parity_smoke.json

166.11 Pass criteria

- Site parity smoke passes (matching UI hash, health OK).
- No PyArrow vs SQL parity mismatch per cycle (counts + batch analytics).
- No unrecoverable auth failures during the run.
- Auth refresh events are allowed and not counted as FDD failures.

scripts/smoke_sites_parity.sh delegates to this harness.

167 REST API reference

168 REST API reference

HTTP and WebSocket surface for **openfdd-bridge** (default http://127.0.0.1:8765). Production: Caddy on :80 proxies to the bridge.

Dashboard SPA paths: `Dashboard routes({{ "/appendix/dashboard-routes/" | relative_url }})`.

OpenAPI: GET /docs and GET /redoc when the bridge runs.

Auth: JWT via POST /api/auth/login. Most routes require Authorization: Bearer <token>. Roles: viewer, operator, integrator, commission, agent, admin. BACnet writes require commission role + `write safety({{ "/bacnet/write-safety/" | relative_url }})`. Dev: OFDD_AUTH_DISABLED on loopback only.

WebSocket: WS /ws/dashboard — POST /api/auth/ws-ticket, then pass the ticket via Sec-WebSocket-Protocol: ofdd.ws, <ticket>. Query ?ticket= is dev-only when OFDD_WS_ALLOW_QUERY_TICKET=1. Tickets are short-lived and not interchangeable with the Bearer token.

168.1 Health & audit

Method	Path	Role	Description
GET	/health	public	Minimal liveness (ok, service, version, auth_required)
GET	/health/stack	auth	Stack traffic-light (verbose URLs/bind with OFDD_DEBUG_DIAGNOSTICS)
POST	/api/auth/ws-ticket	auth	Short-lived WebSocket ticket
GET	/api/audit/summary	auth	Audit counters
GET	/api/audit/events	auth	Audit log (query params)
GET	/api/audit/errors	auth	Recent API errors

*Exact public set depends on OFDD_AUTH_DISABLED and middleware.

168.2 Auth

Method	Path	Description
POST	/api/auth/login	Username/password → JWT
GET	/api/auth/me	Current user
GET	/api/auth/status	Auth mode / lockout hints

168.3 Rules & FDD

Method	Path	Role	Description
GET	/api/rules/saved	user	Rule index
POST	/api/rules/save	operator+	Create/update rule metadata
GET	/api/rules/saved/{id}/source	user	Python source

Method	Path	Role	Description
PUT	/api/rules/saved/{id}/source	operator+	Write .py file
GET	/api/rules/assignments	user	Point bindings
POST	/api/rules/bind	integrator+	Bind rule to point
POST	/api/rules/bindings	integrator+	Bulk bindings
POST	/api/rules/batch	operator+	Run all enabled rules → fdd_results.json
GET/POST	/api/rules/drafts	user	Draft rules

168.4 Rule Lab playground

Method	Path	Description
POST	/api/playground/lint	AST lint Python rule
POST	/api/playground/test-rule	Run apply_faults_arrow() on feather PyArrow window
POST	/api/playground/run-script	Execute script mode
GET	/api/playground/sample-frame	Sample DataFrame for editor

168.5 BRICK / data model

Method	Path	Role	Description
GET	/api/model/sites	user	Site list
GET	/api/model/tree	user	Equipment tree (SPARQL)
GET	/api/model/graph	user	RDF graph fragment
GET		user	Scoped query

Method	Path	Role	Description
	/api/model/scope		
GET	/api/model/export	user	Export TTL/JSON
GET	/api/model/commissioning-export	user	Full commissioning bundle (points + fdd_rule_ids + fdd_rules)
POST	/api/model/commissioning-import	integrator+	Bulk import; validate in UI dry-run first
GET	/api/model/health	user	Model point/equipment counts
GET	/api/model/ttl	user	TTL metadata
POST	/api/model/sync-ttl	integrator+	Regenerate TTL from JSON
POST	/api/model/import	integrator+	Import model payload
POST	/api/model/sites	integrator+	Create site
GET/POST	/api/model/bacnet-sync	integrator+	BACnet ↔ model sync state
GET	/api/model/fdd-query-presets	user	Saved FDD query presets
GET	/api/model/fdd-coverage	user	Rule coverage summary
POST	/api/model/sparql	user	SPARQL query (scoped)

168.6 Rule Lab (dashboard)

Method	Path	Role	Description
POST	/api/rules/lab/validate	operator+	Validate rule source
POST	/api/rules/lab/preview	operator+	Preview rule on historian window
POST	/api/rules/lab/compare	operator+	Compare two rule versions

Method	Path	Role	Description
POST	/api/rules/lab/lint-sql	operator+	DataFusion SQL safety lint
POST	/api/rules/lab/save	operator+	Save Rule Lab draft

Mounted at /api/rules/lab — see OpenAPI /docs.

168.7 Bench validation (read-only)

Method	Path	Role	Description
GET	/api/bench/health	operator+	Bench harness health
GET	/api/bench/mapping	operator+	Semantic cross-source mapping
GET	/api/bench/poll-status	operator+	BACnet + Niagara poll heartbeat
POST	/api/bench/validate/bacnet-vs-niagara	operator+	Dual-source equivalence check
POST	/api/bench/long-fdd/evaluate	operator+	Long FDD smoke evaluate (PyArrow/DataFusion)

See [Bench 5007 long FDD smoke](#)({{ "/operations/bench-5007-long-fdd-smoke/" | relative_url }}).

168.8 Niagara baskStream

Method	Path	Role	Description
GET	/api/niagara/health	read	Connector health
GET		read	Configured stations

Method	Path	Role	Description
POST	/api/niagara/stations/{id}/discover	read	Discover points
POST	/api/niagara/stations/{id}/poll/once	poll	One poll cycle
GET	/api/niagara/stations/{id}/poll/status	read	Poll worker status
GET	/api/niagara/driver/tree	read	Driver commissioning tree

Read-only by default. See [Niagara baskStream connector](#)({ { “/niagara-baskstream-connector/” | relative_url } }).

168.9 BACnet

Method	Path	Role	Description
GET	/config/bacnet	read	Commission URL / config
GET	/api/bacnet/commission/status	read	Commission agent health
GET	/api/bacnet/server/points	read	Server point table
GET	/api/bacnet/inventory	read	Discovered inventory
POST		commission	Who-Is

Method	Path	Role	Description
	/api/ bacnet/ whois		
POST	/api/ bacnet/ discover	commission	Discover devices
POST	/api/ bacnet/read	commission	Read property
POST	/api/ bacnet/ read- multiple	commission	RPM read
POST	/api/ bacnet/ write	write	Supervisory write (gated)
POST	/api/ bacnet/ import-to- model	integrator	Inventory → BRICK
POST	/api/ bacnet/ driver/ sync- discovery	commission	Sync driver tree
POST	/api/ bacnet/ poll/once	commission	Trigger poll cycle
GET	/api/ bacnet/ poll/status	read	Poll worker status
POST	/ingest/ bacnet	read	Ingest samples into feather

Bind address: BACNET_BIND in commission env — see [BACnet network setup]({{ "/bacnet/network-setup/" | relative_url }}).
 Operator guide: Discover and read({{ "/bacnet/discover-read/" | relative_url }}).

PATCH | /api/bacnet/driver/point | commission | Enable poll + interval on point |
 PATCH | /api/bacnet/driver/device | commission | Enable poll for all points on device |
 PATCH | /api/bacnet/driver/device/remap | commission | Change instance and/or address |
 DELETE | /api/bacnet/driver/point/{point_id} | commission | Remove point from registry |
 DELETE | /api/bacnet/driver/device/{device_instance} |

commission | Remove device |
DELETE | /api/bacnet/driver/registry | commission | Clear CSVs +
model sync |

168.10 Faults (check-engine)

Method	Path	Description
GET	/api/faults/catalog	Fixed fault codes
GET	/api/faults/status	GREEN/YELLOW/RED aggregate
GET	/api/faults/tree	Fault tree by equipment
GET	/api/faults/code/ {code}	Code metadata

168.11 Timeseries & sites

Method	Path	Description
GET	/api/timeseries/sites	Feather sites
GET	/api/timeseries/series	Series metadata
GET	/api/timeseries/plot	Plot payload
GET	/api/timeseries/readings	Raw readings
GET	/api/sites	Site list
GET	/api/sites/{site_id}/frame	Wide frame for site

168.12 Building & host

Method	Path	Description
GET	/api/building/status	Building summary
GET	/api/building/alerts	Active alerts
GET	/api/host/stats	CPU/mem/disk

168.13 Local agent (Ollama)

Prefix **/openfdd-agent**. Role **agent** for tool execution.

Method	Path	Description
GET	/openfdd-agent/context	Composed prompt context
GET	/openfdd-agent/tools	Tool manifest
POST	/openfdd-agent/tool	Execute tool (e.g. rules.save)
GET	/openfdd-agent/ollama/health	Ollama reachability
GET	/openfdd-agent/building-insight	Check-engine narrative context
GET	/openfdd-agent/zone-temps	Zone temperature insight
POST	/openfdd-agent/chat	Chat completion (catalog-bound)

Optional host Ollama for building insight — not required for core FDD. Configure via compose profile or host service; see [Architecture overview]({{ "/architecture/overview/" | relative_url }}).

168.14 Modbus

Method	Path	Role	Description
GET	/api/modbus/driver/tree	operator+	Commissioning tree
GET	/api/modbus/poll/status	operator+	Poll worker status
POST	/api/modbus/poll/once	integrator+	Force poll cycle
POST	/api/modbus/read_registers	operator+	One-shot register read
POST	/api/modbus/read_and_store	operator+	Read + CSV + feather (source=modbus)
POST	/api/modbus/refresh	operator+	Re-read one register
PATCH		integrator+	Enable poll interval

Method	Path	Role	Description
	/api/modbus/ register/poll		

See [Driver framework](#)({{ "/drivers/" | relative_url }}).

168.15 JSON API (HTTP/HTTPS)

Method	Path	Role	Description
GET	/api/json-api/ env/status	operator+	Env var configured flags (no secret values)
POST	/api/json-api/ presets/ openweather	integrator+	Register 3-point OpenWeather bundle + optional poll
GET	/api/json-api/ driver/tree	operator+	Endpoints grouped by host
GET	/api/json-api/ poll/status	operator+	Poll worker status
POST	/api/json-api/ poll/once	integrator+	Force poll cycle
POST	/api/json-api/ request	operator+	One-shot GET/POST (no store)
POST	/api/json-api/ read_and_store	operator+	Request + CSV + feather (source=json_api)
POST	/api/json-api/ refresh	operator+	Re-fetch endpoint
PATCH	/api/json-api/ endpoint/poll	integrator+	Enable poll interval
DELETE	/api/json-api/ endpoint/ {point_id}	integrator+	Remove endpoint

Outbound auth: auth_type = none | bearer | basic; optional verify_tls (default true). See [JSON API driver](#)({{ "/drivers/json-api/" | relative_url }}).

168.16 PyPI package (separate)

Library-only HTTP is **not** bundled. See `Python package({{ "/appendix/python-package/" | relative_url }})` for `open_fdd.arrow_runtime` (Arrow rules offline).

168.17 Errors

JSON error bodies; stack traces hidden unless `OFDD_DEBUG_TRACEBACKS=1`. See `LAN hardening({{ "/security/lan-hardening/" | relative_url }})`.

169 Dashboard routes

170 Dashboard routes

Single-page app served by **openfdd-bridge** at / (production: Caddy on :80). All paths below are relative to the site root unless noted.

Auth: Routes under `RequireAuth` need a valid JWT (login at /login). Role gates are enforced on API calls from each tab.

170.1 Public

Path	Page	Notes
/login	Login	JWT session; redirects to / when authenticated

170.2 Main workspace (authenticated)

Path	Tab / page	Purpose
/	Home	Overview, fault analytics, equipment, model health (hash sections)
/rule-lab	Rule Lab	

Path	Tab / page	Purpose
		Author, lint, preview, compare PyArrow vs SQL rules
/model	Data model	BRICK model, point bindings, FDD assignments
/plot	Plot	Timeseries plot from feather historian
/bacnet	BACnet	Commission UI, inventory, driver tree
/modbus	Modbus	Modbus driver (when enabled)
/niagara	Niagara	baskStream connector stations & poll
/json-api	JSON API	External JSON ingest driver
/agent	Building agent	Ollama / agent check-in
/host	Host stats	Container and host metrics

170.3 Analytics & RCx

Path	Destination	Notes
/analytics	/#overview	Redirect to home overview
/analytics/faults	/#fault-analytics	Fault analytics hash
/analytics/equipment	/#equipment	Equipment hash
/analytics/health	/#model-health	Model health hash
/analytics/rcx	RCx report builder	Generate, list, preview, delete DOCX reports

170.4 Legacy redirects

Path	Redirects to
/algorithms	/
/data-model	/model
/fdd-assignments	/model
/fdd	/model
/faults	/#fault-analytics

170.5 Home hash sections

The home page uses URL fragments for deep links:

Hash	Section
#overview	Dashboard overview
#fault-analytics	Fault tree and analytics
#equipment	Equipment rollup
#model-health	Model / data health

170.6 Static assets

Path	Content
/assets/*	Built dashboard JS/CSS (Vite)
/docs, /redoc	OpenAPI (bridge process, not SPA)

API calls from the dashboard use paths documented in [REST API reference](#)({ { "/appendix/bridge_api/" | relative_url } }).

171 Python package

172 Python package (open-fdd)

Install from PyPI (embeddable FDD runtime — not the Docker edge stack):

```
pip install open-fdd
pip install "open-fdd[datafusion]" # optional DataFusion SQL
```

rule backend

```
pip install "open-fdd[analytics]" # optional NumPy helpers
pip install "open-fdd[ml]"       # optional sklearn for
```

offline experiments

172.1 Modules

Module	Purpose
open_fdd.arrow_runtime	Default — apply_faults_arrow on PyArrow Tables, cookbook masks, column maps
open_fdd.playground	Rule Lab lint/compile helpers for Arrow rules

Retired in 3.0.1+: open_fdd.engine (YAML/pandas RuleRunner) is **not** shipped on PyPI. Use Operator Bridge Rule Lab or import rules from workspace/data/rules_py/.

172.2 When to use package-only

Scenario	Use
Lint/test Arrow rules offline	open_fdd.arrow_runtime + open_fdd.playground
Graph ML experiments (Layer B)	open-fdd[ml] in experiments/ (see issue #211)
Full building operator stack	Docker Operator Bridge

172.3 Versioning

Publish: git tag vX.Y.Z (or legacy open-fdd-vX.Y.Z) → GitHub Actions **Publish open-fdd to PyPI**. See [Release process](#)({{ "/developer/release-process/" | relative_url }}).

```
import open_fdd
print(open_fdd.__version__)
```

172.4 Offline Arrow example

```
import pyarrow as pa
import pyarrow.compute as pc
from open_fdd.arrow_runtime import run_arrow_rule
```

```
code = '''
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["SAT"], float(cfg["high"]))
'''

table = pa.table({"SAT": [70.0, 90.0, 88.0]})
result = run_arrow_rule(code, table, {"high": 85})
print(result.flagged_count)
```

API surface: `open_fdd/arrow_runtime` docstrings and [Arrow recipes] (`{{ "/rule-cookbook/arrow-recipes/" | relative_url }}`).

173 Configuration reference

174 Configuration reference

174.1 Bridge / auth

Variable	Default	Notes
OFDD_BRIDGE_HOST	127.0.0.1	0.0.0.0 for LAN
OFDD_AUTH_SECRET	—	Required for LAN
OFDD_AUTH_DISABLED	0	Localhost dev only
OFDD_INSECURE_LAN_DEV	0	Lab only with auth disabled

174.2 BACnet

Variable	Notes
BACNET_BIND	ip/mask:47808 in <code>commission.env</code>
OFDD_BACNET_WRITE_ENABLED	Default off — site-specific name may vary

174.3 Docker edge

Variable	Notes
OPENFDD_IMAGE_TAG	GHCR tag for Ansible deploy
openfdd_docker_pull_from_ghcr	Ansible <code>host_var</code>

Rule Lab Python rules use `rules_store.json + workspace/data/rules_py/*.py`. Legacy YAML schema (`docs/config_schema.json`) is archived — not used by Operator Bridge 3.0.1+.

175 Glossary

176 Glossary

Term	Definition
Operator Bridge	FastAPI service serving API + static dashboard
Rule Lab	UI and APIs for Arrow <code>apply_faults_arrow()</code> rules
Feather store	On-disk columnar historian under <code>workspace/data/feather_store/</code>
fdd_input	Brick tag linking a point to a rule binding role
Check-engine	Building-level GREEN/YELLOW/RED fault summary
Commission	BACnet discovery/read (and optional write) agent container
Poll driver	Scheduled BACnet RPM reads into historian
GHCR	GitHub Container Registry — <code>ghcr.io/bbartling/openfdd-*</code>

177 Workspace env files

178 Workspace env files

All live under `workspace/`. Copy from `*.example` templates; committed `*.local` files are gitignored.

File	Loaded by	Purpose
<code>auth.env.local</code>	<code>run_local.sh</code> , Docker compose, Ansible	JWT secret (OFDD_AUTH_SECRET), integrator/operator/agent passwords,

File	Loaded by	Purpose
		OFDD_AUTH_DISABLED for localhost dev only
data.env.local	run_local.sh, bridge, feather maintain	Feather historian caps (OFDD_FEATHER_*), audit/error log rotation (OFDD_LOG_*), BACnet discovery mutation toggle
json_api.env.local	bridge, Ansible edge sync	API keys for JSON API driver \${ENV:VAR} substitution (e.g. OPENWEATHER_API_KEY)
ollama.env.local	run_local.sh, bridge	Ollama base URL, model name, RAM tier, GPU mode, timeouts — see Ollama and analytics ({{ "/operator-bridge/ollama-analytics/"
mcp.env.local	run_local.sh, MCP sidecar	Enable MCP RAG REST (OFDD_MCP_*), index path
caddy.env.local	run_local.sh, Ansible edge	Reverse proxy mode (http / tls / off), TLS CN, port overrides
pentest.env.local	pentest_production_stack.sh	Production-like LAN scan stack: CORS, auth, passive BACnet defaults

178.1 Ansible / edge extras

Path	Purpose
infra/ansible/secrets/<host>.env.local	SSH deploy credentials, GHCR pull tokens, site-specific BACnet bind — maintainers only

178.2 Related docs

- [Secrets and auth](#)({{ "/security/secrets-auth/" | relative_url }}) — required production variables
- [Logging and audit](#)({{ "/ops/logging/" | relative_url }}) — OFDD_LOG_* detail
- [JSON API driver](#)({{ "/drivers/json-api/" | relative_url }}) — \${ENV:VAR} pattern

- [Configuration reference](#)({{ "/appendix/configuration/" | relative_url }}) — bridge runtime flags not in env files

179 Appendix

180 Appendix

Reference material — **REST API** and **dashboard routes** for integrators. Narrative guides live in the main sections above.

Page	Content
REST API reference ({{ "/appendix/bridge_api/" relative_url }})	relative_url }}
Dashboard routes ({{ "/appendix/dashboard-routes/" relative_url }})	relative_url }}
Configuration reference ({{ "/appendix/configuration/" relative_url }})	relative_url }}
Workspace env files ({{ "/appendix/workspace-env-files/" relative_url }})	relative_url }}
Python package ({{ "/appendix/python-package/" relative_url }})	relative_url }}
Glossary ({{ "/appendix/glossary/" relative_url }})	relative_url }}

181 Arrow rule contract (v1)

182 Arrow rule contract (v1)

Stable v1 batch FDD, Rule Lab preview, and validation smokes share one execution contract.

182.1 Entrypoint

```
def apply_faults_arrow(table: pa.Table, cfg: dict | None = None,
context: dict | None = None):
    ...
```

Parameter	Type	Notes
table	pyarrow.Table	Historian telemetry for one site/window
cfg	dict	Rule thresholds, column names, confirmation settings
context	dict None	Optional bridge metadata (site_id, equipment hints)

182.2 Return type (authoring)

apply_faults_arrow must return a **boolean** pyarrow.Array or pyarrow.ChunkedArray:

- Length **must equal** table.num_rows
- True = raw fault sample (before confirmation)
- Non-boolean returns are cast to bool by the backend
- Null slots in the mask are counted; prefer explicit False for missing inputs

The backend normalizes via open_fdd.arrow_runtime.backend.normalize_fault_mask.

182.3 Runtime result (ArrowRuleResult)

After execution, run_arrow_rule() produces ArrowRuleResult:

Field	Meaning
rule_id	Saved rule id
backend	"arrow" or "datafusion_sql"
row_count	Input table rows
true_count / false_count / null_count	Mask statistics
duration_ms	Wall time
fault_mask	

Field	Meaning
	Confirmed boolean mask (see below)
warnings / errors	Lint/runtime messages
summary	Execution evidence, site id, backend path
preview	Sample fault rows for Rule Lab UI

Use `.as_dict()` for API JSON. Both PyArrow and DataFusion SQL backends normalize to this shape.

182.4 Raw mask vs confirmation

1. Rule returns **raw** boolean mask (`apply_faults_arrow` or SQL fault column).
2. Backend applies **fault confirmation** when `cfg` includes `min_true_rows` and/or `min_elapsed_minutes` (see [Fault confirmation](#) (`{{ "/rule-cookbook/fault-confirmation/" | relative_url }}`)).
3. `ArrowRuleResult.fault_mask` is the **post-confirmation** mask used for batch counts and dashboard alerts.

Confirmation is backend-owned — rules should not implement debounce logic unless documenting a special case.

182.5 DataFusion SQL rules

SQL modules still expose `apply_faults_arrow`; the body delegates to `open_fdd.arrow_runtime.datafusion_backend`. Metadata:

```
backend: datafusion_sql
fault_column: fault
sql: |
    SELECT *, "stat_zn-t" > 76.0 AS fault FROM telemetry
```

Same `ArrowRuleResult` after SQL execution and confirmation.

182.6 Lint & forbidden imports

Edge fault rules allow: `pyarrow`, `pyarrow.compute`, `open_fdd.arrow_runtime.*`, `datetime`, `math`.

Forbidden in production fault rules: `pandas`, `numpy` (for rule logic), file I/O, network calls.

See `lint_arrow_rule()` in `open_fdd.arrow_runtime.backend`.

182.7 Chunked execution

Large tables may run via `run_arrow_rule_chunked()` — masks are concatenated; contract unchanged.

182.8 Related

- [Data types & units]({{ "/rule-authoring/data-types-and-units/" | relative_url }})
- [ADR: Rust-ready Arrow FDD contract](#)({{ "/adr/adr-rust-ready-arrow-fdd-contract/" | relative_url }})
- [Developer: Arrow-native runtime]({{ "/developer/arrow-native-runtime/" | relative_url }})

183 Arrow-native runtime

184 Arrow-native runtime

Open-FDD 3.0 executes Rule Lab rules on **PyArrow tables** end to end.

```
feather_store.read_site_table()
→ fdd_runner / playground.run_arrow_table()
→ apply_faults_arrow(table, cfg, context)
→ pyarrow.compute.masks
```

184.1 Authoring contract

```
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["zone_temp"], cfg["max_zone_temp"])
```

Cookbook helpers: `open_fdd.arrow_runtime.cookbook` (flatline, spread, OOB, after-hours fan).

v1 contract: [ArrowRuleResult]({{ "/rule-authoring/arrow-rule-contract/" | relative_url }}) — input `pa.Table`, output boolean mask, confirmation applied by backend.

Script-mode analytics rules receive table (PyArrow) and `cfg` in the sandbox — not pandas DataFrames.

184.2 PyArrow-only policy (edge + Rule Lab)

Allowed	Forbidden
pyarrow (pa), pyarrow.compute (pc)	import pandas, pd.DataFrame, to_pandas()
open_fdd.arrow_runtime.cookbook helpers	import numpy for rule logic
Script mode globals: table, cfg, out	Legacy df DataFrame scripts

Rule Lab **lint** (POST /api/playground/lint) and rule **save** reject pandas patterns with agent-facing errors, e.g. *use table (PyArrow), not df*.

Fault rules: `apply_faults_arrow(table, cfg, context=None)`. Analytics scripts: top-level code on `table/cfg`, set `out = {"events": [...], "metrics": {...}}`.

184.3 Package layout

Module	Role
<code>open_fdd.arrow_runtime.backend</code>	Execute rule code, batch chunks
<code>open_fdd.arrow_runtime.cookbook</code>	Shared fault masks
<code>open_fdd.arrow_runtime.windows</code>	Rolling min/max, consecutive-true
<code>open_fdd.playground.arrow_templates</code>	Rule Lab starter templates

185 Data types and units

186 Data types and units

Rules read **columnar historian tables** produced by BACnet / Niagara / JSON API ingest. Author against **Feather column names** (`oa-t`, `stat_zn-t`, ...) or BRICK `fdd_input` aliases resolved in the bridge.

186.1 Required table shape

Column / convention	Expectation
timestamp	UTC or site-local consistent ordering; monotonic within a batch window
Point columns	One float (or castable) column per bound historian point
site_id	Optional in table; often passed via cfg / context
Row count	One row per poll sample — irregular spacing is normal

Missing columns: cookbook helpers typically return an all-False mask or zero-filled stand-ins — check helper docs before assuming a fault.

186.2 Numeric casting and nulls

Older off-repo gist	Arrow in Open-FDD
HelperUtils.convert_to_float	pc.cast(col, pa.float64()) or ingest-time typing
HelperUtils.check_datatype_float	Rule Lab lint + bridge ingest validation
HelperUtils.check_datatype_int	Validate integer cfg fields (int(cfg["window_samples"]))
np.maximum / np.minimum	pc.max / pc.min
series.rolling(n).min()	arrow_rolling_min in open_fdd.arrow_runtime.windows
series.diff()	arrow_diff / arrow_abs_diff
df.resample("H")	Not ported 1:1 — use sample-window rolling helpers instead

Null behavior: comparisons involving null produce null; backend cast may treat null as false in the final mask. Prefer `pc.fill_null` or explicit guards when null should mean “no fault.”

186.3 Command normalization

BACnet analog outputs often arrive as **0-100 %**. Open-FDD normalizes with `norm_cmd_array()`:

- Values > 1.0 $\rightarrow$ divide by 100 (0-1 fraction)
- Values ≤ 1.0 $\rightarrow$ used as-is (already fractional)

Legacy gist: `check_range_less_than_one / float_max_check_err` (command max ≤ 1.0 after conversion).

Configure per rule:

```
cfg = {"min_active_command": 0.02, "min_command_delta": 0.03}
```

186.4 Boolean / status columns

Signal	Typical handling
Fan status / run command	Normalize as command; gate with <code>pc.greater(fan, 0.01)</code>
Occupancy	Boolean column or hour-of-week mask via cookbook schedule helpers
Enable / alarm state	Cast to bool; combine with <code>pc.and_</code>
Binary points	<code>pc.equal(col, 1)</code> or cast from 0/1 floats

186.5 Sensor profiles and units

Default bounds, flatline tolerance, and rate limits live in `open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES`.

Profile key	Quantity	Default unit	Example fault codes
zone_temp	Temperature	°F	VAV-C
supply_air_temp	Temperature	°F	AHU-C, RTU-C
outdoor_air_temp	Temperature	°F	BLD-B
mixed_air_temp	Temperature	°F	AHU-D
return_air_temp	Temperature	°F	AHU-D
duct_static_pressure	Pressure	in H ₂ O	AHU-A
zone_humidity	Relative humidity	% RH	BLD-B
co2	CO ₂	ppm	BLD-B

Override via rule cfg (bounds_low, bounds_high, flatline_tolerance, window_samples, max_per_hour). Metric sites: set cfg["temp_unit"] = "metric" or scale constants (see [Expression cookbook]({{ "/rule-cookbook/expression-cookbook/" | relative_url }}#metric-equivalents)).

186.6 Time and windows

Setting	Meaning
window_samples	Rolling window in poll samples (not wall-clock unless derived)
rolling_avg_minutes	Converted to samples using poll_interval_s / median poll interval
hunting_window_hours	Used by PID hunting helpers to derive sample count
min_true_rows	Confirmation: consecutive true samples required
min_elapsed_minutes	Confirmation: minimum wall time with fault true

Irregular timestamps: rolling helpers operate on row order within the loaded table — ensure lookback windows include enough samples for your poll cadence.

186.7 Raw fault vs confirmed fault

Stage	Where
Raw mask	Return value of apply_faults_arrow
Confirmed mask	ArrowRuleResult.fault_mask after apply_fault_confirmation_from_cfg

Document both when tuning — a rule may fire briefly but not meet confirmation.

186.8 Related

- [GL36 & sensor patterns — sensor validation]({{ "/rule-cookbook/expression-cookbook/" | relative_url }}#sensor-validation-bounds-flatline-rate-of-change)
- [Fault confirmation](#)({{ "/rule-cookbook/fault-confirmation/" | relative_url }})

- [PyArrow & DataFusion SQL]({{ "/rule-cookbook/dual-backend-rules/" | relative_url }})

187 Rust readiness checklist

188 Rust readiness checklist

Open-FDD v1 keeps a **table-in, boolean-mask-out** contract so a future Rust or DataFusion-native executor can replace Python wrappers without changing Rule Lab metadata or driver ingest.

Use this checklist when authoring or reviewing rules for stable v1 and Rust migration prep.

188.1 Rule style

☐

No pandas in edge fault rules — PyArrow columns only

☐

No NumPy for core detection logic (optional in central analytics extras only)

☐

No Python row loops over samples for hot-path fault detection — use `pyarrow.compute` and `open_fdd.arrow_runtime.windows`

☐

Arrow arrays in, boolean mask out — length equals input row count

☐

Explicit config schema — document `cfg` keys in rule module docstring

☐

Deterministic null behavior — document how missing columns / nulls are treated

☐

No hidden global state — pure functions of `(table, cfg, context)`

☐

Stable fault-code metadata — letter codes from `[fault catalog]({{ "/fault-codes/" | relative_url }})`

☐

Small fixture tests — fixed `pa.table(...)` inputs with expected mask slices



DataFusion SQL rules stay restricted — single SELECT from t
elemetry, boolean fault column, no DDL/DML

188.2 Backend choice

Prefer PyArrow	Prefer DataFusion SQL
Rolling windows, hunting, flatline	Simple threshold / CASE
Schedule / occupancy gating	Operator-readable SQL preview
Sensor profile helpers	Parity proof before Rust port
ML feature prep (future)	Stateless row-wise logic

See [PyArrow vs DataFusion decision table]({{ "/rule-cookbook/expression-cookbook/" | relative_url }}#pyarrow-vs-datafusion-sql).

188.3 Do not use DataFusion SQL for

- Complex rolling windows or resample-style logic
- PID / control hunting with stateful window counts (use PyArrow helpers)
- Custom helper-heavy HVAC logic
- ML feature preparation pipelines
- Arbitrary file joins or external tables
- Browser-side execution
- Pandas-style row iteration

188.4 Confirmation and evidence



Raw mask produced by rule; confirmation via min_true_rows / min_elapsed_minutes in cfg



Batch results include execution evidence (computation_path, backend id) for smokes

188.5 Packaging



Edge Docker image includes PyArrow runtime



DataFusion optional: `pip install 'open-fdd[datafusion]'` —
matches `pyproject.toml [project.optional-dependencies]`
`datafusion`

188.6 Related ADRs

- [ADR: Rust-ready Arrow FDD contract](#)(`{{ "/adr/adr-rust-ready-arrow-fdd-contract/" | relative_url }}`)
- [\[ADR: DataFusion Rust-ready rule backend\]](#)(`{{ "/adr/adr-datafusion-rust-ready-rule-backend/" | relative_url }}`)

189 Release process

190 Release process

Open-FDD has **two publish paths**:

Artifact	Trigger	Registry
PyPI <code>open-fdd</code>	Git tag <code>vX.Y.Z</code> (or legacy <code>open-fdd-vX.Y.Z</code>)	pypi.org/project/open-fdd
Docker <code>edge stack</code>	Same release tag (or manual workflow)	ghcr.io/bbartling/openfdd-*

Do **not** publish from ordinary branch pushes. Merge to master, bump version, tag, push tag.

190.1 1. Prepare version

```
git checkout master
git pull --ff-only
# Edit pyproject.toml + open_fdd/__init__.py → same X.Y.Z
python scripts/release/check_version.py
```

190.2 2. Merge and tag

```
git tag -a v3.0.30 -m "open-fdd 3.0.30"
git push origin v3.0.30
```

Tag push runs:

- `.github/workflows/publish-open-fdd.yml` — build, test, twine check, PyPI via **Trusted Publishing**
- `.github/workflows/docker-publish.yml` — GHCR images with tags 3.0.30, 3.0, latest

Legacy tag `open-fdd-v3.0.30` is still accepted during transition.

190.3 3. PyPI Trusted Publishing setup (one-time)

On pypi.org → project **open-fdd** → Publishing → **Add a new trusted publisher**:

Field	Value
PyPI project name	open-fdd
Owner	bbartling
Repository	open-fdd
Workflow name	Publish open-fdd to PyPI
Environment	pypi (optional; enables approval gate)

No long-lived API token in the repo.

190.4 4. Verify release

```
pip install open-fdd==3.0.30
python -c "import open_fdd; print(open_fdd.__version__)"

docker pull ghcr.io/bbartling/openfdd-bridge:3.0.30
docker manifest inspect ghcr.io/bbartling/openfdd-bridge:3.0.30
```

190.5 5. Manual workflow_dispatch

- **PyPI:** Publish open-fdd to PyPI with `dry_run=true` (default) — build/test only
- **Docker:** Publish Docker images to GHCR — publishes `pyproject.toml` version + minor alias + latest (optional `image_tag` override)

190.6 Package vs Docker

See `PyPI package({{ "/developer/pypi-package/" | relative_url }})`
and `[Docker images]({{ "/ops/docker-images/" | relative_url }})`.

191 PyPI package

192 PyPI package (open-fdd)

192.1 What it is

Embeddable **Arrow-native FDD compute** for consultants, cloud pipelines, and test benches:

- Lint and test `apply_faults_arrow(table, cfg, context)` rules
- Run rules on PyArrow Tables, Feather, or Parquet
- Optional NumPy analytics helpers (`pip install "open-fdd[analytics]"`)

192.2 What it is not

The PyPI wheel does **not** include:

- Operator Bridge / FastAPI app
- React dashboard
- BACnet commission/poll service
- MCP server
- Ansible / Tailscale deploy tooling

Use **GHCR Docker images** for the full edge stack.

192.3 Install

```
pip install open-fdd
pip install "open-fdd[analytics]" # optional NumPy helpers
pip install "open-fdd[ml]"        # offline sklearn experiments
```

192.4 CLI

```
open-fdd version
open-fdd lint-rule path/to/rule.py
```

```

    open-fdd test-rule path/to/rule.py --input sample.feather --
config cfg.json
    open-fdd run-arrow path/to/rule.py --input in.feather --output
out.feather
    open-fdd validate-rule-pack ./rules

```

192.5 Rule contract

```

def apply_faults_arrow(table, cfg, context=None):
    # return PyArrow bool mask, length == table.num_rows
    ...

```

Use `open_fdd.arrow_runtime.run_arrow_rule` for structured ArrowRuleResult (counts, errors, preview).

192.6 Modules shipped on PyPI

Module	Purpose
<code>open_fdd.arrow_runtime</code>	Rule execution, cookbook masks, column maps
<code>open_fdd.playground</code>	Lint/compile helpers
<code>open_fdd.faults</code>	Fault catalog YAML
<code>open_fdd.schema</code>	Result schemas

`open_fdd.engine` (pandas/YAML) is **not** in the wheel.

192.7 Examples

Repo examples/ — [arrow_minimal](#), Feather batch, generic IoT pipeline.

192.8 Local validation

```

pip install -e ".[test,dev,analytics]"
python scripts/release/check_version.py
pytest open_fdd/tests -q
python -m build && twine check dist/*

```

193 MCP server

194 Open-FDD MCP server

FastMCP agent interface over the **operator bridge** — not a parallel FDD stack.

194.1 Modes

Mode	OFDD_MCP_MODE	Site resolution
Edge	edge	OPENFDD_BRIDGE_BASE_URL (default http:// 127.0.0.1:8765)
Portfolio	portfolio	portfolio/sites.json + optional site_id per tool

RCx Central (:8050 Dash, :8060 API) is a **separate HTTP stack** — not the MCP server. Agents doing RCx overview, FDD preset tables, or DOCX reports should call Central API (/api/central/*) or use the Dash; see [RCx Central agent guide](#).

Acme production edge runs **without** MCP (enable_mcp: false). Run MCP centrally and call http://100.122.106.124 via Tailscale, or use RCx Central with Edge Connections registry.

194.2 Transport

Transport	Env	Use
Streamable HTTP	OFDD_MCP_TRANSPORT=streamable-http (default in Docker)	http:// 127.0.0.1:8090/ mcp
Stdio	OFDD_MCP_TRANSPORT=stdio	Cursor / Claude Desktop

Legacy bridge doc search: POST /tools/search_docs on :8090 (unchanged).

194.3 Human approval

These tools require `human_approved=true`:

- `save_rule`
- `apply_fdd_tuning`
- `run_fdd_batch`

MCP does not command BACnet writes — only Open-FDD bridge APIs.

194.4 Client examples

Cursor (`~/ .cursor/mcp.json`):

```
{
  "mcpServers": {
    "open-fdd": {
      "command": "python",
      "args": ["-m", "mcp_server.run"],
      "env": {
        "PYTHONPATH": "/path/to/open-fdd/workspace:/path/to/open-fdd/workspace/api",
        "OPENFDD_REPO_ROOT": "/path/to/open-fdd",
        "OFDD_MCP_TRANSPORT": "stdio",
        "OFDD_MCP_MODE": "portfolio",
        "OFDD_PORTFOLIO_SITES_PATH": "/path/to/open-fdd/portfolio/sites.json"
      }
    }
  }
}
```

HTTP (bensserver dev stack):

```
docker compose -f docker/compose.dev.yml up -d mcp-rag
curl -fsS http://127.0.0.1:8090/health
```

194.5 Site registry example

```
{
  "sites": [
    {
      "site_id": "acme",
      "name": "Acme GL36 Lab",
      "base_url": "http://100.122.106.124",
      "username": "integrator"
    }
  ]
}
```

```
]
}
```

Store passwords in env or OS keychain — not in git.

194.6 RCx Central (not MCP)

Service	Default URL	Agent docs
Central API	http://127.0.0.1:8060	central-api.md
RCx Dash	http://127.0.0.1:8050	rcx-central-dash-agent.md

MCP portfolio_rollup hits Edge /api/building/portfolio-rollup per site. RCx GET /api/central/overview/{site_id} adds Dash KPIs, fault pie (with [short fault descriptions](#)), and mechanical narrative from FDD presets.

194.7 Fault code labels (agents)

Use fixed lookup — do not paraphrase codes in UI copy:

- Docs: [fault-codes/short-lookup.md](#)
- Python: portfolio/central/fault_code_lookup.py
- Edge API: GET /api/faults/catalog

194.8 Doc RAG refresh

After editing docs/ consumed by MCP search:

```
./scripts/build_mcp_rag_index.sh
```

- workspace/mcp_server/ — FastMCP tools, resources, prompts
- workspace/mcp_rag/retrieval.py — RAG index (rebuild separately)
- Docker image openfdd-mcp-rag — compatibility name; runs unified server

194.9 Module layout

195 Adr Datafusion Rust Ready Rule Backend

196 ADR: DataFusion as optional Rust-ready rule backend

196.1 Status

Accepted — Open-FDD 3.1.0

196.2 Context

Open-FDD v3 executes FDD rules over **Apache Arrow** tables. The primary contract is:

```
apply_faults_arrow(table, cfg, context=None) -> BooleanArray |
ChunkedArray
```

Operators and integrators author rules in Python/PyArrow. BACnet polling, bridge APIs, dashboard, MCP, and Docker deployment all consume normalized ArrowRuleResult payloads.

We want a migration path toward Rust/DataFusion components **without** rewriting the project in Rust or weakening the PyArrow contract.

196.3 Decision

Add an **optional** second rule backend:

Backend	Entry
arrow	apply_faults_arrow(...) (default, first-class)
datafusion_sql	Restricted SELECT over registered telemetry

DataFusion is Apache Arrow-native and implemented in Rust. The Python datafusion package gives Open-FDD a clean seam where a future Rust service can replace or accelerate the wrapper while preserving the same result schema.

196.4 Why

- Same Arrow memory model end-to-end
- SQL expressions are easy to read for simple threshold/CASE rules
- Compare mode can prove SQL masks match PyArrow before migration
- Optional extra — normal `pip install open-fdd` unchanged

196.5 Non-goals (this milestone)

- No Rust rewrite
- No replacing PyArrow rules
- No pandas on edge runtime
- No arbitrary SQL console (injection-safe subset only)
- No changes to BACnet, bridge auth, or MCP unless required for tests

196.6 Future path

Phase A — PyArrow + optional DataFusion Python backend (this ADR)

Phase B — Shared Arrow rule result contract across batch, Rules Lab, and reporting

Phase C — Optional Rust DataFusion crate/service replaces Python wrapper

Phase D — ML/graph analytics consume Arrow/Parquet outputs without changing the FDD rule contract

196.7 Consequences

- CI adds a job with `pip install -e ".[test,dev,datafusion]"` for backend validation
- Rule store persists `sql` and `fault_column` separately from Python code
- Rules Lab gains backend selector; SQL executes server-side only

197 Adr Rust Ready Arrow Fdd Contract

198 ADR: Rust-ready Arrow FDD contract

198.1 Status

Accepted

198.2 Context

Open-FDD runs supervisory FDD over historian telemetry from BACnet, Niagara, JSON API, and future connectors. PyArrow is the primary rule path; DataFusion SQL is an optional backend for SQL-authored rules and Rust migration prep.

198.3 Decision

1. **PyArrow remains primary.** Rules implement `apply_faults_arrow(table, cfg, context=None)` and return Arrow boolean masks normalized to `ArrowRuleResult`.
2. **DataFusion SQL is optional.** Install via `open-fdd[datafusion]`. Rules declare backend: `datafusion_sql`, execute server-side only, read the registered telemetry Arrow table, and normalize to the same `ArrowRuleResult` shape.
3. **Common historian contract.** Drivers ingest timestamp, `site_id`, point columns, and `metadata.source` into Feather shards. Point metadata (`fdd_input`, `cross_source_semantic`, `equipment_id`) drives rule binding — not vendor-specific code in rule engines.
4. **Future Rust.** A Rust execution service may replace the Python Arrow/DataFusion wrappers behind the same table-in, boolean-mask-out contract. UI, commissioning, and driver orchestration stay in Python until migrated.
5. **Drivers must not bypass the contract.** No connector may emit a custom FDD result shape or row-loop rule path for production batch evaluation.

- 6. **No browser-side SQL.** DataFusion runs in the bridge process only.
- 7. **No pandas row-loop runtime on edge.** PyArrow columnar masks only for production FDD.
- 8. **Confirmation.** May use correctness-first Python streak loops today; future vectorized/window implementations must keep the same raw vs confirmed mask contract.

Authoring guide: [Rule authoring (v1)]({{ "/rule-authoring/" | relative_url }}) · [Arrow rule contract]({{ "/rule-authoring/arrow-rule-contract/" | relative_url }}).

198.4 Consequences

- Rule Lab, batch FDD, and validation smokes share one mask contract.
- Confirmation window filters apply after raw masks for all backends.
- Vectorizing confirmation (window functions / Arrow kernels) is planned; correctness-first Python streak loops are acceptable for bench scale today.

199 Bench Validation Agent

200 Bench validation agent (local bensever)

Local bench only. Read-only.

200.1 Topology

Item	Value
Open-FDD host	bensever
BACnet device	instance 5007 , router 2000:7 (MS/TP)
Niagara station	https://192.168.204.11
Niagara BACnet folder	slot:/Drivers/BacnetNetwork/BENS\$20BENCHTEST\$20BOX/points
Password env	

Item	Value
	OPENFDD_NIAGARA_ADMIN_PASSWORD (never commit)

/stream/health returns **302** before login (reachability). **200** after SCRAM.

200.2 One-command validation

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='...'
python3 scripts/bootstrap_bench_dual_source.py
python3 scripts/bench_validate_bacnet_vs_niagara.py --write-report
```

200.3 API tools (bridge must be running)

Action	Endpoint
Bench health	GET /api/bench/health
Cross-source validate	POST /api/bench/validate/bacnet-vs-niagara
Poll status	GET /api/bench/poll-status
Poll cadence	GET /api/bench/poll-cadence
Niagara test	POST /api/niagara/stations/bench9065/test
Niagara discover	POST /api/niagara/stations/bench9065/discover
Start/stop Niagara poll	POST .../poll/start / .../poll/stop
BACnet driver tree	GET /api/bacnet/driver/tree

200.4 MCP / agent notes

- Browser → Open-FDD API only (no direct Niagara WebSocket from React).
- Preserve Niagara ORDs exactly (\$20, \$2d, \$23). PowerShell: single-quote ORDs.
- Default Niagara discovery excludes proxyExt (10 points, not 20).
- Branch: fix/3.0.34-bugs; push and watch PR #300 CI + CodeRabbit.

200.5 Overnight

```
python3 scripts/run_overnight_bench_smoke.py
```

See [overnight-bench-smoke.md](#) and [bench-bacnet-vs-niagara.md](#).

201 Rcx Central Dash Agent

202 RCx Central Dash — agent guide

For Cursor/Codex agents working on **OpenFDD RCx Central** (portfolio/) — analyst Dash + Central API over read-only Edge REST.

202.1 Runtime stack (benserver / analyst workstation)

Service	Script	Port	Bind
RCx Central API	./scripts/run_central_api.sh	8060	0.0.0.0 (OPENFDD_CENTRAL_API_HOST)
RCx Central Dash	./scripts/run_portfolio_dash.sh	8050	0.0.0.0 (OPENFDD_RCX_DASH_HOST)

Env: OPENFDD_CENTRAL_API_URL=http://127.0.0.1:8060 (Dash → API).
LAN access: sudo ./scripts/open_rcx_central_lan_ports.sh (UFW 8050+8060).

Not the operator bridge (:8765) — RCx Central polls Edge sites via Tailscale/LAN URLs in portfolio/config/sites.json.

202.2 Dash UI (2026-06)

Single **Dashboard** tab (faults, building summary, FDD rules, preset buttons, RCx report) + **Edge Connections** tab.

Area	Data source
Fault pie + legend	fault_code_lookup.py + live/CSV faults

Area	Data source
Building summary	build_mechanical_narrative() → Edge FDD presets
FDD rules table	GET /api/central/fdd-analytics/{site_id}
FDD preset buttons	GET /api/central/fdd-preset/{site_id}/{preset_id} → same as Edge Data Model tab
P8 overrides	KPI count always; chart only when overrides present
RCx DOCX	POST /api/central/rcx/report

202.3 Edge parity (Data Model tab)

RCx Central must **not** scrape React. Use Edge REST:

- GET /api/model/fdd-query-presets + GET /api/model/fdd-query-presets/{id}
- GET /api/model/sparql/predefined + POST /api/model/sparql
- GET /api/model/health, GET /api/model/tree

Preset ids: rules_to_equipment, rules_to_sensors, ahus_vavs_zones, missing_rule_bindings, orphan_points, rule_coverage_by_equipment_type, etc. — see portfolio/central/fdd_preset_catalog.py.

202.4 BRICK equipment typing (common bug)

Acme and many sites store types in **brick_type** (AHU, VAV), not equipment_type.

- Count/classify with portfolio/central/equipment_classify.py (mirrors workspace/api/openfdd_bridge/equipment_classify.py).
- TTL sync maps short types → Brick classes (AHU → Air_Handling_Unit, VAV → Variable_Air_Volume_Box) in ttl_service.py.
- After model.json edits on Edge: **sync TTL** before SPARQL HVAC buttons return counts.
- Packaged rooftop serving VAVs: prefer **brick_type: AHU** and display name AHU 01 (not Rtu 01) when it is a full air handler.

202.5 Fault codes (stable labels)

Layer	Path
Docs table	

Layer	Path
	docs/fault-codes/short-lookup.md
Dash lookup	portfolio/central/fault_code_lookup.py
Edge catalog	GET /api/faults/catalog · workspace/api/openfdd_bridge/fault_catalog.py

Never invent codes. Letter suffix only (VAV-C, not VAV-03).

202.6 Central API routes (agent smoke)

```
curl -s http://127.0.0.1:8060/health
curl -s http://127.0.0.1:8060/api/central/overview/acme
curl -s "http://127.0.0.1:8060/api/central/fdd-analytics/acme?hours=24"
curl -s http://127.0.0.1:8060/api/central/fdd-preset/acme/ahus_vavs_zones
```

202.7 Tests

```
pytest tests/portfolio/ tests/scripts/
test_rcx_overnight_patch_cycle.py
pytest tests/workspace_bridge/test_fdd_query_presets.py
```

202.8 PRs (typical split)

- **Edge bridge + TTL + model fixes** → fix/edge-dashboard-bridge (#299)
- **RCx Dash / portfolio only** → feature/rcx-central-overview-ui (#298)

202.9 Related

- Overnight patch cycle
- Model queries
- MCP server — portfolio mode vs RCx Central HTTP API

203 Rcx Central Overnight Patch Cycle

204 RCx Central overnight patch-cycle (agent runbook)

See also: [RCx Central Dash agent guide](#) for UI/API modules and BRICK typing.

Mission: validate **OpenFDD RCx Central** against local services and (optionally) live **ACME Edge** read-only, up to **10 patch cycles** with **2-hour** spacing.

204.1 Safety (non-negotiable)

- **Read-only** on BACnet and live BAS equipment.
- Never commit secrets, .env, tokens, or raw building dumps.
- Merge, publish, and deploy only when RCX_ALLOW_MERGE=1, RCX_ALLOW_PUBLISH=1, or RCX_ALLOW_DEPLOY=1 and the owner authorizes.

204.2 One patch cycle

1. `git pull --ff-only` on feature/rcx-central-overview-ui (or follow-up branch).
2. `gh pr checks 298` and `gh run view <id> --log-failed` for failures.
3. Classify CodeRabbit comments: must-fix / nice-to-have / false positive.
4. Run tests: `python3 -m pytest tests/portfolio tests/workspace_bridge -q`.
5. Start RCx Central: `./scripts/run_central_api.sh`, `./scripts/run_portfolio_dash.sh` (bind 0.0.0.0; LAN: `sudo ./scripts/open_rcx_central_lan_ports.sh`).
6. Run `python3 scripts/rcx_central_overnight_patch_cycle.py` (try-out or overnight env).
7. Patch bugs, commit, push, confirm CI green.
8. Write `reports/rcx_central_overnight_logs/cycle_NN.md`.

204.3 GH Actions

```
gh pr checks 298
gh run list --branch feature/rcx-central-overview-ui --limit 10
gh run view <RUN_ID> --log-failed
```

Fix order: unit tests → operator-bridge → docs → Docker supervisor.

204.4 RCx Central local

```
./scripts/run_central_api.sh # :8060
OPENFDD_CENTRAL_API_URL=http://127.0.0.1:8060 ./scripts/
run_portfolio_dash.sh # :8050
curl -s http://127.0.0.1:8060/health
curl -s -o /dev/null -w '%{http_code}\n' http://127.0.0.1:8050/
```

Edge connections: save URL + credentials on **Edge Connections** tab (local portfolio/config/).

204.5 Live ACME (gated)

```
export OPENFDD_LIVE_ACME=1
export RCX_EDGE_SITE=acme
export RCX_CENTRAL_API_URL=http://127.0.0.1:8060
```

Requires ACME credentials in local registry — never log passwords.

204.6 Try-out vs overnight

Mode	RCX_PATCH_CYCLES	RCX_CYCLE_SLEEP_MINUTES
Try-out	2	0
Overnight	10	120

204.7 Rev bump / deploy

Only when tests and CI pass and owner sets RCX_ALLOW_PUBLISH=1 / RCX_ALLOW_DEPLOY=1. Use existing scripts/upgrade_edge_full.sh — do not invent new deploy paths.

204.8 Branch cleanup after merge

```
git checkout master && git pull --ff-only
git branch -d feature/rcx-central-overview-ui
git push origin --delete feature/rcx-central-overview-ui
```

205 Bench Bacnet Vs Niagara

206 BACnet direct vs Niagara bench validation

Local **benserver** read-only cross-check: same physical BACnet device via two paths.

206.1 Topology

```
BENS BENCHTEST BOX (BACnet device 5007, MS/TP 2000:7)
├─ Open-FDD native BACnet poll (commission agent → samples.csv →
feather source=bacnet)
└─ Niagara Windows station → baskStream WebSocket → Open-FDD
Niagara driver (source=niagara_baskstream)
```

206.2 Quick start

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='...' # never commit
python3 scripts/bootstrap_bench_dual_source.py
python3 scripts/bench_validate_bacnet_vs_niagara.py --write-report
```

Or via API (bridge running):

```
curl -X POST http://127.0.0.1:8765/api/bench/validate/bacnet-vs-
niagara \
-H "Authorization: Bearer $TOKEN" -H "Content-Type: application/
json" \
-d '{"write_report": true}'
```

206.3 Mapping config

Edit workspace/data/bench_bacnet_vs_niagara.yaml for semantic point mapping, tolerances, and Niagara ORDs (\$20 / \$2d preserved).

206.4 Brick model

Import dual-source semantics:

```
# Included in bootstrap; or manually:
python3 -c "
import json, os, sys
from pathlib import Path
REPO = Path('.').resolve()
sys.path.insert(0, str(REPO/'workspace/api'))
os.environ['OFDD_DESKTOP_DATA_DIR'] = str(REPO/'workspace/data')
from openfdd_bridge.model_service import ModelService
from openfdd_bridge.ttl_service import TtlService
ModelService().import_json(json.loads((REPO/'workspace/data/
bench_dual_source_model.json').read_text()), replace=False)
TtlService().sync()
"
```

206.5 Normalized data model

All drivers normalize through driver_point_contract.py:

- **bacnet** → canonical bacnet_direct
- **niagara_baskstream, modbus, json_api**

Semantic identity (fdd_input / cross_source_semantic) links both sources to one Brick point role.

206.6 Poll intervals

- BACnet: poll_interval_s in points.csv (standard: 60, 300, ...)
- Niagara: poll_interval_seconds on station config; worker re-reads each cycle (no restart required)
- Lab minimum Niagara interval: 15s (API validates ≥ 15)

Validate cadence: GET /api/bench/poll-cadence?
source=bacnet_direct&expected_interval_s=60

206.7 Overnight smoke

```
python3 scripts/run_overnight_bench_smoke.py --dry-run # one
checkpoint
python3 scripts/run_overnight_bench_smoke.py # 12h,
checkpoint every 2h
```

Reports: reports/overnight_bench/<timestamp>/

206.8 Read-only rules

No BACnet writes, no Niagara overrides, no alarm ack. Relay booleans are read-only observations only.

207 Dashboard Analytics

208 Dashboard analytics

Engineering analytics pages live under **Analytics** in the operator dashboard (/analytics).

208.1 Pages

Route	Purpose
/analytics	Overview KPIs, fault-hour charts, top faults table
/analytics/faults	Time-range fault analytics
/analytics/equipment	AHU / VAV fault-hour summary
/analytics/health	BACnet / BRICK model health
/analytics/rcx	RCx Report Builder (preview + DOCX download)

208.2 API (read-only)

- GET /api/analytics/overview
- GET /api/analytics/faults?hours=24
- GET /api/analytics/model-health
- POST /api/reports/rcx/preview

- POST /api/reports/rcx/generate → DOCX download

208.3 Fault hours

Elapsed fault hours are estimated from FDD batch run analytics.estimated_fault_duration_sec when present, otherwise from flagged/total sample ratio. See open_fdd/reports/fault_hours.py.

208.4 Theme

All Plotly charts use getPlotlyThemeLayout(theme) from workspace/dashboard/src/lib/plotly-theme.ts.

208.5 Tests

```
pytest tests/workspace_bridge/test_dashboard_analytics.py
open_fdd/tests/test_fault_hours.py
cd workspace/dashboard && npm test
```

Live BACnet is not required. Gate live ACME validation with OPENFDD_LIVE_ACME=1.

209 Dashboard Rcx Makeover Inventory

210 Dashboard + RCx Makeover — Inventory

Branch: feature/dashboard-rcx-mega-makeover
Date: 2026-05-30

210.1 What exists

Area	Location	Notes
React operator dashboard	workspace/dashboard/	Vite + React 19, theme via contexts/theme-

Area	Location	Notes
		context.tsx, CSS vars
Plotly charts	lib/plot-chart.ts, PlotPage.tsx, HostStatsPage.tsx	Partial theme handling; not centralized
Building check-engine	building_status.py, FaultsPage, BuildingInsightDashboard	Single source for alerts
FDD batch + analytics	fdd_results.py, fdd_fault_analytics.py	Per-run sample analytics, equipment enrichment
Poll / model health	poll_throughput.py, model_health.py, device_poll_health.py	Used by agent, not dashboard analytics pages
Zone temp analytics	zone_temp_analytics.py	VAV comfort metrics via agent routes
Central RCx DOCX	portfolio/central/rcx_report.py, fault_hours.py	Central-only (POST /api/central/rcx/report)
Generic open_fdd reports	open_fdd/reports/docx_generator.py, fault_viz.py	Equipment FDD reports, optional deps
Portfolio desk	portfolio/dash/app.py, portfolio/central/api.py	Separate Dash + FastAPI stack
Frontend unit tests	14 Vitest files under workspace/dashboard/src/lib/	Not in CI
Bridge Python tests	tests/workspace_bridge/, tests/portfolio/	Strong coverage for building insight, central

210.2 What partially exists

- **Theme toggle** — app shell updates; Plotly uses inline colors in plot-chart.ts / HostStatsPage.tsx (inconsistent).
- **Fault display** — equipment names in check-engine alerts when model maps columns; catalog page is not analytics-focused.
- **RCx DOCX** — portfolio central builder with basic sections; not exposed on edge bridge; no chart/section checkboxes.

- **Fault hours** — `portfolio/central/fault_hours.py` estimates from rollups; FDD run analytics has duration estimates but no REST aggregation.
- **Demo data** — `workspace/data/acme_gl36_model.json`, `fdd_results.json`, `samples/demo_site.csv`; no unified analytics fixture.

210.3 What is missing (this branch targets)

- Shared `getPlotlyThemeLayout()` helper + all charts using it
- Analytics pages: Overview, Fault Analytics, Equipment, BACnet/Model Health, RCx Report Builder
- Bridge REST: `/api/analytics/overview`, `/faults`, `/equipment/{id}`, `/model-health`, `/reports/rcx/preview`, `/reports/rcx/generate`
- Edge RCx DOCX with selectable sections/charts (`open_fdd/reports/rcx_docx.py`)
- Fault-hour aggregation module with tests
- Demo analytics fixture for CI
- Docs: `docs/dashboard-analytics.md`, `docs/rcx-report-builder.md`

210.4 Files inspected

`AGENTS.md`, `README.md`, `pyproject.toml`
`workspace/dashboard/src/App.tsx`, `AppLayout.tsx`, `theme-context.tsx`, `plot-chart.ts`
`workspace/api/openfdd_bridge/main.py`, `building_status.py`, `fdd_results.py`, `fdd_fault_analytics.py`
`workspace/api/openfdd_bridge/routes/analytics_routes.py`
`portfolio/central/rcx_report.py`, `fault_hours.py`, `api.py`
`open_fdd/reports/`
`tests/workspace_bridge/`, `tests/portfolio/test_central.py`

210.5 Recommended implementation path

1. **Foundation** — `open_fdd/reports/fault_hours.py`, `charts.py`, `rcx_docx.py`; bridge `dashboard_analytics.py` + routes; deps in bridge requirements.
2. **Theme** — `plotly-theme.ts`; refactor `plot-chart.ts` + `HostStatsPage.tsx`.
3. **UI** — Analytics nav section + pages wired to new APIs; RCx Report Builder with preview/generate.
4. **Fixtures + tests** — demo JSON, backend `pytest`, `Vitest` for theme + report builder.
5. **Docs** — operator-facing how-to.

210.6 Known risks

- **Scope** — full prompt is multi-PR; this branch delivers edge dashboard + bridge RCx; Central Dash remains separate.
- **Live BACnet** — analytics read persisted FDD/model/poll data only (read-only).
- **Trend charts in RCx** — need timeseries API + point roles; may be partial when historian empty.
- **ACME coupling** — use demo fixture + model-driven roles, not hard-coded point names.

211 Rcx Central Current State

212 OpenFDD RCx Central — current state

Updated: 2026-05-30
Branches: feature/rcx-central-overview-ui (#298)
Status: Local TTL mirror + SPARQL on Central; benserver sign-off before Docker GHCR publish.

212.1 Product names

User-facing	Internal path
OpenFDD Edge	workspace/, docker/, GHCR images
OpenFDD RCx Central	portfolio/ (Dash + Central API)

212.2 Entrypoints

Service	Command	Port	Bind default
RCx Central Dash	./scripts/run_portfolio_dash.sh	8050	0.0.0.0
RCx Central API	./scripts/run_central_api.sh	8060	0.0.0.0
Docker (both)	docker/rcx-central/ docker-compose.yml	8050, 8060	

LAN firewall (benserver): sudo ./scripts/
open_rcx_central_lan_ports.sh

212.3 Central API

- GET/POST/PUT/DELETE /api/central/edges, POST /api/central/edges/test
- GET /api/central/overview/{site_id} — KPIs, fault pie, mechanical narrative, P8
- GET /api/central/fdd-analytics/{site_id} — rules table + descriptions
- GET /api/central/fdd-preset/{site_id}/{preset_id} — Edge FDD preset proxy + Central enrichment
- POST /api/central/model/remediate/{site_id} — equipment types, BACnet metadata, Edge TTL sync, Central mirror
- POST /api/central/model/sync-ttl/{site_id} — pull data_model.ttl to portfolio/data/sites/{site_id}/model/
- GET /api/central/model/sparql/validate/{site_id} — local AHU/VAV/site SPARQL counts
- POST /api/central/model/sparql/{site_id} — read-only SPARQL on mirrored TTL
- GET /api/central/mechanical-summary/{site_id}
- GET /api/central/model-tree/{site_id} — on-demand BRICK tree (lazy; not on auto-refresh)
- GET /api/central/rcx/points/{site_id} — BACnet point catalog for Report Builder
- POST /api/central/rcx/preview, /charts/preview, /report

212.4 Edge REST (read-only, shared with Data Model tab)

Model: /api/model/tree, /api/model/health, SPARQL, /api/model/fdd-query-presets/{id}

Trends: /api/timeseries/readings

Analytics: /api/analytics/faults

Faults: /api/faults/status, /api/faults/catalog

212.5 Dash UI tabs

Dashboard · Report Builder · Edge Connections

Dashboard: fault mix + legend, building summary, P8 KPI, FDD rules (lazy load), FDD/BRICK preset buttons, **local SPARQL** (TTL mirror).

Report Builder: 3-step wizard — building/time → charts & sections
→ custom points → preview DOCX.
Edge data fetched on demand (not with dashboard load).

212.6 Pre-Docker owner checklist (benserver :8050)

1. Dashboard loads in under 5s without hanging on model tree.
2. **Load full BRICK model** and **Load FDD rules** buttons work when clicked.
3. Report Builder: load catalog → preview charts → generate DOCX for Acme.
4. Fault overlays visible on trend previews when enabled.
5. Sign off here before docker/rcx-central image rebuild or RCX_ALLOW_PUBLISH=1.

212.7 BRICK model notes (agents)

- Classify equipment with brick_type **and** equipment_type — see equipment_classify.py
- TTL maps AHU/VAV → Brick schema classes for SPARQL HVAC queries
- Acme roof unit: brick_type: AHU, name AHU 01 (not Rtu 01 in new models)

212.8 Fault codes

- Stable short labels: docs/fault-codes/short-lookup.md, portfolio/central/fault_code_lookup.py
- Full catalog: GET /api/faults/catalog on Edge

212.9 Agent docs

- [RCx Central Dash agent guide](#)
- [AI agent workflow](#)
- [MCP server](#)

212.10 Tests

pytest tests/portfolio/ · tests/workspace_bridge/
test_fdd_query_presets.py

212.11 Remaining gaps

- **Owner Dash sign-off** on benserver before Docker GHCR publish
- GHCR publish for openfdd-rcx-central image (gated: RCX_ALLOW_PUBLISH=1)
- Optional OpenAI insights hook (templates used today)
- True Edge start/end bounded timeseries (hours lookback used for now)
- Summarize HVAC SPARQL buttons on Dash — **done** (local TTL mirror on Central)

213 Short Lookup

214 Fault code short descriptions

Stable lookup for pie charts, tables, and RCx reports. Titles match `workspace/api/openfdd_bridge/fault_catalog.py`.

Code	Short description
AHU-A	Supply fan performance degradation
AHU-B	Simultaneous heating and cooling
AHU-C	Supply air temperature sensor fault
AHU-D	Mixed-air temperature inconsistent with OAT/RAT
AHU-E	Economizer not economizing
AHU-F	Damper/valve command vs feedback mismatch
AHU-G	PID hunting / excessive control oscillation
VAV-A	Reheat active during cooling demand
VAV-B	Airflow not meeting setpoint
VAV-C	Zone temperature sensor fault
VAV-D	Damper command vs airflow mismatch
VAV-E	Rogue zone (chronic reheat/overcooling)
VAV-F	VAV actuator PID hunting
BLD-A	Whole-building energy deviation
BLD-B	Outdoor-air temperature sensor fault
BLD-C	Equipment running outside occupancy schedule
BLD-D	Point data dropout (stale points)
RTU-A	Supply fan performance degradation

Code	Short description
RTU-B	Simultaneous heating and cooling
RTU-C	Discharge air temperature sensor fault
RTU-D	Damper/valve command vs feedback mismatch
RTU-E	Cooling capacity PID hunting
CH-G	Pump / variable-speed PID hunting

Full catalog: [index.md](#) and live GET /api/faults/catalog on Edge.

Python mirror: `portfolio/central/fault_code_lookup.py` (RCx Central Dash).

215 Niagara Baskstream Connector

216 Niagara baskStream connector (read-only)

Open-FDD integrates with **Niagara Framework** stations through the vendor **baskStream** module. The bridge API performs SCRAM login, opens a persistent WebSocket, and reads point values via MessagePack — the browser never connects directly to Niagara.

216.1 Read-only posture

This connector is **read-only**. It does not implement writes, overrides, alarm acknowledge, or emergency actions. Use Niagara Workbench for OT commands.

216.2 Niagara station setup

1. Install the **baskStream** module on the Niagara station (same module validated by `niagara-baskstream-python-tools` on Linux).
2. Add **BASkStreamService** (or equivalent `baskStream` service) on the station.
3. Ensure HTTPS (port **443**) is reachable from the Open-FDD bridge host.
4. Firewall: allow the bridge server → station on 443.

216.2.1 Health check behavior

- **Before login:** GET /stream/health returns **302** redirect to /login.
- **After SCRAM login:** GET /stream/health returns **200** with JSON including authenticatedUser.

From Linux bench:

```
curl -k -i https://192.168.204.11/stream/health  
# Expect 302 before login
```

216.3 Open-FDD configuration

1. Copy workspace/niagara.env.example → workspace/niagara.env.local (gitignored).
2. Set OPENFDD_NIAGARA_ADMIN_PASSWORD (or your custom env name).
3. In the dashboard **Niagara** tab, add a station:
 - URL: https://<station-ip>
 - Username: Niagara web user
 - **password_env**: OPENFDD_NIAGARA_ADMIN_PASSWORD
 - **verify_tls**: off for self-signed bench certs
 - **default_points_root**: e.g. slot:/Drivers/BacnetNetwork/BENS\$20BENCHTEST\$20BOX/points

216.3.1 ORD encoding

Niagara ORDs use encoded characters (\$20 = space, \$2d = hyphen, \$23 = #). **Preserve these strings exactly** in config, API, and UI. Do not URL-decode or shell-expand them.

PowerShell users: use **single-quoted** strings for ORDs so \$20 is not treated as a variable.

216.3.2 Allowed Origins

The Open-FDD **backend** connector does not require Niagara “Allowed Origins” for browser CORS — traffic is **Browser** → **Open-FDD** /api/niagara → **Niagara**. Direct browser-to-Niagara WebSocket is discouraged.

216.4 Dependencies

Bridge API requires aiohttp and msgpack (see workspace/api/requirements.txt). If missing, the app still boots; /api/niagara/health reports dependencies_ok: false.

216.5 Polling

Enable poll per station in the UI. The worker uses one persistent WebSocket per active station, batched reads, cached discovery metadata, and historian ingest with source=niagara_baskstream.

Disable background worker: OFDD_DISABLE_NIAGARA_POLL_WORKER=1.

216.6 Bench reference (benserver)

Item	Value
Station URL	https://192.168.204.11
BACnet device folder	slot:/Drivers/BacnetNetwork/ BENS\$20BENCHTEST\$20BOX/points
Expected points	~10 BACnet components (proxy extensions excluded by default)

Validate: **Test connection** → **Discover** → **Read selected** / **Poll once**. Values should match the standalone baskstream_cli.py values output (OA-T, DUCT-T, STAT ZN-T, etc.).

216.7 Large sites / multi-building stations (#315)

A single Niagara station may contain many buildings. Use **folder ORDs as building boundaries**:

1. **Station tree browse** (Niagara tab) — preview slot:/ Drivers/... hierarchy before discover.
2. **Right-click** a folder → **Add as building** or **Add as device** (BACnet-style commission tree).
3. Set **default_points_root** to the chosen device .../points folder (auto-suggested when adding a device).
4. **Discover points** under that root only; tune browse_depth and max_nodes in station settings for massive trees.
5. **Poll once** as a dry-run poll; review Activity log and point counts before enabling background poll.
6. Map folders to Open-FDD site/building context via commissioning export/import (open_fdd/validation/niagara_folder_mapping.py helpers).

Truncated browse (max_nodes hit) returns partial trees — narrow the base ORD or increase max_nodes in station config.

217 Data Model

218 Data Model — sync and FDD queries

218.1 Explorer (BACnet poll sync)

The **Explorer** tab focuses on **BACnet poll CSV** ↔ **model.json** sync status and **Sync poll** → **model**. Write TTL lives under **Advanced** (or SPARQL RDF tools).

218.2 FDD query presets

The **SPARQL** tab includes **FDD / BRICK query presets** — composed queries across `model.json`, rule bindings, and BACnet metadata:

- Rules → Equipment / Sensors / BACnet devices
- Equipment → Points, AHUs/VAVs/Zones
- Missing rule bindings, orphan points, sensor classes, coverage by equipment type

API: GET `/api/model/fdd-query-presets`, GET `/api/model/fdd-query-presets/{id}`.

219 Host Resources

220 Host Resources — Easy Pooge reset

Integrators see a **Danger Zone** on the Host Resources tab for job-site redeploys.

- Typed confirmation: POOGE THIS EDGE
- Dry-run preview by default
- Allowlisted path cleanup (historian, BACnet scratch, exports)
- Optional Linux apt upgrade and GHCR image pull

API: POST /api/host/pooge/preview, POST /api/host/pooge/run (integrator only).

221 Trend Plots

222 Trend plots

The **Trend plot** tab reads feather historian data via GET /api/timeseries/readings.

222.1 HTTP / Tailscale

Charts work over plain HTTP on localhost, LAN, and Tailscale — COOP/CORP security headers are omitted on non-HTTPS origins so Plotly is not blocked.

222.2 Troubleshooting

If the chart is blank, open **Trend plot debug** at the bottom of the tab:

- Endpoint called
- Site, equipment, selected keys
- Series and timestamp row counts
- Last error message

Common fixes: enable BACnet polling, pick points with feather columns, reduce FDD overlay scope.

223 Cloud Exporter

224 Cloud exporter sidecar

Optional openfdd-cloud-exporter container polls the bridge API and POSTs JSON to a webhook-compatible URL (PostBin, webhook.site, etc.).

224.1 Enable (edge compose)

```
export OPENFDD_EXPORT_ENDPOINT=https://webhook.site/your-uuid
export OPENFDD_EXPORT_DRY_RUN=1
docker compose --profile cloud-export up -d cloud-exporter
```

Set OPENFDD_EXPORT_DRY_RUN=0 only when you intend live export.

224.2 Security

- Disabled by default (OPENFDD_EXPORT_DRY_RUN=1).
- Does not export operator passwords or auth env files.
- Tokens are redacted in logs.

See workspace/cloud_exporter/README.md for environment variables.

225 Overnight Bench Smoke

226 Overnight bench smoke test

12-hour local validation on **benserver** only. Read-only.

226.1 Command

```
export OPENFDD_NIAGARA_ADMIN_PASSWORD='...'
python3 scripts/run_overnight_bench_smoke.py
```

Options:

Flag	Default	Purpose
--duration-hours	12	Total run length
--checkpoint-hours	2	Wake interval
--dry-run	off	Single checkpoint then exit
--skip-bootstrap	off	Skip BACnet/Niagara setup
--test-poll-freq	off	Run poll interval change test at t=0

226.2 Each checkpoint

1. Cross-source validate (bench_validate_bacnet_vs_niagara)
2. Poll cadence report (BACnet + Niagara)
3. Targeted pytest (contract, validator, Niagara)
4. JSON + Markdown report under reports/overnight_bench/
 <run_id>/

226.3 Final deliverable

final_report.md / final_report.json with PASS/FAIL and next actions.

226.4 GH Actions loop

After code fixes during a wake cycle:

```
git add ... && git commit -m "..." && git push origin fix/3.0.34-bugs  
gh pr checks 300 --watch
```

No empty commits when validation-only cycles pass.

227 Ai Agent Workflow

228 AI agent workflow vs runtime use

228.1 Three agent surfaces

Surface	Port	Use when
OpenFDD Edge (bridge)	8765 / :80 Caddy	Rule Lab, BACnet, Data Model SPARQL, live FDD
OpenFDD RCx Central	8050 Dash, 8060 API	Multi-site analyst dashboard, RCx DOCX, Edge registry
OpenFDD MCP	8090	Cursor/Codex tools over bridge or portfolio/ sites.json

Acme production edge runs **without** MCP (`enable_mcp: false`). Use **RCx Central API** or **portfolio MCP** against `http://100.122.106.124` (Tailscale).

228.2 Runtime analyst (no git clone)

1. `docker compose -f docker/rcx-central/docker-compose.yml up`
or `./scripts/run_central_api.sh + ./scripts/run_portfolio_dash.sh`
2. Open `http://localhost:8050` (LAN: `sudo ./scripts/open_rcx_central_lan_ports.sh` on benserver)
3. **Edge Connections** → add Acme URL → Test → Save
4. **Dashboard** → select building → verify building summary (AHU/VAV counts), FDD preset buttons, fault pie descriptions
5. **RCx report** → Preview → Generate DOCX

Kill containers when done. Registry in Docker volume `rcx-central-config`.

228.3 Developer / coding agent (git working tree)

Use Cursor or Codex against `open-fdd`. Read:

- [RCx Central Dash agent guide](#)
- [Overnight patch cycle](#)

Key modules: `portfolio/dash/overview_tab.py`, `portfolio/central/mechanical_narrative.py`, `portfolio/central/fault_code_lookup.py`, `workspace/api/openfdd_bridge/fdd_query_presets.py`, `ttn_service.py`.

After doc changes consumed by MCP RAG: `./scripts/build_mcp_rag_index.sh` (optional on benserver).

228.4 Model / SPARQL validation (Acme)

1. Confirm `model.json` equipment uses `brick_type` (AHU, VAV) — not only legacy `equipment_type`
2. Sync TTL on Edge (Data Model tab or bridge restart)
3. Edge UI: **Summarize HVAC → AHUs / VAV boxes** must return non-zero for Acme lab
4. RCx Dash: **AHUs / VAVs / Zones** preset → same row counts as Edge

228.5 Secrets

- Never commit `portfolio/config/credentials.json` or passwords in `sites.json`
- Env: `OFDD_AGENT_PASSWORD`, per-site credentials in config store
- Live ACME tests: `OPENFDD_LIVE_ACME=1` only when explicitly requested

228.6 Model routing (test triage)

See `AGENTS.md` — classify CI failures as SIMPLE vs COMPLEX before processing.

229 Docker Desktop Windows

230 RCx Central on Docker Desktop (Windows)

230.1 Prerequisites

- Docker Desktop for Windows
- Edge reachable from the host (e.g. Tailscale)

230.2 Run

```
cd open-fdd
docker compose -f docker/rcx-central/docker-compose.yml up --build
```

Open:

- Dash: `http://localhost:8050`
- API health: `http://localhost:8060/health`

No git clone required at runtime if you pull `ghcr.io/bbartling/openfdd-rcx-central` (when published). Build locally with `OPENFDD_IMAGE_TAG=local`.

230.3 Volumes

Volume	Purpose
rcx-central-data	Rollups, reports (portfolio/data/reports/)
rcx-central-config	sites.json, masked credentials

230.4 Tailscale networking

If Edge is reachable on the Windows host but not from inside the Linux container:

1. Test from host: `curl https://<tailscale-ip>:8765/health`
2. Test from container: `docker compose exec rcx-central-api curl -s http://<tailscale-ip>:8765/health`
3. If container cannot route, use the host's Tailscale IP (Docker Desktop often routes LAN correctly) or document `host.docker.internal` proxy for dev.

230.5 Smoke test

```
./scripts/test_rcx_central_docker.sh
```

Windows PowerShell variant: run `compose up`, then `curl` health URLs manually per steps above.

231 Edge Auth

232 Edge authentication (RCx Central)

232.1 Browser flow

1. Open RCx Central → **Edge Connections**
2. Enter `site_id`, name, Edge base URL (`https://<tailscale-ip>:8765`)
3. Enter username/password (or bearer token via API)
4. **Test Connection** — probes `/health`, `/api/model/health`, `/api/faults/status`

5. **Save Edge** — writes to portfolio/config/sites.json + credentials.json (Docker volume)

232.2 API

```
GET /api/central/edges
POST /api/central/edges
POST /api/central/edges/test
PUT /api/central/edges/{site_id}
DELETE /api/central/edges/{site_id}
```

Secrets are **never** returned in API responses or committed to git. Passwords are masked in UI after save.

232.3 Auth types

- password — POST /api/auth/login on Edge
- bearer — static token in credentials store
- none — public Edge endpoints only (dev)

232.4 Rejected URLs

file://, ftp://, and hosts without scheme are rejected.

233 Index

234 OpenFDD RCx Central

OpenFDD RCx Central is the local analyst / RCx engineer dashboard. It runs on an analyst workstation or **Docker Desktop (Windows/Linux)** — not on the OT edge by default.

Product	Role
OpenFDD Edge	Arrow-native, no-pandas OT-LAN FDD machine at the building
OpenFDD RCx Central	Read-only multi-edge analytics, chart preview, DOCX reports

Internal source path: portfolio/ (package name unchanged).

234.1 Quick start (host Python)

```
pip install -r portfolio/requirements.txt
./scripts/run_central_api.sh # :8060
./scripts/run_portfolio_dash.sh # :8050
```

234.2 Docker Desktop

```
./scripts/run_rcx_central_docker.sh
# Dash http://localhost:8050 API http://localhost:8060/health
```

See [docker-desktop-windows.md](#).

234.3 Pages

- Overview — collected rollup trends (Plotly, light/dark theme)
- Edge Connections — add/test/save Edge URLs (secrets in local volume)
- Mechanical Summary — BRICK equipment counts from Edge APIs
- FDD Rules & Analytics — configured rules, chart packs
- Trend Explorer — pointer to Overview Plotly charts (local collect CSVs)
- RCx Report Builder — matplotlib preview, fault overlays, DOCX download
- Validation Runs — one-off read-only Edge probes
- Settings — API URL and volume paths

Model query details: [model-queries.md](#)

234.4 Safety

RCx Central is **read-only** toward Edge and BACnet. No writes, commands, or setpoint changes.

235 Model Queries

236 Model queries (RCx Central → OpenFDD Edge + local TTL mirror)

OpenFDD RCx Central reuses the **same read-only Edge REST APIs** as the Edge operator React **Data model** tab. RCx Central does not scrape the React UI.

Local BRICK SPARQL: POST /api/central/model/sync-ttl/{site_id} fetches Edge model.json, builds TTL with the same ttl_service as Edge, and stores it under portfolio/data/sites/{site_id}/model/data_model.ttl. Central runs SPARQL locally (rdflib). Optionally calls Edge POST /api/model/sync-ttl first so on-site TTL stays aligned when the Edge image is current.

236.1 Primary endpoints

Method	Edge path	Purpose
GET	/api/model/tree	Equipment + points (BRICK roles, timeseries columns)
GET	/api/model/health	Point/equipment counts, model issues
GET	/api/model/sparql/predefined	Prebuilt SPARQL query list
POST	/api/model/sparql	Run SPARQL ({"query": "..."})
GET	/api/model/fdd-query-presets	FDD commissioning query presets
GET	/api/model/fdd-query-presets/{id}	Run preset by id
GET	/api/model/ttl?save=false	Turtle text (used by Central TTL mirror)
POST	/api/model/sync-ttl	

Method	Edge path	Purpose
		Regenerate TTL on Edge from model.json

236.2 Trends (matplotlib preview + fault overlays)

Method	Edge path	Purpose
GET	/api/timeseries/series?site_id=	Plottable columns per site
GET	/api/timeseries/readings?site_id=&columns=&hours=	Trend samples + optional FDD fault flags

RCx Central maps BRICK **point roles** (e.g. supply_air_temperature) to historian columns via /api/model/tree, then calls /api/timeseries/readings with include_faults=true for overlay bands.

236.3 Analytics

Method	Edge path	Purpose
GET	/api/analytics/overview	KPIs, fault hours by severity/equipment
GET	/api/analytics/faults?hours=	Fault rows for selected window
GET	/api/analytics/model-health	Stale/missing roles

236.4 RCx Central API wrappers

Method	Central path
GET	/api/central/overview/{site_id}
GET	/api/central/mechanical-summary/{site_id}
GET	/api/central/fdd-analytics/{site_id}
GET	/api/central/fdd-preset/{site_id}/{preset_id}
POST	/api/central/model/remediate/{site_id}
POST	/api/central/model/sync-ttl/{site_id}
GET	/api/central/model/ttl/{site_id}

Method	Central path
GET	/api/central/model/sparql/predefined/{site_id}
POST	/api/central/model/sparql/{site_id}
GET	/api/central/model/sparql/validate/{site_id}
POST	/api/central/rcx/preview

Edge model mutations use integrator auth via remediate; SPARQL on Central is read-only.

236.5 FDD preset ids (Edge parity)

Same as React DataModelSparqlPanel FDD buttons:

rules_to_equipment, rules_to_sensors, rules_to_bacnet_devices,
equipment_to_points, ahus_vavs_zones, missing_rule_bindings,
points_by_bacnet_device, sensor_classes_used_by_fdd,
rule_coverage_by_equipment_type, orphan_points

Dash **Local BRICK SPARQL** buttons call Central paths above (mirrored TTL).

236.6 BRICK typing (do not mis-count AHUs/VAVs)

Many sites (including Acme) set **brick_type** (AHU, VAV) without `equipment_type`. Presets and mechanical narrative use `equipment_classify` — not `equipment_type` alone.

After `model.json` changes on Edge, **sync TTL** (POST `/api/central/model/sync-ttl/{site_id}` or remediate) so SPARQL predefined queries (Air_Handling_Unit, Variable_Air_Volume_Box) match equipment counts.

Packaged rooftop air handlers feeding VAVs should be modeled as **brick_type: AHU** even when the legacy BACnet name contains “RTU”.

237 Overnight Acme Validation

238 Overnight ACME validation via RCx Central

RCx Central drives read-only validation of the ACME OpenFDD Edge over Tailscale.

238.1 Prerequisites

- Local RCx Central API (:8060) and Dash (:8050).
- ACME Edge registered under **Edge Connections** with credentials in portfolio/config/ (gitignored).
- gh CLI authenticated for PR/CI inspection.

238.2 Try-out (2 cycles, no sleep)

```
OPENFDD_LIVE_ACME=1 \  
RCX_CENTRAL_API_URL=http://127.0.0.1:8060 \  
RCX_CENTRAL_DASH_URL=http://127.0.0.1:8050 \  
RCX_EDGE_SITE=acme \  
RCX_PATCH_CYCLES=2 \  
RCX_CYCLE_SLEEP_MINUTES=0 \  
python3 scripts/rcx_central_overnight_patch_cycle.py
```

238.3 Overnight (10 cycles, 2 h spacing)

Set RCX_PATCH_CYCLES=10 and RCX_CYCLE_SLEEP_MINUTES=120.

Reports land in reports/rcx_central_overnight_logs/cycle_NN.md.

238.4 Read-only scope

Collects health, model tree counts, fault status, and overview data. No BACnet writes.

239 Patch Cycle Release Process

240 Patch-cycle release process

A **patch cycle** completes when CI is green, local smoke passes, optional ACME read-only validation passes, changes are committed, and the cycle log is written.

240.1 Authorization flags

Flag	Action
RCX_ALLOW_MERGE=0 (default)	Do not merge PRs
RCX_ALLOW_PUBLISH=0	Do not push GHCR images
RCX_ALLOW_DEPLOY=0	Do not deploy to ACME

Set to 1 only when the repo owner authorizes.

240.2 Image bump (when authorized)

1. Tests and `gh pr` checks green.
2. Bump patch tag only.
3. `./scripts/docker_build.sh` and publish via existing GHCR workflow.
4. `./scripts/upgrade_edge_full.sh --limit <host>` when deploy authorized.
5. Re-run overnight validator with `OPENFDD_LIVE_ACME=1`.

241 Report Builder

242 RCx Report Builder (RCx Central)

242.1 Flow

1. Select saved Edge site
2. Choose date range (2h / 24h / 7d)
3. **Collect Data / Preview** — POST /api/central/rcx/preview
4. **Preview Charts** — matplotlib PNG as base64 in browser
5. Select chart/section checkboxes
6. **Generate DOCX** — POST /api/central/rcx/report

Reports save to portfolio/data/reports/ when save_to_volume=true.

242.2 Fault overlays

Preview charts include severity-colored bands when fault rows exist and overlay toggle is on.

242.3 DOCX

Uses open_fdd/reports/rcx_docx.py when available (selectable sections/charts). Falls back to portfolio/central/rcx_report.py.

Read-only safety note is included in every report.

243 Rcx Report Builder

244 RCx Report Builder

The **RCx Report Builder** (/analytics/rcx) collects a read-only snapshot from the edge bridge and generates a Word document.

244.1 Workflow

1. Choose report window (2 h / 24 h / 7 d).
2. Click **Collect Data / Preview** — calls `POST /api/reports/rcx/preview`.
3. Review available vs disabled charts (disabled charts show missing point roles or no fault data).
4. Select sections and charts.
5. Click **Generate DOCX** — `POST /api/reports/rcx/generate` returns `application/vnd.openxmlformats-officedocument.wordprocessingml.document` and saves to `workspace/reports/rcx/`.
6. Use **Saved RCx reports** on the same tab to preview, download, or delete prior DOCX files (UI or agent-generated).

244.2 Implementation

- Preview/readiness: `workspace/api/openfdd_bridge/rcx/chart_preview.py`
- Placeholder hints (BRICK roles → columns → paste instructions): `open_fdd/reports/rcx_placeholders.py`
- DOCX builder: `open_fdd/reports/rcx_docx.py`
- Charts: `open_fdd/reports/charts.py` (matplotlib → BytesIO, no temp PNGs)

Each DOCX placeholder includes system type (AHU/VAV/zone/plant), historian column names from the BRICK model, and disabled-chart gaps in the appendix. FDD rule sections list bound sensor columns per Rule Lab assignment.

244.3 Safety

Report generation is **read-only**. No BACnet writes, commands, overrides, or setpoint changes.

244.4 Dependencies

Bridge API requires `python-docx` and `matplotlib` (see `workspace/api/requirements.txt`).

244.5 Central vs edge

Portfolio Central (portfolio/central/rcx_report.py, port 8060) supports multi-site rollups. The edge builder uses local FDD results and building status for single-site RCx exports from the operator UI.

245 3.1.2

246 Open-FDD 3.1.2

Release date: 2026-06-15

246.1 Highlights

- **DataFusion / Rust-prep validation** — Bench long FDD smoke exercises PyArrow and DataFusion SQL backends with execution evidence and backend equivalence checks.
- **Bench 5007 long FDD smoke** — Model-driven dual-source validation (BACnet MS/TP device 5007 + Niagara bench9065) with 10-minute fault confirmation window.
- **Fault confirmation window** — Confirmed faults require consecutive true samples; reports include raw vs confirmed timing.
- **GHCR tags** — ghcr.io/bbartling/openfdd-bridge:3.1.2, openfdd-commission:3.1.2, openfdd-mcp-rag:3.1.2.

246.2 Known warnings

- `confirmation_engine=python_loop_over_arrow_mask` — confirmation still uses a Python loop over the Arrow fault mask; vectorized confirmation is planned.

246.3 Validation

```
python scripts/smoke_bench_5007_long_fdd.py --synthetic --dry-run
python scripts/smoke_bench_5007_long_fdd.py --live --duration-
minutes 35 --baseline-minutes 10 \
    --confirmation-rows 10 --confirmation-minutes 10 --poll-seconds
60 --strict-live-freshness
```

```
python scripts/inspect_bench_5007_long_fdd_report.py workspace/
reports/bench_5007_long_fdd_*.json
```

Reports now classify **historian_mode** (live_recent, historical_replay, synthetic, dry_run) and refuse to claim live polling was validated when sample timestamps are outside the wall-clock run window. Use --allow-historical-replay only for debugging replay data.

For live bench sign-off, the inspector must show data_is_fresh: true and historian_mode: live_recent, plus non-zero BACnet/Niagara rows.

247 3.1.3

248 Open-FDD 3.1.3

Release date: 2026-06-15

248.1 Patch fixes

- **Bench long FDD evaluate** — when wall-clock run_started_at / lookback filtering would empty a non-empty historian table (demo CSV fallback or stale feather replay), evaluate now uses the full loaded table and sets time_filter_relaxed: true. Strict smoke freshness gates still FAIL/WARN honestly on non-fresh samples.
- **Live smoke lookback** — final matrix evaluate uses the same 24h minimum lookback as cycle evaluates.

248.2 GHCR tags

- ghcr.io/bbartling/openfdd-bridge:3.1.3
- ghcr.io/bbartling/openfdd-commission:3.1.3
- ghcr.io/bbartling/openfdd-mcp-rag:3.1.3

248.3 Validation

```
python scripts/smoke_bench_5007_long_fdd.py --synthetic --dry-run
python scripts/smoke_bench_5007_long_fdd.py --live --duration-
minutes 35 \
    --baseline-minutes 10 --confirmation-rows 10 --confirmation-
minutes 10 \
```

```
--poll-seconds 60 --strict-live-freshness  
python scripts/inspect_bench_5007_long_fdd_report.py workspace/  
reports/bench_5007_long_fdd_*.json
```

249 Release 3.1.5

250 Release 3.1.5

250.1 Dashboard & operator UX

- BAS-style latched alarms with **Clear** for authenticated operators (/api/faults/clear)
- Optimistic BACnet / Modbus / JSON API driver tree deletes (no JS confirm except Pooge nuclear reset)
- Rule Lab SQL CodeMirror editor; Host Stats Pooge + Ollama disabled styling
- Public read-only building snapshot for wall displays

250.2 Rule authoring docs (v1)

- New docs/rule-authoring/ — PyArrow vs DataFusion decision table, Arrow contract, legacy pandas parity, FC4 modernization, Rust checklist

250.3 Fixes

- CI: rename duplicate test_cookbook.py pytest module collision
- WebSocket redacted dashboard includes fault families (not empty)

250.4 Smoke

Detached paired FDD harness default for short / standard / overnight — see [paired FDD smoke](#).

251 Release 3.1.6

252 Release 3.1.6

252.1 Dashboard & operator UX

- Building summary: BACnet P8 override table (top 8), per-device coverage, hourly scan health
- Rule Lab: column preview for PyArrow + DataFusion SQL; SQL actions above editor
- Niagara tab: test URL/password before save, folder tree checkboxes, saved-folder discover, schedules panel

252.2 BACnet & RCx

- Override scan health probes (scan_health staleness, errors)
- RCx DOCX: override tables, per-device breakdown, scan status in model health

252.3 Platform

- GHCR Docker images: **linux/amd64** + **linux/arm64** (Raspberry Pi edge)
- Niagara commission_profile persistence; discover-profile and test-draft APIs

252.4 Smoke

Paired FDD harness and override scan validation unchanged — see paired FDD smoke.

253 Scope

254 SCOPE — open-fdd, rebuilt on a Brick-conformant SurrealDB substrate

Status: **draft / design**. This is the scope for the next-generation open-fdd: an open-source **supervisory fault detection & diagnosis (FDD)** platform for buildings, rebuilt so that the equipment model **is a conformant Brick model**, telemetry is a first-class time-series plane, faults are deterministic, auditable, offline-capable rules, and a future **agent mode** (Rig over SurrealDB vectors) reasons over the same store. The same binary runs on one building/station at the edge and across a portfolio in the cloud.

The domain (equipment, points, telemetry, faults, RCx) is the product. The substrate ideas (single SurrealDB store; an inject plane; a Rhai-decision + DataFusion-compute rule engine; an access gate with audit/undo; edge→cloud sync; an agent over vectors) are borrowed, but this doc stands on its own — it is **not** a generic-platform doc with FDD bolted on.

254.1 The one hard requirement: MUST match Brick

The equipment/point/location model is **not** “tags that resemble Brick.” It is a **Brick-conformant model**:

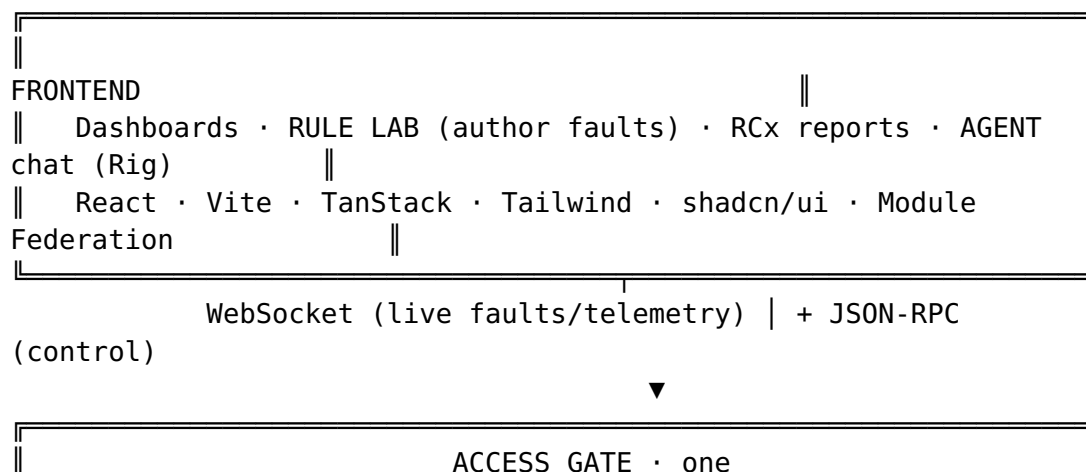
1. Every modelled entity is an instance of a **real Brick class** (e.g. `brick:Air_Handling_Unit`, `brick:Supply_Air_Temperature_Sensor`).
2. Every relationship is a **real Brick object property** (`brick:hasPoint`, `brick:feeds`, `brick:hasPart`, `brick:hasLocation`, ...) with Brick’s inverse semantics preserved.
3. Units are **QUDT** via `brick:hasUnit` (e.g. `unit:DEG_F`).
4. A Point binds to its telemetry the **Brick-native way**:
`ref:hasExternalReference` → `ref:TimeseriesReference` carrying `ref:hasTimeseriesId` and `ref:storedAt` ([Ref Schema](#), [Brick timeseries storage](#)).
5. The model **round-trips to Brick Turtle (.ttl)** and **validates against Brick’s SHACL shapes** — a real Brick model imports and re-exports with no semantic loss, and an invalid model is rejected at the gate.

Conformance is a contract, not a nicety: it is what lets open-fdd interoperate with existing Brick models, SPARQL tooling, and other Brick-aware systems, and it is what makes “bind a rule to every brick:Supply_Air_Temperature_Sensor that feeds this AHU” a portable query instead of a bespoke one. **Where this doc and the Brick ontology disagree, Brick wins.**

254.2 Principles

1. **Brick-conformant or it doesn't ship.** The graph is a faithful Brick model (classes, relationships, units, timeseries refs), TTL-round-trippable and SHACL-validated. See *The Brick model* below.
2. **One binary, edge to cloud.** Same build on a building controller, a Windows box, or the cloud. Difference is configuration (single station vs. portfolio).
3. **Faults fire offline.** The rule/fault runtime is embedded and needs no cloud.
4. **SurrealDB is the single store.** Graph (the Brick model), time-series (telemetry), vector (RAG/agent memory), document (config), auth, live queries, and change feeds in one engine.
5. **Telemetry is data-plane, append-only.** Millions of samples never enter the audit/undo/command path; they append and age out.
6. **Commands cross the gate; reads are SurrealDB-native.** Every mutation is audited/undoable with a correlation id; reads run on a scoped SurrealDB session enforced by row-level permissions. **The agent is a scoped principal too** — its writes cross the same gate.
7. **Python is the oracle, not the runtime.** Existing open-fdd Python rules are the golden reference; each Rust/Rhai/SQL rule is validated against them with fixtures. Python never runs on the edge hot path.

254.3 Architecture



Technology	Use case in this design	What it was in open-fdd (Python)	Reused / migration path
			SPARQL parity becomes a CI check
DataFusion + Arrow	Vectorised window rollups; SQL fault rules; the compute engine	PyArrow compute + optional DataFusion SQL rules	DataFusion SQL rules port almost directly; PyArrow rule logic re-expressed and parity-tested
Rhai	Embedded, sandboxed fault <i>decision</i> + rule composition (offline)	Python rule functions (def high_sat(table, cfg)...))	Python is the oracle; G36 decision logic re-authored in Rhai, validated vs Python fixtures
Fault confirmation (min_elapsed_minutes/min_true_rows)	Debounce/hysteresis before a fault is recorded	Same config mechanics, in Python	Mechanics carried over 1:1; validated against Python outputs
Inject pipeline (decimate→filter→enrich)	Field→edge acquisition + pre-sync gate	openfdd-commission (BACnet discovery/poll), MQTT, edge bridge	BACnet/Niagara drivers reused as inject sources pointed at the append endpoint
SurrealDB CHANGEFEED + shipper / native sync	Edge→cloud replication, partition-by-station	Deferred in open-fdd (#334)	New capability; no Python equivalent to migrate

Technology	Use case in this design	What it was in open-fdd (Python)	Reused / migration path
Access gate (audit/undo/correlation-id)	One mutation chokepoint; SHACL validation; “why did this fire” trace	Ad-hoc; no unified audit/undo/trace	New; trace consumes the same correlation id ingest/gate mint
Rig (rig-core) + SurrealDB vectors	Future agent mode: Brick-grounded RAG + tool-calling over the store	openfdd-mcp-rag (MCP + doc search)	MCP-RAG capability generalised into the Rig agent; embeddings move into SurrealDB
React/Vite/TanStack/shadcn + Module Federation	Dashboards, Rule Lab, RCx, agent chat	Existing browser Rule Lab + dashboard (TS)	Rule Lab front-end reused; points at the gated rule API
Python (pandas/pyarrow + ML)	Oracle + out-of-process ML extension — never on the edge hot path	The whole runtime	Stays as golden-fixture reference and a scoped ML extension; not embedded in the edge binary

254.5 The Brick model — the heart of the system

Brick is RDF/OWL with a Turtle serialization, a deep **class hierarchy**, **inverse relationships**, **QUDT units**, and **SHACL** validation. SurrealDB is a property/ graph store, not an RDF triple store with a reasoner. “MUST match Brick” therefore means we **carry Brick’s semantics explicitly** rather than assume an RDF engine provides them. Four mappings make the model conformant.

254.5.1 1. Entities are Brick-class instances — SCHEMAFULL, typed class reference

The core Brick tables are **SCHEMAFULL**, not schemaless: a building model is a constrained ontology, and the store should refuse anything that isn't conformant rather than accept free-form documents. `class` is a **typed record link** to a `brick_class` row (not a bare string), so an entity literally cannot carry a class that isn't in the loaded ontology.

```
DEFINE TABLE entity SCHEMAFULL; -- a
Brick instance (equipment/point/location/...)
DEFINE FIELD class ON entity TYPE record<brick_class>; -- typed
link → the ontology, not a loose CURIE
DEFINE FIELD label ON entity TYPE string;
DEFINE FIELD station ON entity TYPE string; -- edge
partition (one building/station)
DEFINE FIELD unit ON entity TYPE option<record<qudt_unit>>; --
QUDT term for Points (brick:hasUnit)
DEFINE INDEX entity_station_class ON entity FIELDS station, class;
```

`class` resolves to a real ontology row by construction, and `unit` to a QUDT term. The instance is a SurrealDB record, but its identity is Brick: it has a class IRI and participates in Brick relationships.

254.5.1.1 A lossless RDF sidecar — so TTL round-trip never drops detail

The operational tables above hold the **projected, queryable subset** open-fdd actually runs on (`class`, `station`, `unit`, relationships, timeseries refs). They are deliberately *not* a complete RDF representation — a SCHEMAFULL projection cannot carry blank nodes, datatype literals, language tags, repeated/annotated refs, or non-core Brick annotations. To keep the “*MUST round-trip to Brick TTL*” contract **lossless**, every import also lands the raw model in a sidecar:

```
DEFINE TABLE brick_assertion SCHEMAFULL; -- one row per
imported RDF triple (lossless)
DEFINE FIELD subject ON brick_assertion TYPE string; -- IRI or
_:blank
DEFINE FIELD predicate ON brick_assertion TYPE string;
DEFINE FIELD object ON brick_assertion TYPE string; -- IRI /
literal (with datatype + lang preserved)
DEFINE FIELD graph ON brick_assertion TYPE string; -- source
model / named graph
DEFINE INDEX brick_assertion_spo ON brick_assertion FIELDS
subject, predicate, object;
```

Flow: **import** writes brick_assertion (lossless) **and** projects the operational subset into entity / relations / timeseries refs; **export** regenerates TTL from brick_assertion, reconciled with any operational edits made since import. This is the split that lets the graph tables stay lean and typed while the round-trip stays faithful — the projection is for *running*, the sidecar is for *fidelity*.

254.5.2 2. The Brick class hierarchy is loaded, so subclass inference works

The single thing a property graph does **not** give you for free is RDFS subclass reasoning — “give me all brick:Temperature_Sensor” must also match brick:Supply_Air_Temperature_Sensor (a subclass). We make this native by loading the Brick ontology’s class graph:

```
DEFINE TABLE brick_class SCHEMALESS;          -- one row per
Brick class
DEFINE FIELD curie ON brick_class TYPE string; --
"brick:Temperature_Sensor"
-- subClassOf as graph edges, straight from the Brick ontology:
--  brick_class:Supply_Air_Temperature_Sensor ->subClassOf->
brick_class:Air_Temperature_Sensor -> ...
RELATE brick_class:$child->subClassOf->brick_class:$parent;
```

A query for “all Temperature_Sensors under AHU-1” walks the class closure and the Brick relationship in one SurrealQL graph traversal — the property-graph equivalent of a SPARQL query over an inferred model. For speed, the transitive subclass closure is materialised (a subclass_of_closure table refreshed when the ontology version changes), so rule binding is an index lookup, not a recursive walk per evaluation.

254.5.3 3. Relationships are Brick object properties, inverses preserved

Each Brick object property is a **typed, enforced RELATION table** — not an untyped edge — with in/out constrained to entity and ENFORCED so a dangling or mistyped edge is rejected at write. The relation table definitions are **generated from the Brick ontology** (one per supported object property), not hand-listed, so the closed set tracks the pinned Brick version automatically:

```
-- generated per Brick object property (hasPoint, feeds, hasPart,
hasLocation, ...):
DEFINE TABLE hasPoint TYPE RELATION IN entity OUT entity ENFORCED;
DEFINE TABLE feeds    TYPE RELATION IN entity OUT entity ENFORCED;
-- domain/range narrowing (e.g. hasPoint OUT must be a brick:Point
subclass) is
-- checked at the gate against the ontology + SHACL; see
Conformance below.
```

```

RELATE entity:ahul->hasPoint->entity:sat_sensor;    --
brick:hasPoint
RELATE entity:ahul->feeds->entity:vav_3;            -- brick:feeds

```

Brick relationships are inverse-paired (hasPoint/isPointOf, feeds/isFedBy, hasPart/isPartOf, hasLocation/isLocationOf). We store the **canonical direction once** as the enforced relation and traverse the inverse with SurrealDB's reverse edge syntax (entity:sat_sensor<-hasPoint<-entity), so there is no double-write to keep consistent. On **TTL export** the inverse triples are emitted as Brick expects; on **import** an inverse triple is normalised back to its canonical edge. The closed set of supported relations is the Brick version we target (pinned — see open questions).

254.5.4 4. A Point binds to telemetry the Brick-native way

This is the conformant seam between the **graph** and the **time-series plane**. Brick already standardises it: a Point carries ref:hasExternalReference to a ref:TimeseriesReference whose ref:hasTimeseriesId is the key of the data in some store (timeseries storage).

```

:sat_sensor a brick:Supply_Air_Temperature_Sensor ;
  brick:hasUnit unit:DEG_F ;
  ref:hasTimeseriesReference [ a ref:TimeseriesReference ;    #
specific sub-property
  ref:hasTimeseriesId "ahul.sat" ;
  ref:storedAt :openfdd_surreal ] .

```

The reference is a **first-class record**, not an embedded blob — so it can be queried, indexed, and shared (a BACnet object reference and a timeseries reference can coexist on the same point), and import/export preserves the exact property used:

```

DEFINE TABLE ts_ref SCHEMAFULL;                -- a
ref:TimeseriesReference instance
DEFINE FIELD timeseries_id ON ts_ref TYPE string; --
ref:hasTimeseriesId – the telemetry series key
DEFINE FIELD stored_at    ON ts_ref TYPE option<string>;--
ref:storedAt (database/source)
DEFINE INDEX ts_ref_tsid  ON ts_ref FIELDS timeseries_id UNIQUE;
-- point → ref as an enforced relation; the property used is
preserved for round-trip:
DEFINE TABLE hasExternalReference TYPE RELATION IN entity OUT
ts_ref ENFORCED;
DEFINE FIELD property ON hasExternalReference TYPE string; --
"ref:hasTimeseriesReference" | "ref:hasExternalReference"

```

Brick exposes a **generic** ref:hasExternalReference and a **specific** ref:hasTimeseriesReference (a sub-property of it) (Ref Schema). We store which property the source used (property) so export emits

the same one — generic stays generic, specific stays specific — and a BACnet/IFC external reference uses the same table with a different target type.

The `ts_ref.timeseries_id` **is** the series key of the telemetry plane. So the Brick model says *what a point means* and *where its data lives*, and the telemetry table holds the samples under that same id — no parallel point registry, no drift.

254.5.5 Conformance: validation + round-trip (the contract)

- **SHACL validation at the gate.** Modelling/editing a Brick entity is a command; the gate validates the write against Brick's SHACL shapes (class exists; the relationship's domain/range are legal for the two entities' classes; required properties present; unit is a QUDT term). Invalid models are **rejected**, not stored. Full SHACL runs in CI/import; a fast native subset runs on the hot per-write path (open question 4).
- **Turtle round-trip.** `import` parses a Brick `.ttl` into entity rows + relationship edges + timeseries refs; `export` regenerates `.ttl` (canonical + inverse triples, units, refs). A conformance test asserts `export(import(ttl)) == ttl` semantically for the Brick reference models — this is the objective “matches Brick” gate in CI.
- **SPARQL-style queries.** Rule binding and dashboards query the graph by class + relationship + subclass closure. We do **not** run a SPARQL engine inside SurrealDB; we compile the small set of query shapes we need to SurrealQL graph traversals. (A full SPARQL endpoint over an exported model is an optional cloud add-on.)

254.6 Telemetry plane — first-class time-series, keyed by the Brick timeseries id

A reading is not a document. Telemetry is a dedicated append-only plane, keyed by the Point's `ref:hasTimeseriesId`:

```
DEFINE TABLE telemetry SCHEMALESS;           -- data plane,
append-only
DEFINE FIELD series ON telemetry TYPE string;  -- ==
ref:hasTimeseriesId of the Brick Point
DEFINE FIELD at ON telemetry TYPE datetime;  -- MEASUREMENT
instant (not write time)
DEFINE FIELD value ON telemetry TYPE number;
DEFINE FIELD station ON telemetry TYPE string; -- edge partition
DEFINE INDEX telemetry_station_series_at ON telemetry FIELDS
station, series, at;
```

-- scoped read perms; create/update/delete NONE on scoped sessions
(appends on owner handle)

- **Bucket on at** (measurement time), never write time — the trend-collapse bug that bites any “bucket on row-created” design.
- **Deterministic id from (series, at)** so re-poll, backfill, and sync-replay are idempotent no-ops.
- **Rollups behind a historian boundary** are the first scale lever (pre-aggregated `telemetry_rollup` per grain); an external TSDB is an optional datasource behind the same boundary, never required.
- **Unit/quantity live on the Brick Point** (`brick:hasUnit`), not on every sample — rows stay `{series, at, value}` lean.

254.6.1 The historian boundary — SurrealDB rows → Arrow/DataFusion batches

open-fdd is **Arrow-native**, but the store is SurrealDB. The seam that bridges them is the **historian boundary**, defined early because it’s where a lot of performance and correctness lives. It is a single read interface — *“give me series S (or a Brick-resolved set), window $[t_0, t_1]$, grain G ”* — that returns **Arrow RecordBatches** ready for DataFusion, and decides behind the caller’s back how to produce them:

- **Paged range-scan → Arrow.** For raw/live-tail reads it issues a filtered SurrealQL range scan (`WHERE station=$s AND series=$x AND at BETWEEN $t0 AND $t1 ORDER BY at`) over the `telemetry_station_series_at` index, **paged**, and assembles columnar Arrow batches (not row-by-row into the engine).
- **Rollup table for coarse grains.** Above a threshold it reads `telemetry_rollup` instead of raw rows; epoch-aligned to the engine’s grain so a chart bucket and a rule bucket line up exactly.
- **Optional derived Arrow/Parquet cache.** Hot windows (a rule’s recent evaluation window, a dashboard’s visible range) can be materialised to a **local Arrow/Parquet cache** so repeat reads skip SurrealDB entirely — and this is the same seam an external TSDB (TimescaleDB as a DataFusion TableProvider) plugs into if SurrealDB’s time-series read path stops keeping up. No rule or chart changes when the source swaps.

Defining this boundary now (not after volume bites) keeps the rest of the system talking **Arrow**, not SurrealQL, so the swap surface is one interface.

254.7 FDD engine — rules bind to Brick, decide in Rhai, confirm, record a fault

The decisive design choice: **a rule does not name point ids — it names a Brick query**. A G36 rule says “for each brick:Air_Handling_Unit, take its brick:Supply_Air_Temperature_Sensor and brick:Mixed_Air_Temperature_Sensor via brick:hasPoint...”. The engine resolves that query (with subclass closure) to concrete points and their hasTimeseriesIds, then runs the math. This is what makes a rule **portable across buildings** — the whole point of modelling in Brick.

Three stages, mirroring open-fdd’s existing mechanics:

1. **DataFusion + Arrow** computes window rollups over the bound series (avg/min/max over N minutes). Heavy vectorised math lives here. Pure-threshold faults may be expressed directly as **DataFusion SQL rules** (ported from today’s open-fdd).
2. **Rhai** makes the deterministic decision (G36 logic, economizer/SAT/valve-hunting faults), composes sub-rules via `invoke()`, and is sandboxed (bounded ops, no I/O, fail-closed).
3. **Fault confirmation** applies open-fdd’s `min_elapsed_minutes / min_true_rows` plus debounce/hysteresis, then records a **fault** as an audited insight with a **per-fault trace** (“which points, which window values, why it fired”).

Rules do not fire per sample. At telemetry volume, running rule evaluation on every appended row would melt the engine. Instead ingest **marks dirty series/windows** as samples land (a cheap (series, bucket) dirty-set), and a **scheduler batches** DataFusion runs over those dirty windows on a tick — so evaluation cost scales with *changed windows*, not sample count, and DataFusion sees real Arrow batches (via the historian boundary) rather than one-row calls. A rule may also be run on demand (dry-run, manual re-check). The after-write hook path is reserved for **low-volume config** changes (e.g. a rule edit triggers a re-evaluation), never for telemetry.

Faults are recorded back to SurrealDB and published on live queries (the dashboard’s realtime feed); a **dry-run** path runs a draft rule against real history with no side effects (the Rule Lab preview).

254.8 Faults, RCx, and traces

- A **fault** is an insight record linked to the Brick entity it concerns (so “all faults for AHU-1” / “all economizer faults across the portfolio” are graph queries).
- **RCx reports** are documents assembled from fault history + Brick context, the cloud-side retro-commissioning surface.
- **Traces** answer “why did this fire”: the Rhai evaluation emits a span tree (sub-rules, window values, decision), correlated by id from the same gate/ingest correlation id that stamps audit and faults.

254.9 Inject plane — field to edge store, and Brick discovery

Acquisition (BACnet/Niagara discovery + polling, MQTT) lands samples on the edge telemetry table after a preprocess pipeline (decimate→filter→enrich). The mapping turns a sample into { series = the point's hasTimeseriesId, at, value }. **Discovery also proposes Brick entities**: a BACnet discovery run suggests candidate equipment and points with candidate Brick classes; an operator confirms/edits the class (the one human step Brick conformance requires), and the confirmed entities + timeseries refs land in the graph. The capability decision is taken once (at subscribe / per batch), not per sample; appends bypass the command gate but stay on the station partition.

254.10 Sync plane — station to portfolio

Edge→cloud movement replicates the **store**, not the field. Telemetry and faults are append-only and **partitioned by station**, so two edges never write the same row and reconciliation is ordering+dedup, not merge. The Brick model and rule definitions are the real conflict surface, handled by ownership + last-write-wins + audit tiebreak. Capture uses SurrealDB **CHANGEFEED**; transport is a shipper. (Native SurrealDB edge-sync, if/when confirmed GA for the self-hosted engine, can replace the shipper — but the partition-by-station conflict model stays either way, because sync moves bytes and does not resolve domain conflicts.)

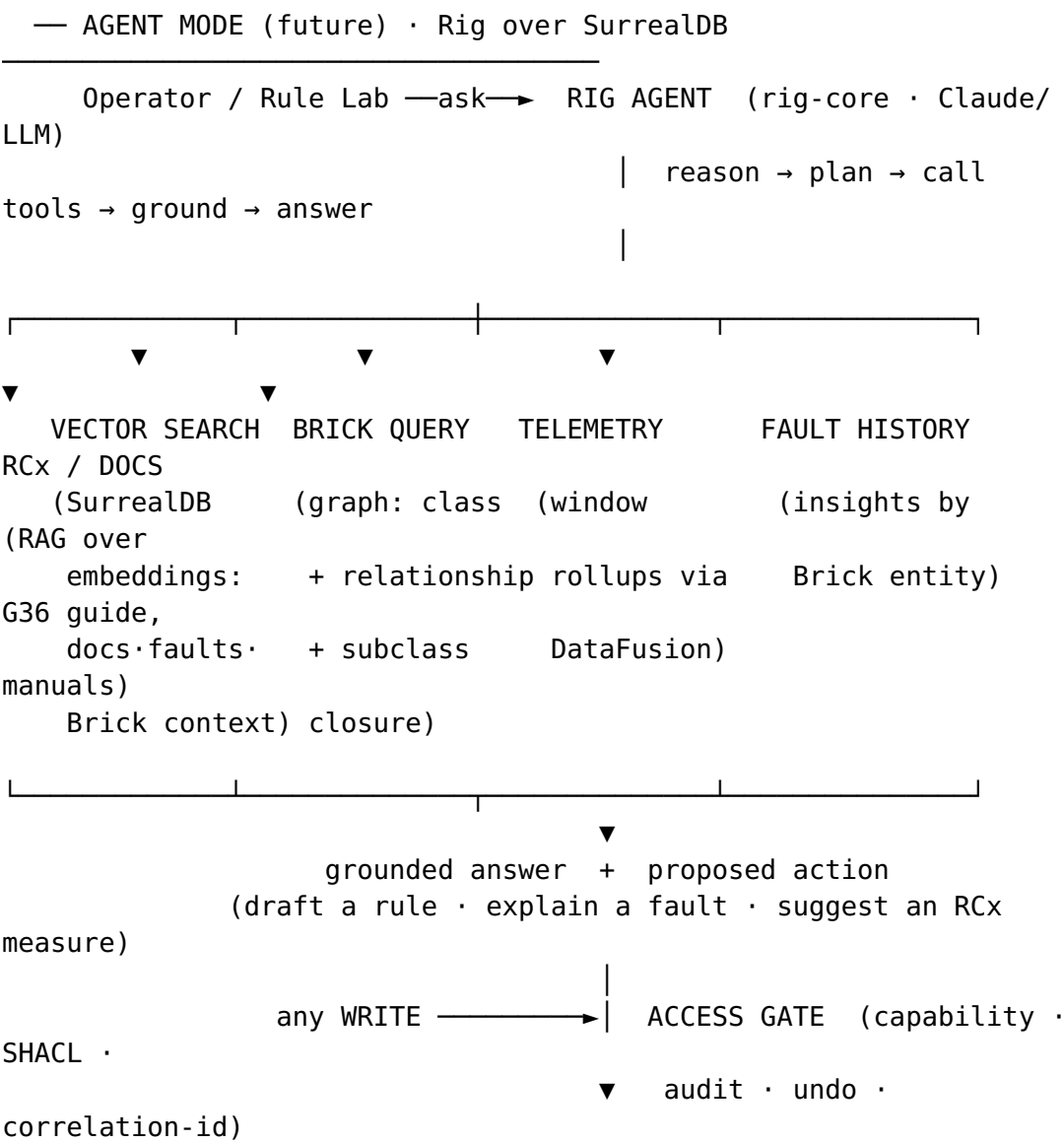
254.11 Access gate, audit, undo

Every command (author a rule, model/edit a Brick entity, write config, **apply an agent proposal**) crosses the gate: authenticate → capability grant → **Brick/SHACL validation** → capture before/after for audit + undo → mint correlation id → apply. Reads

(charts, live faults, Brick queries, agent retrieval) run on a scoped SurrealDB session under row-level permissions. Telemetry appends are data-plane and never produce audit/undo per sample.

254.12 Agent mode (future) — Rig over SurrealDB vectors

A forward-looking mode: an LLM agent an operator (or the Rule Lab) can ask in natural language — “*why is AHU-3 faulting this week?*”, “*draft a stuck-damper rule for the rooftop units*”, “*summarise RCx findings for site 7*” — that reasons over **the same SurrealDB store** the rest of the system uses. Built on **Rig** (rig-core), a Rust LLM-agent framework with a **SurrealDB vector-store integration** (rig-surrealdb), so embeddings live **beside** the Brick model, telemetry, and faults — no second vector database (principle 4).



the agent is a SCOPED PRINCIPAL, like a user

Design points, each load-bearing:

- **Embeddings live in SurrealDB beside the data.** Vector search is SurrealDB-native, so retrieval can be **Brick-filtered** (only this site's AHUs; only economizer faults) by joining a vector hit back to its Brick entity — RAG that knows the building's structure, not just text similarity.
- **The agent's tools are the system's existing read surfaces.** Brick graph query, the telemetry historian, fault history, RCx/doc RAG — the agent calls the same capabilities the FDD engine and dashboards use. No privileged backdoor.
- **Reads are free; writes cross the gate.** The agent answers and *proposes* (a rule draft, a model edit, an RCx note); applying any of it is a gated, audited, undoable command — the agent holds capability grants like any principal, and a proposed rule still runs **dry-run against real history** before a human commits it.
- **Edge or cloud.** Lightweight retrieval/grounding can run at the edge; heavier multi-step reasoning is a cloud add-on. Either way it reads the same synced store.

This is where today's MCP-RAG capability lands, generalised from doc search to a Brick-grounded agent. It is explicitly **future** — the deterministic FDD engine is the product; the agent is an assist layer over it, never the thing that silently changes config.

254.13 Edge and cloud profiles

- **Edge** — one station, single SurrealDB namespace, faults fire offline, local dashboard, lightweight agent retrieval, sync up when connected.
- **Cloud** — portfolio (namespace-per-tenant or per-site), fleet RCx, cross-site rollups, full agent reasoning, optional SPARQL endpoint.

254.14 Build order (smallest load-bearing first)

1. **Brick model core** — SCHEMAFULL entity (typed class) + brick_class hierarchy + ontology-generated **enforced relation** tables + first-class ts_ref + the lossless brick_assertion sidecar; TTL import/export; SHACL validation in CI. The conformance gate. *Nothing else is correct until this is.*
2. **Telemetry plane** — telemetry table keyed by hasTimeseriesId, bucket on at, deterministic id, scoped perms.

3. **Inject** — BACnet/Niagara/MQTT acquisition → preprocess → telemetry; discovery proposes Brick entities for operator confirmation.
4. **FDD engine** — DataFusion window + Rhai decision + fault confirmation; rules bind via Brick queries; dry-run; record faults + traces. Validate each rule against the Python oracle with fixtures.
5. **Frontend** — dashboards, Rule Lab, RCx.
6. **Sync + cloud** — CHANGEFEED shipper, portfolio profile, fleet RCx.
7. **Scale levers** — telemetry rollups behind the historian boundary; optional external TSDB / SPARQL endpoint.
8. **Agent mode (Rig)** — embeddings in SurrealDB; Rig agent with Brick-query / historian / fault-history / RAG tools; gated writes for proposals. Built last, over a store that already works.

254.15 Open questions

1. **Brick version pin.** Which Brick release (1.3 / 1.4) defines our closed set of classes, relationships, and the ref: schema — and the ontology-refresh process when it bumps (re-materialise the subclass closure).
2. **SHACL on the hot path.** Full pyshacl-grade validation in CI/import vs. a native fast subset at the gate per write — which checks are mandatory inline vs. deferred.
3. **Subclass-closure materialisation.** Refresh trigger and storage for subclass_of_closure; cost at full Brick class cardinality.
4. **Inverse relationships.** Confirm “store canonical + traverse reverse” is sufficient everywhere, or whether any Brick query shape forces materialising both directions.
5. **TimeseriesReference identity.** ref:hasTimeseriesId = the human key (ahu1.sat) vs. an opaque id; how it relates to telemetry series cardinality and indexing.
6. **Discovery → Brick classification.** How much can be auto-proposed (BACnet object names → candidate Brick classes) vs. mandatory operator confirmation; where the ML/heuristic lives. (An early agent-mode assist candidate.)
7. **SurrealDB time-series at portfolio cardinality** — where rollups stop sufficing and an external historian is warranted.
8. **Native edge-sync vs. CHANGEFEED shipper** — verify the self-hosted SurrealDB sync mechanism, bidirectionality, and conflict semantics before retiring the shipper.
9. **SPARQL surface** — do we need a real SPARQL endpoint (cloud add-on over exported models) or are compiled SurrealQL query shapes enough for rules + dashboards.
10. **Agent embedding model & SurrealDB vector maturity** — which embedding model, edge vs. cloud embedding, and whether SurrealDB’s vector index meets retrieval latency at

portfolio scale (else an external vector store behind a retrieval seam).

11. **Agent write authority** — which proposals an agent may *apply* under a grant vs. always human-in-the-loop (rule authoring, Brick edits, RCx notes).

254.16 Alternative substrate considered — TimescaleDB/Postgres + a graph add-on

The whole design above bets on **one engine** (SurrealDB) covering graph, time-series, vector, document, and auth. The obvious alternative is the **mature, battle-tested** combination most teams reach for: **PostgreSQL + TimescaleDB** for telemetry, plus a **graph add-on** for the Brick model (Apache AGE as a Postgres extension, or a separate graph DB), with **pgvector** for embeddings. This is recorded as a considered option, not the chosen one.

What it buys:

- **Telemetry at scale, proven.** TimescaleDB hypertables, continuous aggregates, and compression are purpose-built for exactly the high-cardinality time-series load that is SurrealDB's least-proven axis here — it directly retires open question 7.
- **Operational maturity.** Backups, replication, HA, monitoring, and hiring are all solved for Postgres in a way they are not yet for SurrealDB.
- **pgvector** is a well-understood vector index for the agent's RAG retrieval.
- **Brick on a real graph.** Apache AGE (openCypher in Postgres) or a dedicated graph store can hold the Brick model; the typed-relation/closure design above maps onto it.

What it costs — and why it is the alternative, not the plan:

- **Edge→cloud sync does not come for free, and largely does not work the way this design needs.** The entire sync plane here relies on **one engine running embedded at the edge and in the cloud** with change-feed-driven, partition-by-station replication. Postgres/TimescaleDB is **not an embedded, offline-first edge engine**: you'd run a real server per building, and edge→cloud movement becomes logical-replication / Debezium-style CDC plumbing that is far heavier to operate at fleet scale, doesn't run truly offline-first the same way, and has its own conflict story to build. **Assume the edge-sync model in this doc does not carry over** — it would need a different, more operationally expensive sync design.

- **Three engines, three operational surfaces.** Postgres + Timescale + AGE/graph + pgvector is several moving parts to deploy, version, and secure on a building controller — against SurrealDB’s single binary. The “one binary, edge to cloud” principle is lost.
- **Two query languages and a join seam.** Brick (Cypher/AGE) and telemetry (SQL/ Timescale) live in different engines; the “resolve a Brick query → its series → window math” path that is one traversal here becomes a cross-engine join.

Net: this alternative is the **stronger choice if telemetry scale or Postgres operational maturity is the dominant risk and the fleet is mostly cloud-side** — but it sacrifices the embedded single-binary edge story and, critically, the cheap edge→cloud sync that is a core goal of this design. It stays on the table as a fallback if SurrealDB’s time-series or sync axes fail to hold at real cardinality (open questions 7 and 8).

254.17 References

- Brick Schema — <https://brickschema.org>
- Ref Schema (external references / timeseries) — <https://ref-schema.brickschema.org/>
- Brick timeseries storage model — <https://docs.brickschema.org/metadata/timeseries-storage.html>
- Rig (Rust LLM-agent framework) — <https://docs.rig.rs>
- open-fdd Python→Rust cutover (issue #334) — <https://github.com/bbartling/open-fdd/issues/334>